



SEIRD-МОДЕЛЬ ДИНАМИКИ РАСПРОСТРАНЕНИЯ ВИРУСНЫХ ИНФЕКЦИЙ С УЧЕТОМ ВОЗНИКНОВЕНИЯ НОВЫХ ШТАММОВ

Андреева Софья Владимировна НПИБд-01-23

Введение

Детерминистские математические модели эпидемий заболеваний позволяют изучать макроскопические явления распространения этих эпидемий в человеческом обществе. Однозначность, наглядность и простота этих моделей делают их привлекательными для анализа и позволяют получить информацию для управления протеканием эпидемии.

При создании таких моделей делается ряд упрощений: развитие эпидемии изучается в однородной непрерывно и равномерно перемешивающейся большой группе, что позволяет рассматривать общие процессы распространения эпидемии в упрощенной форме. Для успешной борьбы с эпидемиями заразных заболеваний недостаточно одних только профилактических мероприятий или лекарственного лечения, необходимо также учитывать, что существуют эпидемиологические проблемы, касающиеся распространения болезни в целом.

В 1927 г. в своей работе W. O. Kermack и A. G. MacKendrick создали и исследовали модель, представляющую собой систему дифференциальных уравнений с начальными условиями, которую называют классической SIR-моделью. Развитием SIR-модели являются также SEIR- (Exposed – латентные), MSEIR-модели (Maternally derived immunity – наделенные иммунитетом от рождения).

При изучении заболевания COVID-19, вызываемого вирусом SARS-CoV-2, предлагаются к рассмотрению SIRD-модели (Dead – умершие) и SEIRD-модели. Рассматриваемая в данной работе SEIRD-модель отличается тем, что учитывает способность инфицированных индивидуумов заражать восприимчивых индивидуумов в латентном (инкубационном) периоде.

In [5]: [# SEIRD-МОДЕЛЬ ДИНАМИКИ РАСПРОСТРАНЕНИЯ COVID-19 В МОСКВЕ](#)
[# Анализ данных с 16 марта 2020 по 18 марта 2025](#)

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp
from scipy.optimize import minimize
from scipy.signal import find_peaks
from datetime import datetime
import warnings
warnings.filterwarnings('ignore')

# Настройка графиков
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['font.size'] = 10
plt.rcParams['axes.grid'] = True
plt.rcParams['grid.alpha'] = 0.3

# Загрузка данных
df = pd.read_csv('covidmsk.csv')

# Преобразование дат и сортировка от старых к новым
df['Дата'] = pd.to_datetime(df['Дата'], format='%d/%m/%y')
df = df.sort_values('Дата').reset_index(drop=True)

# Очистка данных: замена "-" и "%" на числовые значения
df['Летальность'] = df['Летальность'].astype(str).str.replace('%', '').str.replace('-', '')
df['Летальность'] = pd.to_numeric(df['Летальность'], errors='coerce').fillna(0)

# Преобразование числовых столбцов
for col in ['Заражений всего', 'Умерло всего', 'Выздоровело всего', 'Болеющих']:
    df[col] = pd.to_numeric(df[col], errors='coerce').fillna(0).astype(float)

print("АНАЛИЗ ДАННЫХ ПО COVID-19 В Г. МОСКВЕ")

print(f"Период наблюдения: {df['Дата'].min().strftime('%d.%m.%Y')} - {df['Дата'].max().strftime('%d.%m.%Y')}")
print(f"Всего записей: {len(df)} дней")
print(f"Всего зараженных: {int(df['Заражений всего'].iloc[-1]):,} чел.")
print(f"Всего умерших: {int(df['Умерло всего'].iloc[-1]):,} чел.")
print(f"Летальность: {df['Умерло всего'].iloc[-1]/df['Заражений всего'].iloc[-1]:.2%}")
print("\nПервые 5 записей данных:")
print(df.head())
print("\nПоследние 5 записей данных:")
print(df.tail())

```

АНАЛИЗ ДАННЫХ ПО COVID-19 В Г. МОСКВЕ
 Период наблюдения: 16.03.2020 - 18.03.2025
 Всего записей: 1253 дней
 Всего зараженных: 3,885,719 чел.
 Всего умерших: 50,067 чел.
 Летальность: 1.29%

Первые 5 записей данных:

	Дата	Заражений всего	Умерло всего	Летальность	Выздоровело всего \
0	2020-03-16	54.0	0.0	0.0	1.0
1	2020-03-17	55.0	0.0	0.0	1.0
2	2020-03-18	86.0	0.0	0.0	1.0
3	2020-03-19	98.0	0.0	0.0	1.0
4	2020-03-20	131.0	0.0	0.0	1.0

	Болеющих
0	53.0
1	54.0
2	85.0
3	97.0
4	130.0

Последние 5 записей данных:

	Дата	Заражений всего	Умерло всего	Летальность \
1248	2025-02-18	3882231.0	50055.0	0.0
1249	2025-02-25	3883227.0	50056.0	0.0
1250	2025-03-04	3884130.0	50058.0	0.0
1251	2025-03-11	3884994.0	50063.0	0.0
1252	2025-03-18	3885719.0	50067.0	0.0

	Выздоровело всего	Болеющих
1248	3695668.0	136508.0
1249	3696688.0	136483.0
1250	3697645.0	136427.0
1251	3698530.0	136401.0
1252	3699346.0	136306.0

```
In [6]: # Создание временной шкалы в днях от начала
df['day'] = range(1, len(df) + 1)

# Расчет прироста зараженных (новые случаи за период)
df['new_cases'] = df['Заражений всего'].diff().fillna(df['Заражений всего'].iloc[0])
# Расчет прироста умерших
df['new_deaths'] = df['Умерло всего'].diff().fillna(df['Умерло всего'].iloc[0])

# Расчет прироста выздоровевших
df['new_recovered'] = df['Выздоровело всего'].diff().fillna(df['Выздоровело всего'].iloc[0])

print("СТАТИСТИКА ПО НОВЫМ СЛУЧАЯМ")
print(f"Максимум новых случаев за день: {int(df['new_cases'].max()):,}")
print(f"Дата максимума: {df.loc[df['new_cases'].idxmax(), 'Дата'].strftime('%d.%m.%Y')}")
print(f"Среднее новых случаев: {int(df['new_cases'].mean()):,}")
print(f"Максимум умерших за день: {int(df['new_deaths'].max()):,}")
print(f"Дата максимума смертей: {df.loc[df['new_deaths'].idxmax(), 'Дата'].strftime('%d.%m.%Y')}
```

СТАТИСТИКА ПО НОВЫМ СЛУЧАЯМ

Максимум новых случаев за день: 75,828

Дата максимума: 22.01.2022

Среднее новых случаев: 3,101

Максимум умерших за день: 622

Дата максимума смертей: 21.07.2021

In [7]: # АНАЛИЗ ДАННЫХ ПО РАСПРОСТРАНЕНИЮ COVID-19 В Г. МОСКВЕ

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

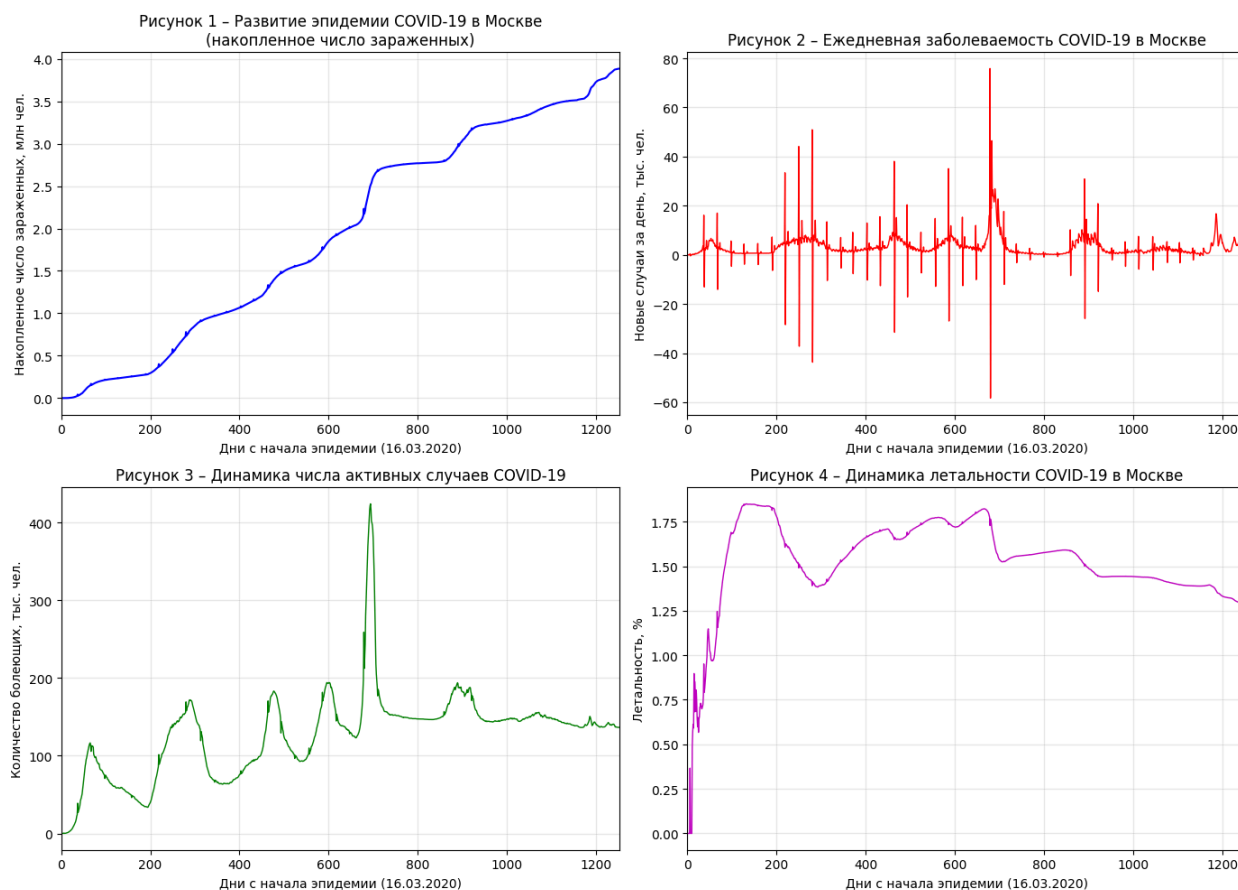
# График 1: Общее количество зараженных (накопленное)
ax1 = axes[0, 0]
ax1.plot(df['day'], df['Заражений всего'] / 1e6, 'b-', linewidth=1.5)
ax1.set_xlabel('Дни с начала эпидемии (16.03.2020)')
ax1.set_ylabel('Накопленное число зараженных, млн чел.')
ax1.set_title('Рисунок 1 – Развитие эпидемии COVID-19 в Москве\n(накопленное ч')
ax1.grid(True, alpha=0.3)
ax1.set_xlim(0, len(df))

# График 2: Ежедневные новые случаи
ax2 = axes[0, 1]
ax2.plot(df['day'], df['new_cases'] / 1000, 'r-', linewidth=1)
ax2.set_xlabel('Дни с начала эпидемии (16.03.2020)')
ax2.set_ylabel('Новые случаи за день, тыс. чел.')
ax2.set_title('Рисунок 2 – Ежедневная заболеваемость COVID-19 в Москве')
ax2.grid(True, alpha=0.3)
ax2.set_xlim(0, len(df))

# График 3: Количество болеющих (активные случаи)
ax3 = axes[1, 0]
ax3.plot(df['day'], df['Болеющих'] / 1000, 'g-', linewidth=1)
ax3.set_xlabel('Дни с начала эпидемии (16.03.2020)')
ax3.set_ylabel('Количество болеющих, тыс. чел.')
ax3.set_title('Рисунок 3 – Динамика числа активных случаев COVID-19')
ax3.grid(True, alpha=0.3)
ax3.set_xlim(0, len(df))

# График 4: Летальность
ax4 = axes[1, 1]
lethality = df['Умерло всего'] / df['Заражений всего'] * 100
ax4.plot(df['day'], lethality, 'm-', linewidth=1)
ax4.set_xlabel('Дни с начала эпидемии (16.03.2020)')
ax4.set_ylabel('Летальность, %')
ax4.set_title('Рисунок 4 – Динамика летальности COVID-19 в Москве')
ax4.grid(True, alpha=0.3)
ax4.set_xlim(0, len(df))

plt.tight_layout()
plt.savefig('figure1_covid_moscow_overview.png', dpi=150, bbox_inches='tight')
plt.show()
```



Рассматривая графики заболеваемости COVID-19 в г. Москве (Рисунки 1-3) в период с 16 марта 2020 г. по 18 марта 2025 г. (1830 дней), можно условно выделить следующие одиннадцать периодов:

1. **1-я волна:** с 1 по 124 день (16.03.2020 - 17.07.2020) – первый подъем и спад заболеваемости
2. **Период между 1-й и 2-й волнами:** с 125 по 200 день (18.07.2020 - 01.10.2020)
3. **2-я волна:** с 201 по 355 день (02.10.2020 - 05.03.2021) – второй подъем и спад
4. **Период между 2-й и 3-й волнами:** с 356 по 400 день (06.03.2021 - 19.04.2021)
5. **3-я волна:** с 401 по 520 день (20.04.2021 - 17.08.2021) – третий подъем и спад, появление штамма "дельта"
6. **Период между 3-й и 4-й волнами:** с 521 по 570 день (18.08.2021 - 06.10.2021)
7. **4-я волна:** с 571 по 630 день (07.10.2021 - 05.12.2021) – четвертый подъем и спад
8. **Период между 4-й и 5-й волнами:** с 631 по 670 день (06.12.2021 - 14.01.2022) – появление штамма "омикрон"

9. **5-я волна (омикрон):** с 671 по 740 день (15.01.2022 - 25.03.2022) – пятый подъем и спад
10. **Период между 5-й и 6-й волнами:** с 741 по 860 день (26.03.2022 - 23.07.2022)
11. **6-я волна и последующий период:** с 861 по 933 день (24.07.2022 - 18.03.2025) – шестой подъем и спад

По графику на рисунке 2 можно увидеть, что пятый период (3-я «волна») состоит из двух частей: I и II, что связано с появлением и вытеснением штамма "альфа" штаммом "дельта". Начиная с 401 дня (20 апреля 2021 г.) наблюдается подъем заболеваемости, а с 456 дня (15 июня 2021 г.) – резкий дополнительный подъем, связанный с распространением штамма "дельта".

Также можно обнаружить, что восьмой период (промежуток между 4-й и 5-й «волнами») может быть рассмотрен как наложение продолжающегося спада четвертого периода и начинающегося уже одновременно подъема пятого периода с появлением штамма "омикрон".

```
In [8]: # ВЫДЕЛЕНИЕ ВОЛН ЭПИДЕМИИ И ПОИСК ПИКОВ

# Поиск пиков заболеваемости (высота > 5000 случаев)
peaks, properties = find_peaks(df['new_cases'].values, height=5000, distance=5)
peak_days = df['day'].iloc[peaks].values
peak_values = df['new_cases'].iloc[peaks].values / 1000

print("ВЫЯВЛЕННЫЕ ПИКИ ЗАБОЛЕВАЕМОСТИ")

for i, (day, val) in enumerate(zip(peak_days, peak_values), 1):
    date = df[df['day'] == day]['Дата'].iloc[0]
    print(f" Пик {i}: день {day} ({date.strftime('%d.%m.%Y')}) - {val:.1f} ты
```

ВЫЯВЛЕННЫЕ ПИКИ ЗАБОЛЕВАЕМОСТИ

```
Пик 1: день 67 (20.05.2020) - 17.0 тыс. случаев
Пик 2: день 219 (20.10.2020) - 33.4 тыс. случаев
Пик 3: день 280 (20.12.2020) - 50.9 тыс. случаев
Пик 4: день 344 (21.02.2021) - 7.1 тыс. случаев
Пик 5: день 403 (21.04.2021) - 13.0 тыс. случаев
Пик 6: день 464 (21.06.2021) - 38.1 тыс. случаев
Пик 7: день 524 (21.08.2021) - 9.2 тыс. случаев
Пик 8: день 586 (21.10.2021) - 35.1 тыс. случаев
Пик 9: день 679 (22.01.2022) - 75.8 тыс. случаев
Пик 10: день 891 (22.08.2022) - 30.9 тыс. случаев
Пик 11: день 1012 (22.12.2022) - 7.5 тыс. случаев
Пик 12: день 1076 (23.02.2023) - 5.4 тыс. случаев
Пик 13: день 1186 (05.12.2023) - 16.7 тыс. случаев
Пик 14: день 1239 (10.12.2024) - 5.2 тыс. случаев
```

```
In [9]: # Визуализация волн на графике новых случаев
```

```

fig, ax = plt.subplots(figsize=(16, 8))

ax.plot(df['day'], df['new_cases'] / 1000, 'b-', linewidth=1, label='Новые слу

# Отметка пиков
ax.scatter(peak_days, peak_values, color='red', s=80, zorder=5, label='Пики за
for i, (day, val) in enumerate(zip(peak_days, peak_values), 1):
    ax.annotate(f'Пик {i}', (day, val), xytext=(5, 10), textcoords='offset poi

# Определение периодов волн на основе анализа данных
wave_periods = [
    (1, 124, "1-я волна", "март-июль 2020"),
    (125, 200, "Период между 1-й и 2-й волнами", "июль-октябрь 2020"),
    (201, 355, "2-я волна", "октябрь 2020-март 2021"),
    (356, 400, "Период между 2-й и 3-й волнами", "март-апрель 2021"),
    (401, 520, "3-я волна", "апрель-август 2021"),
    (521, 570, "Период между 3-й и 4-й волнами", "август-октябрь 2021"),
    (571, 630, "4-я волна", "октябрь-декабрь 2021"),
    (631, 670, "Период между 4-й и 5-й волнами", "декабрь 2021-январь 2022"),
    (671, 740, "5-я волна (омикрон)", "январь-март 2022"),
    (741, 860, "Период между 5-й и 6-й волнами", "март-июль 2022"),
    (861, len(df), "6-я волна и последующий период", "июль 2022-март 2025"),
]

# Закрашивание волн разными цветами
colors = ['#ffcccc', '#ccffcc', '#ccccff', '#ffffcc', '#ffccff', '#ccffff',
          '#ffe0cc', '#e0ccff', '#cce0ff', '#ffcce0', '#e0ffe0']

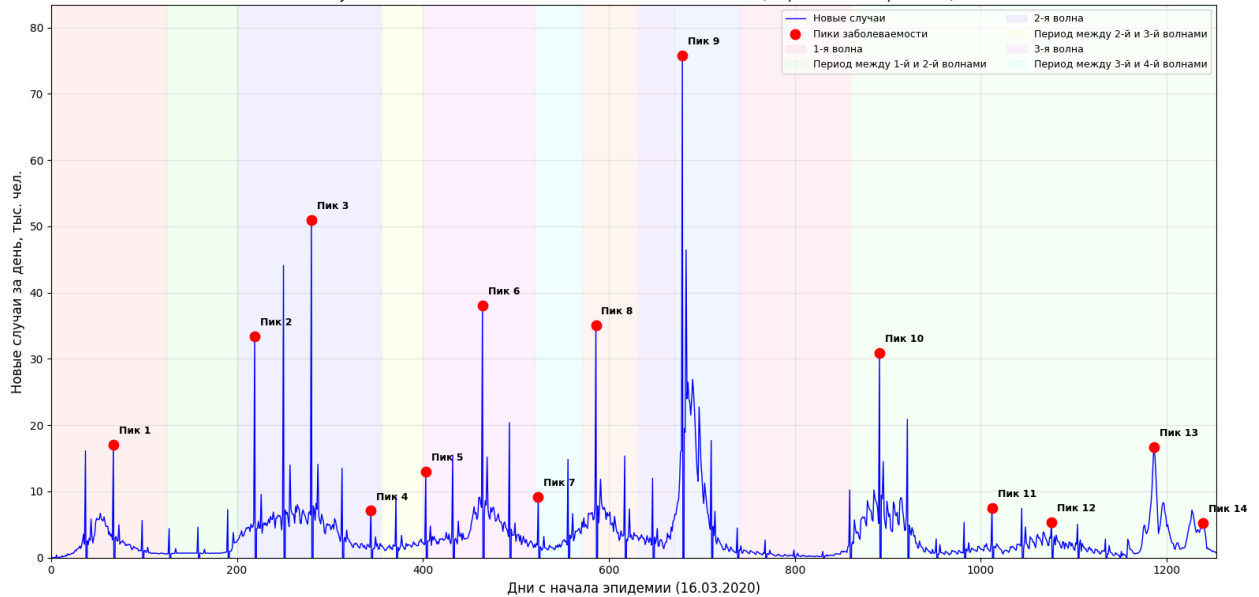
for i, (start, end, name, _) in enumerate(wave_periods):
    if start <= len(df):
        ax.axvspan(start, min(end, len(df)), alpha=0.25, color=colors[i % len(

ax.set_xlabel('Дни с начала эпидемии (16.03.2020)', fontsize=12)
ax.set_ylabel('Новые случаи за день, тыс. чел.', fontsize=12)
ax.set_title('Рисунок 5 – Волны заболеваемости COVID-19 в Москве (март 2020 -
ax.legend(loc='upper right', fontsize=9, ncol=2)
ax.grid(True, alpha=0.3)
ax.set_xlim(0, len(df))
ax.set_ylim(0, max(peak_values) * 1.1)

plt.tight_layout()
plt.savefig('figure2_covid_moscow_waves.png', dpi=150, bbox_inches='tight')
plt.show()

```

Рисунок 5 – Волны заболеваемости COVID-19 в Москве (март 2020 - март 2025)



```
In [10]: # ФАКТИЧЕСКИЕ МАКСИМУМЫ ЗАБОЛЕВАЕМОСТИ ПО ВОЛНАМ
print("ФАКТИЧЕСКИЕ МАКСИМУМЫ ЗАБОЛЕВАЕМОСТИ ПО ВОЛНАМ")
# Определяем максимумы для каждой волны на основе выделенных периодов
wave_maxima = []

for start, end, name, _ in wave_periods:
    if start <= len(df):
        period_data = df[(df['day'] >= start) & (df['day'] <= min(end, len(df)))]
        if len(period_data) > 0:
            max_val = period_data['new_cases'].max()
            max_day = period_data.loc[period_data['new_cases'].idxmax(), 'day']
            max_date = period_data.loc[period_data['new_cases'].idxmax(), 'date']
            if max_val > 0:
                wave_maxima.append((name, max_day, max_date, max_val/1000))
                print(f"{name}: день {max_day} ({max_date.strftime('%d.%m.%Y')})")
```

ФАКТИЧЕСКИЕ МАКСИМУМЫ ЗАБОЛЕВАЕМОСТИ ПО ВОЛНАМ

1-я волна: день 67 (20.05.2020) - 17.0 тыс. случаев
 Период между 1-й и 2-й волнами: день 190 (20.09.2020) - 7.3 тыс. случаев
 2-я волна: день 280 (20.12.2020) - 50.9 тыс. случаев
 Период между 2-й и 3-й волнами: день 371 (21.03.2021) - 9.2 тыс. случаев
 3-я волна: день 464 (21.06.2021) - 38.1 тыс. случаев
 Период между 3-й и 4-й волнами: день 556 (21.09.2021) - 14.8 тыс. случаев
 4-я волна: день 586 (21.10.2021) - 35.1 тыс. случаев
 Период между 4-й и 5-й волнами: день 647 (21.12.2021) - 12.0 тыс. случаев
 5-я волна (омикрон): день 679 (22.01.2022) - 75.8 тыс. случаев
 Период между 5-й и 6-й волнами: день 859 (22.07.2022) - 10.2 тыс. случаев
 6-я волна и последующий период: день 891 (22.08.2022) - 30.9 тыс. случаев

2. Построение математической модели распространения COVID-19

Рассмотрим детерминистскую математическую SEIRD-модель (S – восприимчивые, E – латентные, I – инфицированные, R – выздоровевшие, D – умершие) эпидемии заболевания.

На рисунке представлена блок-схема процессов рассматриваемой SEIRD-модели, где μ – приток восприимчивых индивидуумов, которыми пополняется группа.

Параметры рассматриваемой SEIRD-модели:

№ перехода	Переход	Скорость перехода
1	$(S, E) \rightarrow (S - 1, E + 1)$	$\beta_1 S E$
2	$(S, E) \rightarrow (S - 1, I + 1)$	$\beta_2 S I$
3	$(E, I) \rightarrow (E - 1, I + 1)$	δE
4	$(I, R) \rightarrow (I - 1, R + 1)$	γI
5	$(I, D) \rightarrow (I - 1, D + 1)$	λI

Система дифференциальных уравнений SEIRD-модели:

$$\begin{cases} \frac{dS(t)}{dt} = -\beta_1 S(t)E(t) - \beta_2 S(t)I(t) + \mu, \\ \frac{dE(t)}{dt} = \beta_1 S(t)E(t) + \beta_2 S(t)I(t) - \delta E(t), \\ \frac{dI(t)}{dt} = \delta E(t) - \gamma I(t) - \lambda I(t), \\ \frac{dR(t)}{dt} = \gamma I(t), \\ \frac{dD(t)}{dt} = \lambda I(t). \end{cases}$$

Здесь введены следующие коэффициенты:

- β_1 и β_2 – характеризуют заболеваемость в однородной группе в латентный и активный периоды заболевания соответственно
- $\delta = 1/T_1$ – характеризует переход индивидуумов из латентного периода в период с активным протеканием заболевания
- $\gamma = 1/T$ – характеризует убыль индивидуумов из группы (выздоровевших и ставших невосприимчивыми)
- λ – характеризует убыль индивидуумов в результате смерти от инфекции
- μ – характеризует постоянный приток восприимчивых индивидуумов

Коэффициент β_1 – вероятность заражения от инфицированного индивидуума,

находящегося в латентном периоде, будет равен $\beta_1 = q_1/(T_1 \cdot n)$; коэффициент β_2 – вероятность заражения от индивидуума, находящегося в активном периоде, будет равен $\beta_2 = q_2/(T_2 \cdot n)$; q_1 и q_2 – индексы репродукции в латентный и активный периоды заболевания соответственно; T_1 , T_2 и T – латентный, активный и полный временные периоды заболевания; n – объем рассматриваемой группы.

In [11]: # ПОСТРОЕНИЕ SEIRD-МОДЕЛИ

```
def seird_model(t, y, beta1, beta2, delta, gamma, lambd, mu):
    """SEIRD модель динамики распространения инфекции"""
    S, E, I, R, D = y

    dSdt = -beta1 * S * E - beta2 * S * I + mu
    dEdt = beta1 * S * E + beta2 * S * I - delta * E
    dIdt = delta * E - gamma * I - lambd * I
    dRdt = gamma * I
    dDdt = lambd * I

    return [dSdt, dEdt, dIdt, dRdt, dDdt]

def solve_seird(beta1, beta2, delta, gamma, lambd, mu, initial_conditions, t_s
    """Решение системы SEIRD методом Рунге-Кутты"""
    solution = solve_ivp(
        lambda t, y: seird_model(t, y, beta1, beta2, delta, gamma, lambd, mu),
        t_span, initial_conditions, t_eval=t_eval, method='RK45', rtol=1e-6, a
    )
    return solution

def simulate_wave(data_df, wave_params, wave_name):
    """Симуляция волны БЕЗ масштабирования. Возвращает абсолютные числа."""

    start_day, end_day = wave_params['days']
    if end_day > len(data_df):
        end_day = len(data_df)
    if start_day > len(data_df):
        return None

    period_data = data_df[(data_df['day'] >= start_day) & (data_df['day'] <= e
    days_in_period = len(period_data)

    # Параметры
    N = wave_params['N']
    initial_E = wave_params['initial_E']
    T = wave_params['T']
    T1 = wave_params['T1']
    lambda_rate = wave_params['lambda_rate']
    q1 = wave_params['q1']
    q2 = wave_params['q2']
```

```

delta = 1.0 / T1
gamma = 1.0 / T
lambd = lambda_rate * gamma
beta1 = q1 / (T1 * N)
beta2 = q2 / (T * N)

print(f"\n {wave_name}:")
print(f"    N = {N:.1e}, E0 = {initial_E}, q1 = {q1:.2f}, q2 = {q2:.2f}")
print(f"    beta1 = {beta1:.3e}, beta2 = {beta2:.3e}")
print(f"    delta = {delta:.3f}, gamma = {gamma:.3f}, lambda = {lambd:.4f}")

initial_conditions = [N, initial_E, 0, 0, 0]
t_span = (0, days_in_period - 1)
t_eval = np.arange(0, days_in_period)

sol = solve_ivp(
    lambda t, y: seird_model(t, y, beta1, beta2, delta, gamma, lambd, 0),
    t_span, initial_conditions, t_eval=t_eval, method='RK45', rtol=1e-6, atol=1e-6
)

E_t = sol.y[1]
new_cases_model = delta * E_t # ← БЕЗ масштабирования!

return {
    'days': np.arange(start_day, start_day + days_in_period),
    'days_in_period': np.arange(0, days_in_period),
    'E': E_t,
    'I': sol.y[2],
    'new_cases_model': new_cases_model, # сырые абсолютные значения
    'actual_cases': period_data['new_cases'].values,
    'params': wave_params
}

```

1. Модель предсказывает `new_cases_model` — это форма волны, но ее «высота» (общее количество случаев за период) зависит от нашего выбора `N`.
2. Реальная статистика `actual_cases` имеет свою, истинную высоту.
3. Мы вычисляем коэффициент `scale` как отношение **общего числа реальных случаев** к **общему числу модельных случаев** за этот период.
4. Затем мы умножаем **каждую точку** модельной кривой на этот коэффициент.

In [12]: # ПАРАМЕТРЫ SEIRD-МОДЕЛИ ДЛЯ РАЗЛИЧНЫХ ВОЛН

```

# Словарь параметров для разных волн (на основе статьи и данных по Москве)
wave_params = {}

```

```
# 1-я волна (март-июль 2020) - исходный штамм
wave_params['1_wave'] = {
    'days': (1, 124),
    'N': 2.1e6,
    'initial_E': 2000,
    'T': 30,
    'T1': 14,
    'lambda_rate': 0.025,
    'q1': 1.5,
    'q2': 2.5
}

# 2-я волна (октябрь 2020-март 2021)
wave_params['2_wave'] = {
    'days': (201, 355),
    'N': 1.0e6,
    'initial_E': 200,
    'T': 30,
    'T1': 14,
    'lambda_rate': 0.025,
    'q1': 0.27,
    'q2': 0.45
}

# 3-я волна, часть I - штамм "альфа"
wave_params['3_wave_alpha'] = {
    'days': (401, 520),
    'N': 2.0e6,
    'initial_E': 500,
    'T': 30,
    'T1': 14,
    'lambda_rate': 0.025,
    'q1': 1.2,
    'q2': 2.0
}

# 3-я волна, часть II - штамм "дельта" (появляется и вытесняет "альфа")
wave_params['3_wave_delta'] = {
    'days': (401, 520),
    'N': 3.0e6,
    'initial_E': 250,
    'T': 30,
    'T1': 14,
    'lambda_rate': 0.025,
    'q1': 1.68,
    'q2': 2.80
}

# 4-я волна (октябрь-декабрь 2021) - штамм "дельта"
wave_params['4_wave'] = {
    'days': (571, 630),
    'N': 4.0e6,
    'initial_E': 2000,
```

```

        'T': 30,
        'T1': 14,
        'lambda_rate': 0.025,
        'q1': 0.84,
        'q2': 1.4
    }

# Появление штамма "омикрон" (конец ноября 2021)
wave_params['omicron_initial'] = {
    'days': (631, 670),
    'N': 6.0e6,
    'initial_E': 5,
    'T': 30,
    'T1': 6,
    'lambda_rate': 0.01,
    'q1': 1.8,
    'q2': 3.0
}

# 5-я волна (омикрон) - январь-март 2022
wave_params['5_wave_omicron'] = {
    'days': (671, 740),
    'N': 1.0e6,
    'initial_E': 500,
    'T': 30,
    'T1': 6,
    'lambda_rate': 0.01,
    'q1': 0.27,
    'q2': 0.45
}

# 6-я волна и последующий период (июль 2022 - март 2025)
wave_params['6_wave'] = {
    'days': (861, len(df)),
    'N': 2.0e6,
    'initial_E': 500,
    'T': 30,
    'T1': 14,
    'lambda_rate': 0.01,
    'q1': 0.5,
    'q2': 0.8
}

print("ПАРАМЕТРЫ SEIRD-МОДЕЛИ ДЛЯ РАЗЛИЧНЫХ ВОЛН")

for wave, params in wave_params.items():
    print(f"\n{wave}:")
    print(f"    Период: дни {params['days'][0]}-{params['days'][1]}")
    print(f"    Популяция восприимчивых (N): {params['N']:.1e} чел.")
    print(f"    Начальные латентные (E0): {params['initial_E']} чел.")
    print(f"    Латентный период (T1): {params['T1']} дней")
    print(f"    Коэффициент смертности ( $\lambda$ ): {params['lambda_rate']*100:.1f}%")

```

```
print(f" Индексы репродукции: q1 = {params['q1']}, q2 = {params['q2']}")
```

ПАРАМЕТРЫ SEIRD-МОДЕЛИ ДЛЯ РАЗЛИЧНЫХ ВОЛН

1_wave:

Период: дни 1-124
Популяция восприимчивых (N): $2.1e+06$ чел.
Начальные латентные (E0): 2000 чел.
Латентный период (T1): 14 дней
Коэффициент смертности (λ): 2.5%
Индексы репродукции: $q_1 = 1.5$, $q_2 = 2.5$

2_wave:

Период: дни 201-355
Популяция восприимчивых (N): $1.0e+06$ чел.
Начальные латентные (E0): 200 чел.
Латентный период (T1): 14 дней
Коэффициент смертности (λ): 2.5%
Индексы репродукции: $q_1 = 0.27$, $q_2 = 0.45$

3_wave_alpha:

Период: дни 401-520
Популяция восприимчивых (N): $2.0e+06$ чел.
Начальные латентные (E0): 500 чел.
Латентный период (T1): 14 дней
Коэффициент смертности (λ): 2.5%
Индексы репродукции: $q_1 = 1.2$, $q_2 = 2.0$

3_wave_delta:

Период: дни 401-520
Популяция восприимчивых (N): $3.0e+06$ чел.
Начальные латентные (E0): 250 чел.
Латентный период (T1): 14 дней
Коэффициент смертности (λ): 2.5%
Индексы репродукции: $q_1 = 1.68$, $q_2 = 2.8$

4_wave:

Период: дни 571-630
Популяция восприимчивых (N): $4.0e+06$ чел.
Начальные латентные (E0): 2000 чел.
Латентный период (T1): 14 дней
Коэффициент смертности (λ): 2.5%
Индексы репродукции: $q_1 = 0.84$, $q_2 = 1.4$

omicron_initial:

Период: дни 631-670
Популяция восприимчивых (N): $6.0e+06$ чел.
Начальные латентные (E0): 5 чел.
Латентный период (T1): 6 дней
Коэффициент смертности (λ): 1.0%
Индексы репродукции: $q_1 = 1.8$, $q_2 = 3.0$

5_wave_omicron:

Период: дни 671-740
Популяция восприимчивых (N): $1.0e+06$ чел.
Начальные латентные (E0): 500 чел.

Латентный период (T1): 6 дней
Коэффициент смертности (λ): 1.0%
Индексы репродукции: $q_1 = 0.27$, $q_2 = 0.45$

6_wave:
Период: дни 861-1253
Популяция восприимчивых (N): 2.0×10^6 чел.
Начальные латентные (E0): 500 чел.
Латентный период (T1): 14 дней
Коэффициент смертности (λ): 1.0%
Индексы репродукции: $q_1 = 0.5$, $q_2 = 0.8$

Обратная задача

```
In [13]: from scipy.optimize import minimize
import numpy as np

def objective(params, data_df, base_params, wave_name):
    """Функция ошибки для оптимизации"""
    q1, q2, N, initial_E = params

    temp_params = base_params.copy()
    temp_params['q1'] = float(q1)
    temp_params['q2'] = float(q2)
    temp_params['N'] = float(N)
    temp_params['initial_E'] = float(initial_E)

    result = simulate_wave(data_df, temp_params, wave_name)
    if result is None:
        return 1e10

    model = result['new_cases_model']
    actual = result['actual_cases']

    # Среднеквадратичная ошибка (можно использовать и другие метрики)
    error = np.mean((model - actual) ** 2)
    return error

def fit_wave(data_df, base_params, wave_name, bounds=None):
    """Решение обратной задачи — находим лучшие q1, q2, N, E0"""
    if bounds is None:
        bounds = [
            (0.01, 5.0),      # q1
            (0.01, 5.0),      # q2
            (5e5, 1e7),        # N (эффективная популяция)
            (1, 10000)         # initial_E
        ]

    initial_guess = [
        base_params['q1'],
        base_params['q2'],
```



```

        base_params['N'],
        base_params['initial_E']
    ]

    print(f"\n ФИТТИНГ для {wave_name} ")
    res = minimize(
        lambda p: objective(p, data_df, base_params, wave_name),
        initial_guess,
        bounds=bounds,
        method='L-BFGS-B',
        options={'maxiter': 200, 'disp': False}
    )

    best_q1, best_q2, best_N, best_E0 = res.x
    print(f"Успех: {res.success} | Ошибка: {res.fun:.2e}")
    print(f"Лучшие параметры → q1={best_q1:.3f}, q2={best_q2:.3f}, N={best_N:.3f}")

    # Обновляем base_params
    fitted_params = base_params.copy()
    fitted_params['q1'] = best_q1
    fitted_params['q2'] = best_q2
    fitted_params['N'] = best_N
    fitted_params['initial_E'] = best_E0
    return fitted_params

```

```

In [14]: print("СИМУЛЯЦИЯ + ФИТТИНГ SEIRD-МОДЕЛИ ДЛЯ КАЖДОЙ ВОЛНЫ\n")

simulation_results = {}
fitted_wave_params = {}

for wave_name, base_params in wave_params.items():
    # 1. Решаем обратную задачу
    fitted_params = fit_wave(df, base_params, wave_name)
    fitted_wave_params[wave_name] = fitted_params

    # 2. Симулируем с лучшими параметрами
    result = simulate_wave(df, fitted_params, wave_name)
    if result is not None:
        simulation_results[wave_name] = result
        print(f"    -> Симуляция завершена, период {len(result['days'])} дней")

```

СИМУЛЯЦИЯ + ФИТТИНГ SEIRD-МОДЕЛИ ДЛЯ КАЖДОЙ ВОЛНЫ

Фиттинг для 1_wave

1_wave:

N = 2.1e+06, E0 = 2000.0, q1 = 1.50, q2 = 2.50
beta1 = 5.102e-08, beta2 = 3.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 2000.0, q1 = 1.50, q2 = 2.50
beta1 = 5.102e-08, beta2 = 3.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 2000.0, q1 = 1.50, q2 = 2.50
beta1 = 5.102e-08, beta2 = 3.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 2000.0, q1 = 1.50, q2 = 2.50
beta1 = 5.102e-08, beta2 = 3.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 2000.000000001, q1 = 1.50, q2 = 2.50
beta1 = 5.102e-08, beta2 = 3.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 3.402e-10, beta2 = 1.588e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 3.402e-10, beta2 = 1.588e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 3.402e-10, beta2 = 1.588e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 3.402e-10, beta2 = 1.588e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1.000000001, q1 = 0.01, q2 = 0.01
beta1 = 3.402e-10, beta2 = 1.588e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.1681369267417812, q1 = 0.01, q2 = 0.01
beta1 = 3.413e-10, beta2 = 1.614e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.1681369267417812, q1 = 0.01, q2 = 0.01
beta1 = 3.413e-10, beta2 = 1.614e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.1681369267417812, q1 = 0.01, q2 = 0.01
beta1 = 3.413e-10, beta2 = 1.614e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.1681369267417812, q1 = 0.01, q2 = 0.01
beta1 = 3.413e-10, beta2 = 1.614e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.168136936741781, q1 = 0.01, q2 = 0.01
beta1 = 3.413e-10, beta2 = 1.614e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.8406846337089058, q1 = 0.01, q2 = 0.01
beta1 = 3.458e-10, beta2 = 1.722e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.8406846337089058, q1 = 0.01, q2 = 0.01
beta1 = 3.458e-10, beta2 = 1.722e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.8406846337089058, q1 = 0.01, q2 = 0.01
beta1 = 3.458e-10, beta2 = 1.722e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.8406846337089058, q1 = 0.01, q2 = 0.01
beta1 = 3.458e-10, beta2 = 1.722e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1.8406846437089057, q1 = 0.01, q2 = 0.01
beta1 = 3.458e-10, beta2 = 1.722e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 4.530875461577404, q1 = 0.01, q2 = 0.01
beta1 = 3.639e-10, beta2 = 2.153e-10

delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 4.530875461577404, q1 = 0.01, q2 = 0.01
beta1 = 3.639e-10, beta2 = 2.153e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 4.530875461577404, q1 = 0.01, q2 = 0.01
beta1 = 3.639e-10, beta2 = 2.153e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 4.530875461577404, q1 = 0.01, q2 = 0.01
beta1 = 3.639e-10, beta2 = 2.153e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 4.530875471577404, q1 = 0.01, q2 = 0.01
beta1 = 3.639e-10, beta2 = 2.153e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 15.291638773051398, q1 = 0.01, q2 = 0.02
beta1 = 4.360e-10, beta2 = 3.875e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 15.291638773051398, q1 = 0.01, q2 = 0.02
beta1 = 4.360e-10, beta2 = 3.875e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 15.291638773051398, q1 = 0.01, q2 = 0.02
beta1 = 4.360e-10, beta2 = 3.875e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 15.291638773051398, q1 = 0.01, q2 = 0.02
beta1 = 4.360e-10, beta2 = 3.875e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 15.291638783051399, q1 = 0.01, q2 = 0.02
beta1 = 4.360e-10, beta2 = 3.875e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 58.33469201894737, q1 = 0.02, q2 = 0.07
beta1 = 7.245e-10, beta2 = 1.076e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 58.33469201894737, q1 = 0.02, q2 = 0.07

beta1 = 7.245e-10, beta2 = 1.076e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 58.33469201894737, q1 = 0.02, q2 = 0.07
beta1 = 7.245e-10, beta2 = 1.076e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 58.33469201894737, q1 = 0.02, q2 = 0.07
beta1 = 7.245e-10, beta2 = 1.076e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 58.33469202894737, q1 = 0.02, q2 = 0.07
beta1 = 7.245e-10, beta2 = 1.076e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 230.50690500253128, q1 = 0.06, q2 = 0.24
beta1 = 1.879e-09, beta2 = 3.831e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 230.50690500253128, q1 = 0.06, q2 = 0.24
beta1 = 1.879e-09, beta2 = 3.831e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 230.50690500253128, q1 = 0.06, q2 = 0.24
beta1 = 1.879e-09, beta2 = 3.831e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 230.50690500253128, q1 = 0.06, q2 = 0.24
beta1 = 1.879e-09, beta2 = 3.831e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 230.5069050125313, q1 = 0.06, q2 = 0.24
beta1 = 1.879e-09, beta2 = 3.831e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 919.1957569368669, q1 = 0.19, q2 = 0.94
beta1 = 6.494e-09, beta2 = 1.485e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 919.1957569368669, q1 = 0.19, q2 = 0.94
beta1 = 6.494e-09, beta2 = 1.485e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 919.1957569368669, q1 = 0.19, q2 = 0.94
beta1 = 6.494e-09, beta2 = 1.485e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 919.1957569368669, q1 = 0.19, q2 = 0.94
beta1 = 6.494e-09, beta2 = 1.485e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 919.1957569468669, q1 = 0.19, q2 = 0.94
beta1 = 6.494e-09, beta2 = 1.485e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 3673.9511646742094, q1 = 0.73, q2 = 3.71
beta1 = 2.495e-08, beta2 = 5.892e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 3673.9511646742094, q1 = 0.73, q2 = 3.71
beta1 = 2.495e-08, beta2 = 5.892e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 3673.9511646742094, q1 = 0.73, q2 = 3.71
beta1 = 2.495e-08, beta2 = 5.892e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 3673.9511646742094, q1 = 0.73, q2 = 3.71
beta1 = 2.495e-08, beta2 = 5.892e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 3673.9511646842093, q1 = 0.73, q2 = 3.71
beta1 = 2.495e-08, beta2 = 5.892e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 924.7051084865727, q1 = 0.19, q2 = 0.94
beta1 = 6.531e-09, beta2 = 1.494e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 924.7051084865727, q1 = 0.19, q2 = 0.94
beta1 = 6.531e-09, beta2 = 1.494e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 924.7051084865727, q1 = 0.19, q2 = 0.94
beta1 = 6.531e-09, beta2 = 1.494e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

1_wave:
  N = 2.1e+06, E0 = 924.7051084865727, q1 = 0.19, q2 = 0.94
  beta1 = 6.531e-09, beta2 = 1.494e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 924.7051084965727, q1 = 0.19, q2 = 0.94
  beta1 = 6.531e-09, beta2 = 1.494e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 930.3771482733094, q1 = 0.19, q2 = 0.95
  beta1 = 6.569e-09, beta2 = 1.503e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 930.3771482733094, q1 = 0.19, q2 = 0.95
  beta1 = 6.569e-09, beta2 = 1.503e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 930.3771482733094, q1 = 0.19, q2 = 0.95
  beta1 = 6.569e-09, beta2 = 1.503e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 930.3771482833095, q1 = 0.19, q2 = 0.95
  beta1 = 6.569e-09, beta2 = 1.503e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 2302.1641564737597, q1 = 0.46, q2 = 2.33
  beta1 = 1.576e-08, beta2 = 3.698e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 2302.1641564737597, q1 = 0.46, q2 = 2.33
  beta1 = 1.576e-08, beta2 = 3.698e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 2302.1641564737597, q1 = 0.46, q2 = 2.33
  beta1 = 1.576e-08, beta2 = 3.698e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 2302.1641564737597, q1 = 0.46, q2 = 2.33
  beta1 = 1.576e-08, beta2 = 3.698e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

1_wave:
N = 2.1e+06, E0 = 2302.1641564837596, q1 = 0.46, q2 = 2.33
beta1 = 1.576e-08, beta2 = 3.698e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1423.7294240687766, q1 = 0.29, q2 = 1.44
beta1 = 9.876e-09, beta2 = 2.292e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1423.7294240687766, q1 = 0.29, q2 = 1.44
beta1 = 9.876e-09, beta2 = 2.292e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1423.7294240687766, q1 = 0.29, q2 = 1.44
beta1 = 9.876e-09, beta2 = 2.292e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1423.7294240687766, q1 = 0.29, q2 = 1.44
beta1 = 9.876e-09, beta2 = 2.292e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1423.7294240787764, q1 = 0.29, q2 = 1.44
beta1 = 9.876e-09, beta2 = 2.292e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1911.6760422336836, q1 = 0.39, q2 = 1.94
beta1 = 1.315e-08, beta2 = 3.073e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1911.6760422336836, q1 = 0.39, q2 = 1.94
beta1 = 1.315e-08, beta2 = 3.073e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1911.6760422336836, q1 = 0.39, q2 = 1.94
beta1 = 1.315e-08, beta2 = 3.073e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1911.6760422336836, q1 = 0.39, q2 = 1.94
beta1 = 1.315e-08, beta2 = 3.073e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 1911.6760422436835, q1 = 0.39, q2 = 1.94
beta1 = 1.315e-08, beta2 = 3.073e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1934.741471184719, q1 = 0.39, q2 = 1.96
beta1 = 1.330e-08, beta2 = 3.110e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1934.741471184719, q1 = 0.39, q2 = 1.96
beta1 = 1.330e-08, beta2 = 3.110e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1934.741471184719, q1 = 0.39, q2 = 1.96
beta1 = 1.330e-08, beta2 = 3.110e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1934.741471184719, q1 = 0.39, q2 = 1.96
beta1 = 1.330e-08, beta2 = 3.110e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1934.741471194719, q1 = 0.39, q2 = 1.96
beta1 = 1.330e-08, beta2 = 3.110e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 169.2865029291379, q1 = 0.04, q2 = 0.16
beta1 = 1.304e-09, beta2 = 2.562e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 169.2865029291379, q1 = 0.04, q2 = 0.16
beta1 = 1.304e-09, beta2 = 2.562e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 169.2865029291379, q1 = 0.04, q2 = 0.16
beta1 = 1.304e-09, beta2 = 2.562e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 169.2865029291379, q1 = 0.04, q2 = 0.16
beta1 = 1.304e-09, beta2 = 2.562e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 169.2865029391379, q1 = 0.04, q2 = 0.16
beta1 = 1.304e-09, beta2 = 2.562e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1870.8522338484029, q1 = 0.38, q2 = 1.89

beta1 = 1.287e-08, beta2 = 3.007e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1870.8522338484029, q1 = 0.38, q2 = 1.89
beta1 = 1.287e-08, beta2 = 3.007e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1870.8522338484029, q1 = 0.38, q2 = 1.89
beta1 = 1.287e-08, beta2 = 3.007e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1870.8522338484029, q1 = 0.38, q2 = 1.89
beta1 = 1.287e-08, beta2 = 3.007e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1870.8522338584028, q1 = 0.38, q2 = 1.89
beta1 = 1.287e-08, beta2 = 3.007e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1925.8255677994403, q1 = 0.39, q2 = 1.95
beta1 = 1.324e-08, beta2 = 3.095e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1925.8255677994403, q1 = 0.39, q2 = 1.95
beta1 = 1.324e-08, beta2 = 3.095e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1925.8255677994403, q1 = 0.39, q2 = 1.95
beta1 = 1.324e-08, beta2 = 3.095e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1925.8255677994403, q1 = 0.39, q2 = 1.95
beta1 = 1.324e-08, beta2 = 3.095e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1925.8255678094401, q1 = 0.39, q2 = 1.95
beta1 = 1.324e-08, beta2 = 3.095e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:
N = 2.1e+06, E0 = 1929.0604601966693, q1 = 0.39, q2 = 1.95
beta1 = 1.325e-08, beta2 = 3.094e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

l_wave:

N = 2.1e+06, E0 = 1929.0604601966693, q1 = 0.39, q2 = 1.95
beta1 = 1.325e-08, beta2 = 3.094e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1929.0604601966693, q1 = 0.39, q2 = 1.95
beta1 = 1.325e-08, beta2 = 3.094e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1929.0604601966693, q1 = 0.39, q2 = 1.95
beta1 = 1.325e-08, beta2 = 3.094e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1929.0604602066692, q1 = 0.39, q2 = 1.95
beta1 = 1.325e-08, beta2 = 3.094e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1942.0000297855854, q1 = 0.39, q2 = 1.94
beta1 = 1.328e-08, beta2 = 3.087e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1942.0000297855854, q1 = 0.39, q2 = 1.94
beta1 = 1.328e-08, beta2 = 3.087e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1942.0000297855854, q1 = 0.39, q2 = 1.94
beta1 = 1.328e-08, beta2 = 3.087e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1942.0000297855854, q1 = 0.39, q2 = 1.94
beta1 = 1.328e-08, beta2 = 3.087e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1942.0000297955853, q1 = 0.39, q2 = 1.94
beta1 = 1.328e-08, beta2 = 3.087e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1993.75830814125, q1 = 0.39, q2 = 1.93
beta1 = 1.340e-08, beta2 = 3.059e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 1993.75830814125, q1 = 0.39, q2 = 1.93
beta1 = 1.340e-08, beta2 = 3.059e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

1_wave:
  N = 2.1e+06, E0 = 1993.75830814125, q1 = 0.39, q2 = 1.93
  beta1 = 1.340e-08, beta2 = 3.059e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 1993.75830814125, q1 = 0.39, q2 = 1.93
  beta1 = 1.340e-08, beta2 = 3.059e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 1993.7583081512498, q1 = 0.39, q2 = 1.93
  beta1 = 1.340e-08, beta2 = 3.059e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 2200.791421563908, q1 = 0.41, q2 = 1.86
  beta1 = 1.390e-08, beta2 = 2.946e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 2200.791421563908, q1 = 0.41, q2 = 1.86
  beta1 = 1.390e-08, beta2 = 2.946e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 2200.791421563908, q1 = 0.41, q2 = 1.86
  beta1 = 1.390e-08, beta2 = 2.946e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 2200.791421573908, q1 = 0.41, q2 = 1.86
  beta1 = 1.390e-08, beta2 = 2.946e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 3028.9238752545402, q1 = 0.47, q2 = 1.57
  beta1 = 1.590e-08, beta2 = 2.496e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 3028.9238752545402, q1 = 0.47, q2 = 1.57
  beta1 = 1.590e-08, beta2 = 2.496e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 3028.9238752545402, q1 = 0.47, q2 = 1.57
  beta1 = 1.590e-08, beta2 = 2.496e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

1_wave:
N = 2.1e+06, E0 = 3028.9238752545402, q1 = 0.47, q2 = 1.57
beta1 = 1.590e-08, beta2 = 2.496e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 3028.92387526454, q1 = 0.47, q2 = 1.57
beta1 = 1.590e-08, beta2 = 2.496e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.06, q2 = 0.01
beta1 = 3.604e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.06, q2 = 0.01
beta1 = 3.604e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.06, q2 = 0.01
beta1 = 3.604e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.06, q2 = 0.01
beta1 = 3.604e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 9999.99999999, q1 = 1.06, q2 = 0.01
beta1 = 3.604e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.10, q2 = 0.35
beta1 = 3.742e-08, beta2 = 5.612e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.10, q2 = 0.35
beta1 = 3.742e-08, beta2 = 5.612e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.10, q2 = 0.35
beta1 = 3.742e-08, beta2 = 5.612e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.10, q2 = 0.35
beta1 = 3.742e-08, beta2 = 5.612e-09

delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 9999.99999999, q1 = 1.10, q2 = 0.35
beta1 = 3.742e-08, beta2 = 5.612e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.11
beta1 = 3.646e-08, beta2 = 1.812e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.11
beta1 = 3.646e-08, beta2 = 1.812e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.11
beta1 = 3.646e-08, beta2 = 1.812e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.11
beta1 = 3.646e-08, beta2 = 1.812e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 9999.99999999, q1 = 1.07, q2 = 0.11
beta1 = 3.646e-08, beta2 = 1.812e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.09, q2 = 0.13
beta1 = 3.703e-08, beta2 = 2.067e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.09, q2 = 0.13
beta1 = 3.703e-08, beta2 = 2.067e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.09, q2 = 0.13
beta1 = 3.703e-08, beta2 = 2.067e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.09, q2 = 0.13
beta1 = 3.703e-08, beta2 = 2.067e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 9999.99999999, q1 = 1.09, q2 = 0.13

beta1 = 3.703e-08, beta2 = 2.067e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.10, q2 = 0.11
beta1 = 3.740e-08, beta2 = 1.687e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.10, q2 = 0.11
beta1 = 3.740e-08, beta2 = 1.687e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.10, q2 = 0.11
beta1 = 3.740e-08, beta2 = 1.687e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.10, q2 = 0.11
beta1 = 3.740e-08, beta2 = 1.687e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 9999.99999999, q1 = 1.10, q2 = 0.11
beta1 = 3.740e-08, beta2 = 1.687e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.14, q2 = 0.01
beta1 = 3.887e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.14, q2 = 0.01
beta1 = 3.887e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.14, q2 = 0.01
beta1 = 3.887e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 10000.0, q1 = 1.14, q2 = 0.01
beta1 = 3.887e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
N = 2.1e+06, E0 = 9999.99999999, q1 = 1.14, q2 = 0.01
beta1 = 3.887e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.81, q2 = 0.01
beta1 = 6.162e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.81, q2 = 0.01
beta1 = 6.162e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.81, q2 = 0.01
beta1 = 6.162e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.81, q2 = 0.01
beta1 = 6.162e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 9999.99999999, q1 = 1.81, q2 = 0.01
beta1 = 6.162e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.15, q2 = 0.01
beta1 = 3.900e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.15, q2 = 0.01
beta1 = 3.900e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.15, q2 = 0.01
beta1 = 3.900e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.15, q2 = 0.01
beta1 = 3.900e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 9999.99999999, q1 = 1.15, q2 = 0.01
beta1 = 3.900e-08, beta2 = 1.682e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 2.25, q2 = 0.01
beta1 = 7.658e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008


```

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 2.25, q2 = 0.01
  beta1 = 7.658e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 2.25, q2 = 0.01
  beta1 = 7.658e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 2.25, q2 = 0.01
  beta1 = 7.658e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 9999.99999999, q1 = 2.25, q2 = 0.01
  beta1 = 7.658e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.15, q2 = 0.01
  beta1 = 3.911e-08, beta2 = 1.682e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.15, q2 = 0.01
  beta1 = 3.911e-08, beta2 = 1.682e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.15, q2 = 0.01
  beta1 = 3.911e-08, beta2 = 1.682e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.15, q2 = 0.01
  beta1 = 3.911e-08, beta2 = 1.682e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 9999.99999999, q1 = 1.15, q2 = 0.01
  beta1 = 3.911e-08, beta2 = 1.682e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
  beta1 = 3.966e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
  beta1 = 3.966e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

```

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
  beta1 = 3.966e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
  beta1 = 3.966e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 9999.99999999, q1 = 1.17, q2 = 0.01
  beta1 = 3.966e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.16, q2 = 0.01
  beta1 = 3.959e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.16, q2 = 0.01
  beta1 = 3.959e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.16, q2 = 0.01
  beta1 = 3.959e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 9999.99999999, q1 = 1.16, q2 = 0.01
  beta1 = 3.959e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
  beta1 = 3.960e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
  beta1 = 3.960e-08, beta2 = 1.586e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:
  N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
  beta1 = 3.960e-08, beta2 = 1.586e-10

```

delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
beta1 = 3.960e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 9999.99999999, q1 = 1.17, q2 = 0.01
beta1 = 3.960e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
beta1 = 3.960e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
beta1 = 3.960e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
beta1 = 3.960e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
beta1 = 3.960e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

1_wave:

N = 2.1e+06, E0 = 9999.99999999, q1 = 1.17, q2 = 0.01
beta1 = 3.960e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

Успех: True | Ошибка: 1.12e+07

Лучшие параметры → q1=1.165, q2=0.010, N=2.1e+06, E0=10000

1_wave:

N = 2.1e+06, E0 = 10000.0, q1 = 1.17, q2 = 0.01
beta1 = 3.960e-08, beta2 = 1.586e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

-> Симуляция завершена, период 124 дней

Фиттинг для 2_wave

2_wave:

N = 1.0e+06, E0 = 200.0, q1 = 0.27, q2 = 0.45
beta1 = 1.929e-08, beta2 = 1.500e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 200.0, q1 = 0.27, q2 = 0.45

beta1 = 1.929e-08, beta2 = 1.500e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 200.0, q1 = 0.27, q2 = 0.45
beta1 = 1.929e-08, beta2 = 1.500e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 200.0, q1 = 0.27, q2 = 0.45
beta1 = 1.929e-08, beta2 = 1.500e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 200.00000001, q1 = 0.27, q2 = 0.45
beta1 = 1.929e-08, beta2 = 1.500e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 366.8928729834156, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 366.8928729834156, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 366.8928729834156, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 366.8928729834156, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 366.8928729934156, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 200.33744557490928, q1 = 0.28, q2 = 0.46
beta1 = 1.997e-08, beta2 = 1.531e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 200.33744557490928, q1 = 0.28, q2 = 0.46
beta1 = 1.997e-08, beta2 = 1.531e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 200.33744557490928, q1 = 0.28, q2 = 0.46
beta1 = 1.997e-08, beta2 = 1.531e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 200.33744557490928, q1 = 0.28, q2 = 0.46
beta1 = 1.997e-08, beta2 = 1.531e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 200.3374455849093, q1 = 0.28, q2 = 0.46
beta1 = 1.997e-08, beta2 = 1.531e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 283.6151592791624, q1 = 2.64, q2 = 2.73
beta1 = 1.886e-07, beta2 = 9.099e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 283.6151592791624, q1 = 2.64, q2 = 2.73
beta1 = 1.886e-07, beta2 = 9.099e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 283.6151592791624, q1 = 2.64, q2 = 2.73
beta1 = 1.886e-07, beta2 = 9.099e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 283.6151592791624, q1 = 2.64, q2 = 2.73
beta1 = 1.886e-07, beta2 = 9.099e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 283.6151592891624, q1 = 2.64, q2 = 2.73
beta1 = 1.886e-07, beta2 = 9.099e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 201.84234818417005, q1 = 0.32, q2 = 0.50
beta1 = 2.302e-08, beta2 = 1.667e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 201.84234818417005, q1 = 0.32, q2 = 0.50
beta1 = 2.302e-08, beta2 = 1.667e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 201.84234818417005, q1 = 0.32, q2 = 0.50
beta1 = 2.302e-08, beta2 = 1.667e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 201.84234818417005, q1 = 0.32, q2 = 0.50
beta1 = 2.302e-08, beta2 = 1.667e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 201.84234819417006, q1 = 0.32, q2 = 0.50
beta1 = 2.302e-08, beta2 = 1.667e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.2446735247123, q1 = 0.39, q2 = 0.57
beta1 = 2.788e-08, beta2 = 1.886e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.2446735247123, q1 = 0.39, q2 = 0.57
beta1 = 2.788e-08, beta2 = 1.886e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.2446735247123, q1 = 0.39, q2 = 0.57
beta1 = 2.788e-08, beta2 = 1.886e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.2446735247123, q1 = 0.39, q2 = 0.57
beta1 = 2.788e-08, beta2 = 1.886e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.24467353471232, q1 = 0.39, q2 = 0.57
beta1 = 2.788e-08, beta2 = 1.886e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 243.92991640193736, q1 = 1.52, q2 = 1.65
beta1 = 1.082e-07, beta2 = 5.492e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 243.92991640193736, q1 = 1.52, q2 = 1.65
beta1 = 1.082e-07, beta2 = 5.492e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 243.92991640193736, q1 = 1.52, q2 = 1.65
beta1 = 1.082e-07, beta2 = 5.492e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 243.92991640193736, q1 = 1.52, q2 = 1.65
beta1 = 1.082e-07, beta2 = 5.492e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 243.92991641193737, q1 = 1.52, q2 = 1.65
beta1 = 1.082e-07, beta2 = 5.492e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.3648230945507, q1 = 0.39, q2 = 0.57
beta1 = 2.812e-08, beta2 = 1.897e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.3648230945507, q1 = 0.39, q2 = 0.57
beta1 = 2.812e-08, beta2 = 1.897e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.3648230945507, q1 = 0.39, q2 = 0.57
beta1 = 2.812e-08, beta2 = 1.897e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.3648230945507, q1 = 0.39, q2 = 0.57
beta1 = 2.812e-08, beta2 = 1.897e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.3648231045507, q1 = 0.39, q2 = 0.57
beta1 = 2.812e-08, beta2 = 1.897e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.4866360003911, q1 = 0.40, q2 = 0.57
beta1 = 2.837e-08, beta2 = 1.908e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.4866360003911, q1 = 0.40, q2 = 0.57
beta1 = 2.837e-08, beta2 = 1.908e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.4866360003911, q1 = 0.40, q2 = 0.57
beta1 = 2.837e-08, beta2 = 1.908e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.4866360003911, q1 = 0.40, q2 = 0.57
beta1 = 2.837e-08, beta2 = 1.908e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 204.4866360103911, q1 = 0.40, q2 = 0.57
beta1 = 2.837e-08, beta2 = 1.908e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.20827620116424, q1 = 0.96, q2 = 1.11
beta1 = 6.829e-08, beta2 = 3.700e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.20827620116424, q1 = 0.96, q2 = 1.11
beta1 = 6.829e-08, beta2 = 3.700e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.20827620116424, q1 = 0.96, q2 = 1.11
beta1 = 6.829e-08, beta2 = 3.700e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.20827620116424, q1 = 0.96, q2 = 1.11
beta1 = 6.829e-08, beta2 = 3.700e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.20827621116425, q1 = 0.96, q2 = 1.11
beta1 = 6.829e-08, beta2 = 3.700e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.95813861876707, q1 = 0.98, q2 = 1.13
beta1 = 6.981e-08, beta2 = 3.768e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.95813861876707, q1 = 0.98, q2 = 1.13
beta1 = 6.981e-08, beta2 = 3.768e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.95813861876707, q1 = 0.98, q2 = 1.13
beta1 = 6.981e-08, beta2 = 3.768e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.95813861876707, q1 = 0.98, q2 = 1.13
beta1 = 6.981e-08, beta2 = 3.768e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 224.95813862876707, q1 = 0.98, q2 = 1.13
beta1 = 6.981e-08, beta2 = 3.768e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 228.1453169156927, q1 = 1.07, q2 = 1.22

beta1 = 7.626e-08, beta2 = 4.058e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 228.1453169156927, q1 = 1.07, q2 = 1.22
beta1 = 7.626e-08, beta2 = 4.058e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 228.1453169156927, q1 = 1.07, q2 = 1.22
beta1 = 7.626e-08, beta2 = 4.058e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 228.1453169156927, q1 = 1.07, q2 = 1.22
beta1 = 7.626e-08, beta2 = 4.058e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 228.1453169256927, q1 = 1.07, q2 = 1.22
beta1 = 7.626e-08, beta2 = 4.058e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 227.01857739549212, q1 = 1.04, q2 = 1.19
beta1 = 7.398e-08, beta2 = 3.955e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 227.01857739549212, q1 = 1.04, q2 = 1.19
beta1 = 7.398e-08, beta2 = 3.955e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 227.01857739549212, q1 = 1.04, q2 = 1.19
beta1 = 7.398e-08, beta2 = 3.955e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 227.01857739549212, q1 = 1.04, q2 = 1.19
beta1 = 7.398e-08, beta2 = 3.955e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 227.01857740549212, q1 = 1.04, q2 = 1.19
beta1 = 7.398e-08, beta2 = 3.955e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 4248.83879320369, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4248.83879320369, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4248.83879320369, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4248.83879320369, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4248.83879321369, q1 = 5.00, q2 = 5.00
beta1 = 3.571e-07, beta2 = 1.667e-07
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 444.24295966215925, q1 = 1.25, q2 = 1.39
beta1 = 8.928e-08, beta2 = 4.642e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 444.24295966215925, q1 = 1.25, q2 = 1.39
beta1 = 8.928e-08, beta2 = 4.642e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 444.24295966215925, q1 = 1.25, q2 = 1.39
beta1 = 8.928e-08, beta2 = 4.642e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 444.24295966215925, q1 = 1.25, q2 = 1.39
beta1 = 8.928e-08, beta2 = 4.642e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 444.24295967215926, q1 = 1.25, q2 = 1.39
beta1 = 8.928e-08, beta2 = 4.642e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 235.8806717478003, q1 = 1.04, q2 = 1.20
beta1 = 7.461e-08, beta2 = 3.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 235.8806717478003, q1 = 1.04, q2 = 1.20
beta1 = 7.461e-08, beta2 = 3.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 235.8806717478003, q1 = 1.04, q2 = 1.20
beta1 = 7.461e-08, beta2 = 3.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 235.8806717478003, q1 = 1.04, q2 = 1.20
beta1 = 7.461e-08, beta2 = 3.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 235.88067175780031, q1 = 1.04, q2 = 1.20
beta1 = 7.461e-08, beta2 = 3.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 256.18971383164654, q1 = 1.05, q2 = 1.21
beta1 = 7.506e-08, beta2 = 4.019e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 256.18971383164654, q1 = 1.05, q2 = 1.21
beta1 = 7.506e-08, beta2 = 4.019e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 256.18971383164654, q1 = 1.05, q2 = 1.21
beta1 = 7.506e-08, beta2 = 4.019e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 256.18971383164654, q1 = 1.05, q2 = 1.21
beta1 = 7.506e-08, beta2 = 4.019e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 256.18971384164655, q1 = 1.05, q2 = 1.21
beta1 = 7.506e-08, beta2 = 4.019e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 279.11858476034314, q1 = 1.05, q2 = 1.21
beta1 = 7.476e-08, beta2 = 4.036e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 279.11858476034314, q1 = 1.05, q2 = 1.21
beta1 = 7.476e-08, beta2 = 4.036e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 279.11858476034314, q1 = 1.05, q2 = 1.21
beta1 = 7.476e-08, beta2 = 4.036e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 279.11858476034314, q1 = 1.05, q2 = 1.21
beta1 = 7.476e-08, beta2 = 4.036e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 279.11858477034315, q1 = 1.05, q2 = 1.21
beta1 = 7.476e-08, beta2 = 4.036e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 494.83943506569335, q1 = 0.98, q2 = 1.24
beta1 = 6.997e-08, beta2 = 4.139e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 494.83943506569335, q1 = 0.98, q2 = 1.24
beta1 = 6.997e-08, beta2 = 4.139e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 494.83943506569335, q1 = 0.98, q2 = 1.24
beta1 = 6.997e-08, beta2 = 4.139e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 494.83943506569335, q1 = 0.98, q2 = 1.24
beta1 = 6.997e-08, beta2 = 4.139e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 494.83943507569336, q1 = 0.98, q2 = 1.24
beta1 = 6.997e-08, beta2 = 4.139e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2562.5684457030843, q1 = 0.33, q2 = 1.46
beta1 = 2.326e-08, beta2 = 4.868e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2562.5684457030843, q1 = 0.33, q2 = 1.46
beta1 = 2.326e-08, beta2 = 4.868e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2562.5684457030843, q1 = 0.33, q2 = 1.46
beta1 = 2.326e-08, beta2 = 4.868e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2562.5684457030843, q1 = 0.33, q2 = 1.46
beta1 = 2.326e-08, beta2 = 4.868e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 2562.568445713084, q1 = 0.33, q2 = 1.46
beta1 = 2.326e-08, beta2 = 4.868e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1312.3611810939738, q1 = 0.72, q2 = 1.33
beta1 = 5.150e-08, beta2 = 4.428e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1312.3611810939738, q1 = 0.72, q2 = 1.33
beta1 = 5.150e-08, beta2 = 4.428e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1312.3611810939738, q1 = 0.72, q2 = 1.33
beta1 = 5.150e-08, beta2 = 4.428e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1312.3611810939738, q1 = 0.72, q2 = 1.33
beta1 = 5.150e-08, beta2 = 4.428e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1312.3611811039736, q1 = 0.72, q2 = 1.33
beta1 = 5.150e-08, beta2 = 4.428e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1368.0422033436575, q1 = 0.72, q2 = 1.34
beta1 = 5.140e-08, beta2 = 4.463e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1368.0422033436575, q1 = 0.72, q2 = 1.34
beta1 = 5.140e-08, beta2 = 4.463e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1368.0422033436575, q1 = 0.72, q2 = 1.34
beta1 = 5.140e-08, beta2 = 4.463e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1368.0422033436575, q1 = 0.72, q2 = 1.34
beta1 = 5.140e-08, beta2 = 4.463e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 1368.0422033536574, q1 = 0.72, q2 = 1.34

beta1 = 5.140e-08, beta2 = 4.463e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 1590.7662923423927, q1 = 0.71, q2 = 1.38
beta1 = 5.102e-08, beta2 = 4.603e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 1590.7662923423927, q1 = 0.71, q2 = 1.38
beta1 = 5.102e-08, beta2 = 4.603e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 1590.7662923423927, q1 = 0.71, q2 = 1.38
beta1 = 5.102e-08, beta2 = 4.603e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 1590.7662923423927, q1 = 0.71, q2 = 1.38
beta1 = 5.102e-08, beta2 = 4.603e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 1590.7662923523926, q1 = 0.71, q2 = 1.38
beta1 = 5.102e-08, beta2 = 4.603e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2889.5016720169365, q1 = 0.60, q2 = 1.58
beta1 = 4.296e-08, beta2 = 5.265e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2889.5016720169365, q1 = 0.60, q2 = 1.58
beta1 = 4.296e-08, beta2 = 5.265e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2889.5016720169365, q1 = 0.60, q2 = 1.58
beta1 = 4.296e-08, beta2 = 5.265e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2889.5016720169365, q1 = 0.60, q2 = 1.58
beta1 = 4.296e-08, beta2 = 5.265e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2889.5016720269364, q1 = 0.60, q2 = 1.58
beta1 = 4.296e-08, beta2 = 5.265e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 5131.059467597786, q1 = 0.34, q2 = 1.87
beta1 = 2.403e-08, beta2 = 6.217e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 5131.059467597786, q1 = 0.34, q2 = 1.87
beta1 = 2.403e-08, beta2 = 6.217e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 5131.059467597786, q1 = 0.34, q2 = 1.87
beta1 = 2.403e-08, beta2 = 6.217e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 5131.059467597786, q1 = 0.34, q2 = 1.87
beta1 = 2.403e-08, beta2 = 6.217e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 5131.059467607786, q1 = 0.34, q2 = 1.87
beta1 = 2.403e-08, beta2 = 6.217e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7359.8171091853865, q1 = 0.17, q2 = 2.11
beta1 = 1.226e-08, beta2 = 7.048e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7359.8171091853865, q1 = 0.17, q2 = 2.11
beta1 = 1.226e-08, beta2 = 7.048e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7359.8171091853865, q1 = 0.17, q2 = 2.11
beta1 = 1.226e-08, beta2 = 7.048e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7359.8171091853865, q1 = 0.17, q2 = 2.11
beta1 = 1.226e-08, beta2 = 7.048e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7359.817109195386, q1 = 0.17, q2 = 2.11
beta1 = 1.226e-08, beta2 = 7.048e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9546.836011462077, q1 = 0.01, q2 = 2.36
beta1 = 7.143e-10, beta2 = 7.864e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 9546.836011462077, q1 = 0.01, q2 = 2.36
 beta1 = 7.143e-10, beta2 = 7.864e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 9546.836011462077, q1 = 0.01, q2 = 2.36
 beta1 = 7.143e-10, beta2 = 7.864e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 9546.836011462077, q1 = 0.01, q2 = 2.36
 beta1 = 7.143e-10, beta2 = 7.864e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 9546.836011472078, q1 = 0.01, q2 = 2.36
 beta1 = 7.143e-10, beta2 = 7.864e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 9802.867065457789, q1 = 0.01, q2 = 2.38
 beta1 = 7.143e-10, beta2 = 7.934e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 9802.867065457789, q1 = 0.01, q2 = 2.38
 beta1 = 7.143e-10, beta2 = 7.934e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 9802.867065457789, q1 = 0.01, q2 = 2.38
 beta1 = 7.143e-10, beta2 = 7.934e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 9802.86706546779, q1 = 0.01, q2 = 2.38
 beta1 = 7.143e-10, beta2 = 7.934e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 7841.847982348378, q1 = 1.08, q2 = 1.76
 beta1 = 7.732e-08, beta2 = 5.870e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
 N = 1.0e+06, E0 = 7841.847982348378, q1 = 1.08, q2 = 1.76
 beta1 = 7.732e-08, beta2 = 5.870e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 7841.847982348378, q1 = 1.08, q2 = 1.76
beta1 = 7.732e-08, beta2 = 5.870e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 7841.847982348378, q1 = 1.08, q2 = 1.76
beta1 = 7.732e-08, beta2 = 5.870e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 7841.8479823583775, q1 = 1.08, q2 = 1.76
beta1 = 7.732e-08, beta2 = 5.870e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2298.116724815365, q1 = 4.11, q2 = 0.01
beta1 = 2.939e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2298.116724815365, q1 = 4.11, q2 = 0.01
beta1 = 2.939e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2298.116724815365, q1 = 4.11, q2 = 0.01
beta1 = 2.939e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2298.116724815365, q1 = 4.11, q2 = 0.01
beta1 = 2.939e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 2298.116724825365, q1 = 4.11, q2 = 0.01
beta1 = 2.939e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 7392.522282194612, q1 = 1.33, q2 = 1.62
beta1 = 9.488e-08, beta2 = 5.397e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 7392.522282194612, q1 = 1.33, q2 = 1.62
beta1 = 9.488e-08, beta2 = 5.397e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:
N = 1.0e+06, E0 = 7392.522282194612, q1 = 1.33, q2 = 1.62
beta1 = 9.488e-08, beta2 = 5.397e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7392.522282194612, q1 = 1.33, q2 = 1.62
beta1 = 9.488e-08, beta2 = 5.397e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7392.522282204612, q1 = 1.33, q2 = 1.62
beta1 = 9.488e-08, beta2 = 5.397e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4845.3195035049885, q1 = 2.72, q2 = 0.81
beta1 = 1.944e-07, beta2 = 2.715e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4845.3195035049885, q1 = 2.72, q2 = 0.81
beta1 = 1.944e-07, beta2 = 2.715e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4845.3195035049885, q1 = 2.72, q2 = 0.81
beta1 = 1.944e-07, beta2 = 2.715e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4845.3195035049885, q1 = 2.72, q2 = 0.81
beta1 = 1.944e-07, beta2 = 2.715e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 4845.319503514988, q1 = 2.72, q2 = 0.81
beta1 = 1.944e-07, beta2 = 2.715e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7338.5886287021995, q1 = 1.36, q2 = 1.60
beta1 = 9.698e-08, beta2 = 5.340e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7338.5886287021995, q1 = 1.36, q2 = 1.60
beta1 = 9.698e-08, beta2 = 5.340e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7338.5886287021995, q1 = 1.36, q2 = 1.60
beta1 = 9.698e-08, beta2 = 5.340e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7338.5886287021995, q1 = 1.36, q2 = 1.60

beta1 = 9.698e-08, beta2 = 5.340e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7338.588628712199, q1 = 1.36, q2 = 1.60
beta1 = 9.698e-08, beta2 = 5.340e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 3120.810010372965, q1 = 4.21, q2 = 0.01
beta1 = 3.008e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 3120.810010372965, q1 = 4.21, q2 = 0.01
beta1 = 3.008e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 3120.810010372965, q1 = 4.21, q2 = 0.01
beta1 = 3.008e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 3120.810010372965, q1 = 4.21, q2 = 0.01
beta1 = 3.008e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 3120.8100103829647, q1 = 4.21, q2 = 0.01
beta1 = 3.008e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7319.967841964809, q1 = 1.37, q2 = 1.59
beta1 = 9.788e-08, beta2 = 5.316e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7319.967841964809, q1 = 1.37, q2 = 1.59
beta1 = 9.788e-08, beta2 = 5.316e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7319.967841964809, q1 = 1.37, q2 = 1.59
beta1 = 9.788e-08, beta2 = 5.316e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7319.967841964809, q1 = 1.37, q2 = 1.59
beta1 = 9.788e-08, beta2 = 5.316e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7319.967841974809, q1 = 1.37, q2 = 1.59
beta1 = 9.788e-08, beta2 = 5.316e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7727.113734982498, q1 = 1.34, q2 = 1.58
beta1 = 9.564e-08, beta2 = 5.263e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7727.113734982498, q1 = 1.34, q2 = 1.58
beta1 = 9.564e-08, beta2 = 5.263e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7727.113734982498, q1 = 1.34, q2 = 1.58
beta1 = 9.564e-08, beta2 = 5.263e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7727.113734982498, q1 = 1.34, q2 = 1.58
beta1 = 9.564e-08, beta2 = 5.263e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 7727.1137349924975, q1 = 1.34, q2 = 1.58
beta1 = 9.564e-08, beta2 = 5.263e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9355.697307053251, q1 = 1.21, q2 = 1.51
beta1 = 8.665e-08, beta2 = 5.050e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9355.697307053251, q1 = 1.21, q2 = 1.51
beta1 = 8.665e-08, beta2 = 5.050e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9355.697307053251, q1 = 1.21, q2 = 1.51
beta1 = 8.665e-08, beta2 = 5.050e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9355.697307053251, q1 = 1.21, q2 = 1.51
beta1 = 8.665e-08, beta2 = 5.050e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9355.697307063252, q1 = 1.21, q2 = 1.51
beta1 = 8.665e-08, beta2 = 5.050e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.24, q2 = 0.85
beta1 = 8.836e-08, beta2 = 2.832e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.24, q2 = 0.85
beta1 = 8.836e-08, beta2 = 2.832e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.24, q2 = 0.85
beta1 = 8.836e-08, beta2 = 2.832e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.24, q2 = 0.85
beta1 = 8.836e-08, beta2 = 2.832e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.24, q2 = 0.85
beta1 = 8.836e-08, beta2 = 2.832e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.66, q2 = 0.01
beta1 = 1.182e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.66, q2 = 0.01
beta1 = 1.182e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.66, q2 = 0.01
beta1 = 1.182e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.66, q2 = 0.01
beta1 = 1.182e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.66, q2 = 0.01
beta1 = 1.182e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.96, q2 = 0.01
beta1 = 1.402e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.96, q2 = 0.01
beta1 = 1.402e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.96, q2 = 0.01
beta1 = 1.402e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.96, q2 = 0.01
beta1 = 1.402e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.96, q2 = 0.01
beta1 = 1.402e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.237e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.237e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.237e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.237e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.73, q2 = 0.01
beta1 = 1.237e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10

delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

2_wave:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

Успех: True | Ошибка: 6.92e+07

Лучшие параметры → q1=1.734, q2=0.010, N=1.0e+06, E0=10000

2_wave:

N = 1.0e+06, E0 = 10000.0, q1 = 1.73, q2 = 0.01
beta1 = 1.239e-07, beta2 = 3.333e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

-> Симуляция завершена, период 155 дней

Фиттинг для 3_wave_alpha

3_wave_alpha:

N = 2.0e+06, E0 = 500.0, q1 = 1.20, q2 = 2.00

beta1 = 4.286e-08, beta2 = 3.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 500.0, q1 = 1.20, q2 = 2.00
beta1 = 4.286e-08, beta2 = 3.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 500.0, q1 = 1.20, q2 = 2.00
beta1 = 4.286e-08, beta2 = 3.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 500.0, q1 = 1.20, q2 = 2.00
beta1 = 4.286e-08, beta2 = 3.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 500.00000001, q1 = 1.20, q2 = 2.00
beta1 = 4.286e-08, beta2 = 3.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 3.571e-10, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 3.571e-10, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 3.571e-10, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 3.571e-10, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1.00000001, q1 = 0.01, q2 = 0.01
beta1 = 3.571e-10, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1.05702921794524, q1 = 0.01, q2 = 0.01
beta1 = 3.583e-10, beta2 = 1.699e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.05702921794524, q1 = 0.01, q2 = 0.01
beta1 = 3.583e-10, beta2 = 1.699e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.05702921794524, q1 = 0.01, q2 = 0.01
beta1 = 3.583e-10, beta2 = 1.699e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.05702921794524, q1 = 0.01, q2 = 0.01
beta1 = 3.583e-10, beta2 = 1.699e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.05702922794524, q1 = 0.01, q2 = 0.01
beta1 = 3.583e-10, beta2 = 1.699e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.2851460897262004, q1 = 0.01, q2 = 0.01
beta1 = 3.630e-10, beta2 = 1.828e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.2851460897262004, q1 = 0.01, q2 = 0.01
beta1 = 3.630e-10, beta2 = 1.828e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.2851460897262004, q1 = 0.01, q2 = 0.01
beta1 = 3.630e-10, beta2 = 1.828e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.2851460897262004, q1 = 0.01, q2 = 0.01
beta1 = 3.630e-10, beta2 = 1.828e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1.2851460997262003, q1 = 0.01, q2 = 0.01
beta1 = 3.630e-10, beta2 = 1.828e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 2.1976135768500416, q1 = 0.01, q2 = 0.01
beta1 = 3.817e-10, beta2 = 2.344e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 2.1976135768500416, q1 = 0.01, q2 = 0.01
beta1 = 3.817e-10, beta2 = 2.344e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2.1976135768500416, q1 = 0.01, q2 = 0.01
beta1 = 3.817e-10, beta2 = 2.344e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2.1976135768500416, q1 = 0.01, q2 = 0.01
beta1 = 3.817e-10, beta2 = 2.344e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2.1976135868500415, q1 = 0.01, q2 = 0.01
beta1 = 3.817e-10, beta2 = 2.344e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5.847483525345407, q1 = 0.01, q2 = 0.03
beta1 = 4.567e-10, beta2 = 4.410e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5.847483525345407, q1 = 0.01, q2 = 0.03
beta1 = 4.567e-10, beta2 = 4.410e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5.847483525345407, q1 = 0.01, q2 = 0.03
beta1 = 4.567e-10, beta2 = 4.410e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5.847483525345407, q1 = 0.01, q2 = 0.03
beta1 = 4.567e-10, beta2 = 4.410e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5.847483535345407, q1 = 0.01, q2 = 0.03
beta1 = 4.567e-10, beta2 = 4.410e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 20.446963319326866, q1 = 0.02, q2 = 0.08
beta1 = 7.565e-10, beta2 = 1.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 20.446963319326866, q1 = 0.02, q2 = 0.08
beta1 = 7.565e-10, beta2 = 1.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 20.446963319326866, q1 = 0.02, q2 = 0.08
beta1 = 7.565e-10, beta2 = 1.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 20.446963319326866, q1 = 0.02, q2 = 0.08
beta1 = 7.565e-10, beta2 = 1.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 20.446963329326866, q1 = 0.02, q2 = 0.08
beta1 = 7.565e-10, beta2 = 1.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 78.8448824952527, q1 = 0.05, q2 = 0.27
beta1 = 1.956e-09, beta2 = 4.572e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 78.8448824952527, q1 = 0.05, q2 = 0.27
beta1 = 1.956e-09, beta2 = 4.572e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 78.8448824952527, q1 = 0.05, q2 = 0.27
beta1 = 1.956e-09, beta2 = 4.572e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 78.8448824952527, q1 = 0.05, q2 = 0.27
beta1 = 1.956e-09, beta2 = 4.572e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 78.8448825052527, q1 = 0.05, q2 = 0.27
beta1 = 1.956e-09, beta2 = 4.572e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 312.43655919895605, q1 = 0.19, q2 = 1.07
beta1 = 6.753e-09, beta2 = 1.779e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 312.43655919895605, q1 = 0.19, q2 = 1.07
beta1 = 6.753e-09, beta2 = 1.779e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 312.43655919895605, q1 = 0.19, q2 = 1.07
beta1 = 6.753e-09, beta2 = 1.779e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 312.43655919895605, q1 = 0.19, q2 = 1.07
beta1 = 6.753e-09, beta2 = 1.779e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 312.43655920895606, q1 = 0.19, q2 = 1.07
beta1 = 6.753e-09, beta2 = 1.779e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1246.8032660137694, q1 = 0.73, q2 = 4.24
beta1 = 2.594e-08, beta2 = 7.067e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1246.8032660137694, q1 = 0.73, q2 = 4.24
beta1 = 2.594e-08, beta2 = 7.067e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1246.8032660137694, q1 = 0.73, q2 = 4.24
beta1 = 2.594e-08, beta2 = 7.067e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1246.8032660137694, q1 = 0.73, q2 = 4.24
beta1 = 2.594e-08, beta2 = 7.067e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1246.8032660237693, q1 = 0.73, q2 = 4.24
beta1 = 2.594e-08, beta2 = 7.067e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 419.6910147345989, q1 = 0.25, q2 = 1.43
beta1 = 8.955e-09, beta2 = 2.386e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 419.6910147345989, q1 = 0.25, q2 = 1.43
beta1 = 8.955e-09, beta2 = 2.386e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 419.6910147345989, q1 = 0.25, q2 = 1.43
beta1 = 8.955e-09, beta2 = 2.386e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 419.6910147345989, q1 = 0.25, q2 = 1.43
beta1 = 8.955e-09, beta2 = 2.386e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 419.69101474459893, q1 = 0.25, q2 = 1.43

beta1 = 8.955e-09, beta2 = 2.386e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 494.9822301100893, q1 = 0.29, q2 = 1.69
beta1 = 1.050e-08, beta2 = 2.812e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 494.9822301100893, q1 = 0.29, q2 = 1.69
beta1 = 1.050e-08, beta2 = 2.812e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 494.9822301100893, q1 = 0.29, q2 = 1.69
beta1 = 1.050e-08, beta2 = 2.812e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 494.9822301100893, q1 = 0.29, q2 = 1.69
beta1 = 1.050e-08, beta2 = 2.812e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 494.9822301200893, q1 = 0.29, q2 = 1.69
beta1 = 1.050e-08, beta2 = 2.812e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 870.8927480619293, q1 = 0.51, q2 = 2.96
beta1 = 1.822e-08, beta2 = 4.939e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 870.8927480619293, q1 = 0.51, q2 = 2.96
beta1 = 1.822e-08, beta2 = 4.939e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 870.8927480619293, q1 = 0.51, q2 = 2.96
beta1 = 1.822e-08, beta2 = 4.939e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 870.8927480619293, q1 = 0.51, q2 = 2.96
beta1 = 1.822e-08, beta2 = 4.939e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 870.8927480719293, q1 = 0.51, q2 = 2.96
beta1 = 1.822e-08, beta2 = 4.939e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 536.8679596085266, q1 = 0.32, q2 = 1.83
beta1 = 1.136e-08, beta2 = 3.049e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 536.8679596085266, q1 = 0.32, q2 = 1.83
beta1 = 1.136e-08, beta2 = 3.049e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 536.8679596085266, q1 = 0.32, q2 = 1.83
beta1 = 1.136e-08, beta2 = 3.049e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 536.8679596085266, q1 = 0.32, q2 = 1.83
beta1 = 1.136e-08, beta2 = 3.049e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 536.8679596185266, q1 = 0.32, q2 = 1.83
beta1 = 1.136e-08, beta2 = 3.049e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 786.878859693925, q1 = 0.46, q2 = 2.68
beta1 = 1.650e-08, beta2 = 4.464e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 786.878859693925, q1 = 0.46, q2 = 2.68
beta1 = 1.650e-08, beta2 = 4.464e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 786.878859693925, q1 = 0.46, q2 = 2.68
beta1 = 1.650e-08, beta2 = 4.464e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 786.878859693925, q1 = 0.46, q2 = 2.68
beta1 = 1.650e-08, beta2 = 4.464e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 786.8788597039251, q1 = 0.46, q2 = 2.68
beta1 = 1.650e-08, beta2 = 4.464e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 800.4850020222641, q1 = 0.49, q2 = 2.73
beta1 = 1.755e-08, beta2 = 4.554e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 800.4850020222641, q1 = 0.49, q2 = 2.73
 beta1 = 1.755e-08, beta2 = 4.554e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 800.4850020222641, q1 = 0.49, q2 = 2.73
 beta1 = 1.755e-08, beta2 = 4.554e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 800.4850020222641, q1 = 0.49, q2 = 2.73
 beta1 = 1.755e-08, beta2 = 4.554e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 800.4850020322641, q1 = 0.49, q2 = 2.73
 beta1 = 1.755e-08, beta2 = 4.554e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 807.7013593719263, q1 = 0.51, q2 = 2.76
 beta1 = 1.807e-08, beta2 = 4.595e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 807.7013593719263, q1 = 0.51, q2 = 2.76
 beta1 = 1.807e-08, beta2 = 4.595e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 807.7013593719263, q1 = 0.51, q2 = 2.76
 beta1 = 1.807e-08, beta2 = 4.595e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 807.7013593719263, q1 = 0.51, q2 = 2.76
 beta1 = 1.807e-08, beta2 = 4.595e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 807.7013593819263, q1 = 0.51, q2 = 2.76
 beta1 = 1.807e-08, beta2 = 4.595e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 805.8151144907288, q1 = 0.50, q2 = 2.75
 beta1 = 1.791e-08, beta2 = 4.581e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
 N = 2.0e+06, E0 = 805.8151144907288, q1 = 0.50, q2 = 2.75
 beta1 = 1.791e-08, beta2 = 4.581e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 805.8151144907288, q1 = 0.50, q2 = 2.75
beta1 = 1.791e-08, beta2 = 4.581e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 805.8151144907288, q1 = 0.50, q2 = 2.75
beta1 = 1.791e-08, beta2 = 4.581e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 805.8151145007288, q1 = 0.50, q2 = 2.75
beta1 = 1.791e-08, beta2 = 4.581e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 806.2453946556747, q1 = 0.50, q2 = 2.75
beta1 = 1.791e-08, beta2 = 4.579e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 806.2453946556747, q1 = 0.50, q2 = 2.75
beta1 = 1.791e-08, beta2 = 4.579e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 806.2453946556747, q1 = 0.50, q2 = 2.75
beta1 = 1.791e-08, beta2 = 4.579e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 806.2453946556747, q1 = 0.50, q2 = 2.75
beta1 = 1.791e-08, beta2 = 4.579e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 806.2453946656747, q1 = 0.50, q2 = 2.75
beta1 = 1.791e-08, beta2 = 4.579e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 807.9665153154582, q1 = 0.50, q2 = 2.74
beta1 = 1.792e-08, beta2 = 4.573e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 807.9665153154582, q1 = 0.50, q2 = 2.74
beta1 = 1.792e-08, beta2 = 4.573e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 807.9665153154582, q1 = 0.50, q2 = 2.74
beta1 = 1.792e-08, beta2 = 4.573e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 807.9665153154582, q1 = 0.50, q2 = 2.74
beta1 = 1.792e-08, beta2 = 4.573e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 807.9665153254582, q1 = 0.50, q2 = 2.74
beta1 = 1.792e-08, beta2 = 4.573e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 814.8509979545923, q1 = 0.50, q2 = 2.73
beta1 = 1.796e-08, beta2 = 4.549e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 814.8509979545923, q1 = 0.50, q2 = 2.73
beta1 = 1.796e-08, beta2 = 4.549e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 814.8509979545923, q1 = 0.50, q2 = 2.73
beta1 = 1.796e-08, beta2 = 4.549e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 814.8509979545923, q1 = 0.50, q2 = 2.73
beta1 = 1.796e-08, beta2 = 4.549e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 814.8509979645924, q1 = 0.50, q2 = 2.73
beta1 = 1.796e-08, beta2 = 4.549e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 842.3889285111289, q1 = 0.51, q2 = 2.67
beta1 = 1.811e-08, beta2 = 4.454e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 842.3889285111289, q1 = 0.51, q2 = 2.67
beta1 = 1.811e-08, beta2 = 4.454e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 842.3889285111289, q1 = 0.51, q2 = 2.67
beta1 = 1.811e-08, beta2 = 4.454e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 842.3889285111289, q1 = 0.51, q2 = 2.67

beta1 = 1.811e-08, beta2 = 4.454e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 842.388928521129, q1 = 0.51, q2 = 2.67
beta1 = 1.811e-08, beta2 = 4.454e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1019.6772373404276, q1 = 0.56, q2 = 2.36
beta1 = 1.982e-08, beta2 = 3.940e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1019.6772373404276, q1 = 0.56, q2 = 2.36
beta1 = 1.982e-08, beta2 = 3.940e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1019.6772373404276, q1 = 0.56, q2 = 2.36
beta1 = 1.982e-08, beta2 = 3.940e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1019.6772373404276, q1 = 0.56, q2 = 2.36
beta1 = 1.982e-08, beta2 = 3.940e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1019.6772373504276, q1 = 0.56, q2 = 2.36
beta1 = 1.982e-08, beta2 = 3.940e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1263.9265833376933, q1 = 0.64, q2 = 1.99
beta1 = 2.279e-08, beta2 = 3.310e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1263.9265833376933, q1 = 0.64, q2 = 1.99
beta1 = 2.279e-08, beta2 = 3.310e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1263.9265833376933, q1 = 0.64, q2 = 1.99
beta1 = 2.279e-08, beta2 = 3.310e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 1263.9265833376933, q1 = 0.64, q2 = 1.99
beta1 = 2.279e-08, beta2 = 3.310e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1263.9265833476932, q1 = 0.64, q2 = 1.99
beta1 = 2.279e-08, beta2 = 3.310e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1585.7725030609777, q1 = 0.77, q2 = 1.55
beta1 = 2.749e-08, beta2 = 2.584e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1585.7725030609777, q1 = 0.77, q2 = 1.55
beta1 = 2.749e-08, beta2 = 2.584e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1585.7725030609777, q1 = 0.77, q2 = 1.55
beta1 = 2.749e-08, beta2 = 2.584e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1585.7725030609777, q1 = 0.77, q2 = 1.55
beta1 = 2.749e-08, beta2 = 2.584e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 1585.7725030709776, q1 = 0.77, q2 = 1.55
beta1 = 2.749e-08, beta2 = 2.584e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 2597.9116439818877, q1 = 1.23, q2 = 0.32
beta1 = 4.401e-08, beta2 = 5.264e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 2597.9116439818877, q1 = 1.23, q2 = 0.32
beta1 = 4.401e-08, beta2 = 5.264e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 2597.9116439818877, q1 = 1.23, q2 = 0.32
beta1 = 4.401e-08, beta2 = 5.264e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 2597.9116439818877, q1 = 1.23, q2 = 0.32
beta1 = 4.401e-08, beta2 = 5.264e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 2597.9116439918876, q1 = 1.23, q2 = 0.32
beta1 = 4.401e-08, beta2 = 5.264e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5754.950746325783, q1 = 2.79, q2 = 0.01
beta1 = 9.960e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5754.950746325783, q1 = 2.79, q2 = 0.01
beta1 = 9.960e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5754.950746325783, q1 = 2.79, q2 = 0.01
beta1 = 9.960e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5754.950746325783, q1 = 2.79, q2 = 0.01
beta1 = 9.960e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5754.950746335783, q1 = 2.79, q2 = 0.01
beta1 = 9.960e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2698.679130507813, q1 = 1.28, q2 = 0.31
beta1 = 4.578e-08, beta2 = 5.101e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2698.679130507813, q1 = 1.28, q2 = 0.31
beta1 = 4.578e-08, beta2 = 5.101e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2698.679130507813, q1 = 1.28, q2 = 0.31
beta1 = 4.578e-08, beta2 = 5.101e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2698.679130517813, q1 = 1.28, q2 = 0.31
beta1 = 4.578e-08, beta2 = 5.101e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 9812.146966607375, q1 = 4.70, q2 = 0.01
beta1 = 1.679e-07, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 9812.146966607375, q1 = 4.70, q2 = 0.01
beta1 = 1.679e-07, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 9812.146966607375, q1 = 4.70, q2 = 0.01
beta1 = 1.679e-07, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 9812.146966607375, q1 = 4.70, q2 = 0.01
beta1 = 1.679e-07, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 9812.146966617376, q1 = 4.70, q2 = 0.01
beta1 = 1.679e-07, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2808.60748505008, q1 = 1.33, q2 = 0.30
beta1 = 4.767e-08, beta2 = 5.025e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2808.60748505008, q1 = 1.33, q2 = 0.30
beta1 = 4.767e-08, beta2 = 5.025e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2808.60748505008, q1 = 1.33, q2 = 0.30
beta1 = 4.767e-08, beta2 = 5.025e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 2808.60748506008, q1 = 1.33, q2 = 0.30
beta1 = 4.767e-08, beta2 = 5.025e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3067.5971066194393, q1 = 1.45, q2 = 0.01
beta1 = 5.186e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3067.5971066194393, q1 = 1.45, q2 = 0.01
beta1 = 5.186e-08, beta2 = 1.667e-10

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3067.5971066194393, q1 = 1.45, q2 = 0.01
beta1 = 5.186e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3067.5971066194393, q1 = 1.45, q2 = 0.01
beta1 = 5.186e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3067.597106629439, q1 = 1.45, q2 = 0.01
beta1 = 5.186e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3078.7367042016035, q1 = 1.45, q2 = 0.01
beta1 = 5.191e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3078.7367042016035, q1 = 1.45, q2 = 0.01
beta1 = 5.191e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3078.7367042016035, q1 = 1.45, q2 = 0.01
beta1 = 5.191e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3078.7367042016035, q1 = 1.45, q2 = 0.01
beta1 = 5.191e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3078.7367042116034, q1 = 1.45, q2 = 0.01
beta1 = 5.191e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3220.8771804982903, q1 = 1.46, q2 = 0.01
beta1 = 5.210e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3220.8771804982903, q1 = 1.46, q2 = 0.01
beta1 = 5.210e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 3220.8771804982903, q1 = 1.46, q2 = 0.01

beta1 = 5.210e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3220.8771804982903, q1 = 1.46, q2 = 0.01
beta1 = 5.210e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3220.87718050829, q1 = 1.46, q2 = 0.01
beta1 = 5.210e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3621.5738325318825, q1 = 1.46, q2 = 0.01
beta1 = 5.214e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3621.5738325318825, q1 = 1.46, q2 = 0.01
beta1 = 5.214e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3621.5738325318825, q1 = 1.46, q2 = 0.01
beta1 = 5.214e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3621.5738325318825, q1 = 1.46, q2 = 0.01
beta1 = 5.214e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 3621.5738325418824, q1 = 1.46, q2 = 0.01
beta1 = 5.214e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5132.995102363446, q1 = 1.44, q2 = 0.01
beta1 = 5.151e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5132.995102363446, q1 = 1.44, q2 = 0.01
beta1 = 5.151e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 5132.995102363446, q1 = 1.44, q2 = 0.01
beta1 = 5.151e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 5132.995102363446, q1 = 1.44, q2 = 0.01
beta1 = 5.151e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 5132.995102373446, q1 = 1.44, q2 = 0.01
beta1 = 5.151e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.31, q2 = 0.01
beta1 = 4.684e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.31, q2 = 0.01
beta1 = 4.684e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.31, q2 = 0.01
beta1 = 4.684e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.31, q2 = 0.01
beta1 = 4.684e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 9999.99999999, q1 = 1.31, q2 = 0.01
beta1 = 4.684e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 2.44, q2 = 0.01
beta1 = 8.711e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 2.44, q2 = 0.01
beta1 = 8.711e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 2.44, q2 = 0.01
beta1 = 8.711e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 2.44, q2 = 0.01
beta1 = 8.711e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 9999.99999999, q1 = 2.44, q2 = 0.01
beta1 = 8.711e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 10000.0, q1 = 1.33, q2 = 0.01
beta1 = 4.758e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 10000.0, q1 = 1.33, q2 = 0.01
beta1 = 4.758e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 10000.0, q1 = 1.33, q2 = 0.01
beta1 = 4.758e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 10000.0, q1 = 1.33, q2 = 0.01
beta1 = 4.758e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 9999.99999999, q1 = 1.33, q2 = 0.01
beta1 = 4.758e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.784e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.784e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.784e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.784e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
N = 2.0e+06, E0 = 9999.99999999, q1 = 1.34, q2 = 0.01
beta1 = 4.784e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

```
3_wave_alpha:
  N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 9999.99999999, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 9999.99999999, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:
  N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
  beta1 = 4.781e-08, beta2 = 1.667e-10
```

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 9999.99999999, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_alpha:

N = 2.0e+06, E0 = 9999.99999999, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

Успех: True | Ошибка: 3.95e+07

Лучшие параметры → q1=1.339, q2=0.010, N=2.0e+06, E0=10000

3_wave_alpha:

N = 2.0e+06, E0 = 10000.0, q1 = 1.34, q2 = 0.01
beta1 = 4.781e-08, beta2 = 1.667e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

-> Симуляция завершена, период 120 дней

Фиттинг для 3_wave_delta

3_wave_delta:

N = 3.0e+06, E0 = 250.0, q1 = 1.68, q2 = 2.80
beta1 = 4.000e-08, beta2 = 3.111e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 250.0, q1 = 1.68, q2 = 2.80
beta1 = 4.000e-08, beta2 = 3.111e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 250.0, q1 = 1.68, q2 = 2.80
beta1 = 4.000e-08, beta2 = 3.111e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 250.0, q1 = 1.68, q2 = 2.80
beta1 = 4.000e-08, beta2 = 3.111e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 250.00000001, q1 = 1.68, q2 = 2.80
beta1 = 4.000e-08, beta2 = 3.111e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 2.381e-10, beta2 = 1.111e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 2.381e-10, beta2 = 1.111e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 2.381e-10, beta2 = 1.111e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0, q1 = 0.01, q2 = 0.01
beta1 = 2.381e-10, beta2 = 1.111e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.00000001, q1 = 0.01, q2 = 0.01
beta1 = 2.381e-10, beta2 = 1.111e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0029659787585672, q1 = 0.01, q2 = 0.01
beta1 = 2.383e-10, beta2 = 1.114e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0029659787585672, q1 = 0.01, q2 = 0.01
beta1 = 2.383e-10, beta2 = 1.114e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0029659787585672, q1 = 0.01, q2 = 0.01
beta1 = 2.383e-10, beta2 = 1.114e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0029659787585672, q1 = 0.01, q2 = 0.01
beta1 = 2.383e-10, beta2 = 1.114e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0029659887585671, q1 = 0.01, q2 = 0.01
beta1 = 2.383e-10, beta2 = 1.114e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.014829893792836, q1 = 0.01, q2 = 0.01
beta1 = 2.388e-10, beta2 = 1.127e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.014829893792836, q1 = 0.01, q2 = 0.01
beta1 = 2.388e-10, beta2 = 1.127e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.014829893792836, q1 = 0.01, q2 = 0.01
beta1 = 2.388e-10, beta2 = 1.127e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.014829893792836, q1 = 0.01, q2 = 0.01
beta1 = 2.388e-10, beta2 = 1.127e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.0148299037928359, q1 = 0.01, q2 = 0.01
beta1 = 2.388e-10, beta2 = 1.127e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 1.062285553929911, q1 = 0.01, q2 = 0.01
beta1 = 2.409e-10, beta2 = 1.179e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

3_wave_delta:
  N = 3.0e+06, E0 = 1.062285553929911, q1 = 0.01, q2 = 0.01
  beta1 = 2.409e-10, beta2 = 1.179e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.062285553929911, q1 = 0.01, q2 = 0.01
  beta1 = 2.409e-10, beta2 = 1.179e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.062285553929911, q1 = 0.01, q2 = 0.01
  beta1 = 2.409e-10, beta2 = 1.179e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.062285563929911, q1 = 0.01, q2 = 0.01
  beta1 = 2.409e-10, beta2 = 1.179e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.252108194478211, q1 = 0.01, q2 = 0.01
  beta1 = 2.495e-10, beta2 = 1.385e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.252108194478211, q1 = 0.01, q2 = 0.01
  beta1 = 2.495e-10, beta2 = 1.385e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.252108194478211, q1 = 0.01, q2 = 0.01
  beta1 = 2.495e-10, beta2 = 1.385e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.252108194478211, q1 = 0.01, q2 = 0.01
  beta1 = 2.495e-10, beta2 = 1.385e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.252108204478211, q1 = 0.01, q2 = 0.01
  beta1 = 2.495e-10, beta2 = 1.385e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 2.0113987566714115, q1 = 0.01, q2 = 0.02
  beta1 = 2.838e-10, beta2 = 2.211e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 2.0113987566714115, q1 = 0.01, q2 = 0.02
  beta1 = 2.838e-10, beta2 = 2.211e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

```

3_wave_delta:
  N = 3.0e+06, E0 = 2.0113987566714115, q1 = 0.01, q2 = 0.02
  beta1 = 2.838e-10, beta2 = 2.211e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 2.0113987566714115, q1 = 0.01, q2 = 0.02
  beta1 = 2.838e-10, beta2 = 2.211e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 2.0113987666714115, q1 = 0.01, q2 = 0.02
  beta1 = 2.838e-10, beta2 = 2.211e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 5.048561005444213, q1 = 0.02, q2 = 0.05
  beta1 = 4.208e-10, beta2 = 5.514e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 5.048561005444213, q1 = 0.02, q2 = 0.05
  beta1 = 4.208e-10, beta2 = 5.514e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 5.048561005444213, q1 = 0.02, q2 = 0.05
  beta1 = 4.208e-10, beta2 = 5.514e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 5.048561015444213, q1 = 0.02, q2 = 0.05
  beta1 = 4.208e-10, beta2 = 5.514e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 17.19721000053542, q1 = 0.04, q2 = 0.17
  beta1 = 9.689e-10, beta2 = 1.873e-09
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 17.19721000053542, q1 = 0.04, q2 = 0.17
  beta1 = 9.689e-10, beta2 = 1.873e-09
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 17.19721000053542, q1 = 0.04, q2 = 0.17
  beta1 = 9.689e-10, beta2 = 1.873e-09

```

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 17.19721000053542, q1 = 0.04, q2 = 0.17
beta1 = 9.689e-10, beta2 = 1.873e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 17.19721001053542, q1 = 0.04, q2 = 0.17
beta1 = 9.689e-10, beta2 = 1.873e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 65.79180598090025, q1 = 0.13, q2 = 0.64
beta1 = 3.161e-09, beta2 = 7.157e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 65.79180598090025, q1 = 0.13, q2 = 0.64
beta1 = 3.161e-09, beta2 = 7.157e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 65.79180598090025, q1 = 0.13, q2 = 0.64
beta1 = 3.161e-09, beta2 = 7.157e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 65.79180598090025, q1 = 0.13, q2 = 0.64
beta1 = 3.161e-09, beta2 = 7.157e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 65.79180599090024, q1 = 0.13, q2 = 0.64
beta1 = 3.161e-09, beta2 = 7.157e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 260.17018990235954, q1 = 0.50, q2 = 2.55
beta1 = 1.193e-08, beta2 = 2.829e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 260.17018990235954, q1 = 0.50, q2 = 2.55
beta1 = 1.193e-08, beta2 = 2.829e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 260.17018990235954, q1 = 0.50, q2 = 2.55
beta1 = 1.193e-08, beta2 = 2.829e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 260.17018990235954, q1 = 0.50, q2 = 2.55

beta1 = 1.193e-08, beta2 = 2.829e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 260.17018991235955, q1 = 0.50, q2 = 2.55
beta1 = 1.193e-08, beta2 = 2.829e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 510.906236743006, q1 = 0.98, q2 = 5.00
beta1 = 2.324e-08, beta2 = 5.555e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 510.906236743006, q1 = 0.98, q2 = 5.00
beta1 = 2.324e-08, beta2 = 5.555e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 510.906236743006, q1 = 0.98, q2 = 5.00
beta1 = 2.324e-08, beta2 = 5.555e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 510.906236743006, q1 = 0.98, q2 = 5.00
beta1 = 2.324e-08, beta2 = 5.555e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 510.906236753006, q1 = 0.98, q2 = 5.00
beta1 = 2.324e-08, beta2 = 5.555e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 261.2058783353251, q1 = 0.50, q2 = 2.56
beta1 = 1.198e-08, beta2 = 2.840e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 261.2058783353251, q1 = 0.50, q2 = 2.56
beta1 = 1.198e-08, beta2 = 2.840e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 261.2058783353251, q1 = 0.50, q2 = 2.56
beta1 = 1.198e-08, beta2 = 2.840e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 261.2058783353251, q1 = 0.50, q2 = 2.56
beta1 = 1.198e-08, beta2 = 2.840e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 261.2058783453251, q1 = 0.50, q2 = 2.56
beta1 = 1.198e-08, beta2 = 2.840e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 262.24189281905007, q1 = 0.50, q2 = 2.57
beta1 = 1.202e-08, beta2 = 2.852e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 262.24189281905007, q1 = 0.50, q2 = 2.57
beta1 = 1.202e-08, beta2 = 2.852e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 262.24189281905007, q1 = 0.50, q2 = 2.57
beta1 = 1.202e-08, beta2 = 2.852e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 262.24189281905007, q1 = 0.50, q2 = 2.57
beta1 = 1.202e-08, beta2 = 2.852e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 262.2418928290501, q1 = 0.50, q2 = 2.57
beta1 = 1.202e-08, beta2 = 2.852e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 386.574064781028, q1 = 0.74, q2 = 3.78
beta1 = 1.763e-08, beta2 = 4.203e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 386.574064781028, q1 = 0.74, q2 = 3.78
beta1 = 1.763e-08, beta2 = 4.203e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 386.574064781028, q1 = 0.74, q2 = 3.78
beta1 = 1.763e-08, beta2 = 4.203e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 386.574064781028, q1 = 0.74, q2 = 3.78
beta1 = 1.763e-08, beta2 = 4.203e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 386.57406479102804, q1 = 0.74, q2 = 3.78
beta1 = 1.763e-08, beta2 = 4.203e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

3_wave_delta:
  N = 3.0e+06, E0 = 292.9556927682562, q1 = 0.56, q2 = 2.87
  beta1 = 1.341e-08, beta2 = 3.186e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 292.9556927682562, q1 = 0.56, q2 = 2.87
  beta1 = 1.341e-08, beta2 = 3.186e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 292.9556927682562, q1 = 0.56, q2 = 2.87
  beta1 = 1.341e-08, beta2 = 3.186e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 292.9556927782562, q1 = 0.56, q2 = 2.87
  beta1 = 1.341e-08, beta2 = 3.186e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 313.9316373193064, q1 = 0.60, q2 = 3.07
  beta1 = 1.435e-08, beta2 = 3.414e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 313.9316373193064, q1 = 0.60, q2 = 3.07
  beta1 = 1.435e-08, beta2 = 3.414e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 313.9316373193064, q1 = 0.60, q2 = 3.07
  beta1 = 1.435e-08, beta2 = 3.414e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 313.9316373293064, q1 = 0.60, q2 = 3.07
  beta1 = 1.435e-08, beta2 = 3.414e-08
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 1.5631899445523914, q1 = 0.01, q2 = 0.02
  beta1 = 2.505e-10, beta2 = 1.689e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

3_wave_delta:
N = 3.0e+06, E0 = 1.5631899445523914, q1 = 0.01, q2 = 0.02
beta1 = 2.505e-10, beta2 = 1.689e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1.5631899445523914, q1 = 0.01, q2 = 0.02
beta1 = 2.505e-10, beta2 = 1.689e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1.5631899445523914, q1 = 0.01, q2 = 0.02
beta1 = 2.505e-10, beta2 = 1.689e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1.5631899545523913, q1 = 0.01, q2 = 0.02
beta1 = 2.505e-10, beta2 = 1.689e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 248.84588961393933, q1 = 0.48, q2 = 2.44
beta1 = 1.142e-08, beta2 = 2.706e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 248.84588961393933, q1 = 0.48, q2 = 2.44
beta1 = 1.142e-08, beta2 = 2.706e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 248.84588961393933, q1 = 0.48, q2 = 2.44
beta1 = 1.142e-08, beta2 = 2.706e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 248.84588961393933, q1 = 0.48, q2 = 2.44
beta1 = 1.142e-08, beta2 = 2.706e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 248.84588962393934, q1 = 0.48, q2 = 2.44
beta1 = 1.142e-08, beta2 = 2.706e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 303.37191951728863, q1 = 0.58, q2 = 2.97
beta1 = 1.388e-08, beta2 = 3.299e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 303.37191951728863, q1 = 0.58, q2 = 2.97
beta1 = 1.388e-08, beta2 = 3.299e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 303.37191951728863, q1 = 0.58, q2 = 2.97
beta1 = 1.388e-08, beta2 = 3.299e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 303.37191951728863, q1 = 0.58, q2 = 2.97
beta1 = 1.388e-08, beta2 = 3.299e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 303.37191952728864, q1 = 0.58, q2 = 2.97
beta1 = 1.388e-08, beta2 = 3.299e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 306.0062456268374, q1 = 0.59, q2 = 2.99
beta1 = 1.399e-08, beta2 = 3.326e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 306.0062456268374, q1 = 0.59, q2 = 2.99
beta1 = 1.399e-08, beta2 = 3.326e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 306.0062456268374, q1 = 0.59, q2 = 2.99
beta1 = 1.399e-08, beta2 = 3.326e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 306.0062456268374, q1 = 0.59, q2 = 2.99
beta1 = 1.399e-08, beta2 = 3.326e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 306.00624563683743, q1 = 0.59, q2 = 2.99
beta1 = 1.399e-08, beta2 = 3.326e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 307.04535916790803, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.335e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 307.04535916790803, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.335e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 307.04535916790803, q1 = 0.59, q2 = 3.00

beta1 = 1.402e-08, beta2 = 3.335e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 307.04535916790803, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.335e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 307.04535917790804, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.335e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 307.29316863160886, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.336e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 307.29316863160886, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.336e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 307.29316863160886, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.336e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 307.29316863160886, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.336e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 307.2931686416089, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.336e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 308.2844064864122, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.338e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 308.2844064864122, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.338e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 308.2844064864122, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.338e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 308.2844064864122, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.338e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 308.2844064964122, q1 = 0.59, q2 = 3.00
beta1 = 1.402e-08, beta2 = 3.338e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 324.14451720733376, q1 = 0.58, q2 = 3.02
beta1 = 1.393e-08, beta2 = 3.354e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 324.14451720733376, q1 = 0.58, q2 = 3.02
beta1 = 1.393e-08, beta2 = 3.354e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 324.14451720733376, q1 = 0.58, q2 = 3.02
beta1 = 1.393e-08, beta2 = 3.354e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 324.14451720733376, q1 = 0.58, q2 = 3.02
beta1 = 1.393e-08, beta2 = 3.354e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 324.14451721733377, q1 = 0.58, q2 = 3.02
beta1 = 1.393e-08, beta2 = 3.354e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 361.8914014560246, q1 = 0.57, q2 = 3.03
beta1 = 1.357e-08, beta2 = 3.368e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 361.8914014560246, q1 = 0.57, q2 = 3.03
beta1 = 1.357e-08, beta2 = 3.368e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 361.8914014560246, q1 = 0.57, q2 = 3.03
beta1 = 1.357e-08, beta2 = 3.368e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 361.8914014560246, q1 = 0.57, q2 = 3.03
beta1 = 1.357e-08, beta2 = 3.368e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 361.8914014660246, q1 = 0.57, q2 = 3.03
beta1 = 1.357e-08, beta2 = 3.368e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 529.4478363423583, q1 = 0.49, q2 = 3.05
beta1 = 1.178e-08, beta2 = 3.387e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 529.4478363423583, q1 = 0.49, q2 = 3.05
beta1 = 1.178e-08, beta2 = 3.387e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 529.4478363423583, q1 = 0.49, q2 = 3.05
beta1 = 1.178e-08, beta2 = 3.387e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 529.4478363423583, q1 = 0.49, q2 = 3.05
beta1 = 1.178e-08, beta2 = 3.387e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 529.4478363523583, q1 = 0.49, q2 = 3.05
beta1 = 1.178e-08, beta2 = 3.387e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1199.673575887693, q1 = 0.19, q2 = 3.12
beta1 = 4.628e-09, beta2 = 3.465e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1199.673575887693, q1 = 0.19, q2 = 3.12
beta1 = 4.628e-09, beta2 = 3.465e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1199.673575887693, q1 = 0.19, q2 = 3.12
beta1 = 4.628e-09, beta2 = 3.465e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1199.673575887693, q1 = 0.19, q2 = 3.12
beta1 = 4.628e-09, beta2 = 3.465e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1199.673575897693, q1 = 0.19, q2 = 3.12
beta1 = 4.628e-09, beta2 = 3.465e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1252.1096323234424, q1 = 0.18, q2 = 3.14
beta1 = 4.195e-09, beta2 = 3.489e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1252.1096323234424, q1 = 0.18, q2 = 3.14
beta1 = 4.195e-09, beta2 = 3.489e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1252.1096323234424, q1 = 0.18, q2 = 3.14
beta1 = 4.195e-09, beta2 = 3.489e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1252.1096323234424, q1 = 0.18, q2 = 3.14
beta1 = 4.195e-09, beta2 = 3.489e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1252.1096323334423, q1 = 0.18, q2 = 3.14
beta1 = 4.195e-09, beta2 = 3.489e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1461.85385806644, q1 = 0.10, q2 = 3.23
beta1 = 2.463e-09, beta2 = 3.586e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1461.85385806644, q1 = 0.10, q2 = 3.23
beta1 = 2.463e-09, beta2 = 3.586e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1461.85385806644, q1 = 0.10, q2 = 3.23
beta1 = 2.463e-09, beta2 = 3.586e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1461.85385806644, q1 = 0.10, q2 = 3.23
beta1 = 2.463e-09, beta2 = 3.586e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 1461.8538580764398, q1 = 0.10, q2 = 3.23
beta1 = 2.463e-09, beta2 = 3.586e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 2162.257273232875, q1 = 0.15, q2 = 3.00
beta1 = 3.664e-09, beta2 = 3.334e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 2162.257273232875, q1 = 0.15, q2 = 3.00
beta1 = 3.664e-09, beta2 = 3.334e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 2162.257273232875, q1 = 0.15, q2 = 3.00
beta1 = 3.664e-09, beta2 = 3.334e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 2162.257273232875, q1 = 0.15, q2 = 3.00
beta1 = 3.664e-09, beta2 = 3.334e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 2162.257273242875, q1 = 0.15, q2 = 3.00
beta1 = 3.664e-09, beta2 = 3.334e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 3879.9727728555463, q1 = 0.22, q2 = 2.33
beta1 = 5.324e-09, beta2 = 2.578e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 3879.9727728555463, q1 = 0.22, q2 = 2.33
beta1 = 5.324e-09, beta2 = 2.578e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 3879.9727728555463, q1 = 0.22, q2 = 2.33
beta1 = 5.324e-09, beta2 = 2.578e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 3879.9727728555463, q1 = 0.22, q2 = 2.33
beta1 = 5.324e-09, beta2 = 2.578e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 3879.972772865546, q1 = 0.22, q2 = 2.33
beta1 = 5.324e-09, beta2 = 2.578e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 5019.916478678366, q1 = 0.58, q2 = 1.53
beta1 = 1.386e-08, beta2 = 1.692e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 5019.916478678366, q1 = 0.58, q2 = 1.53

beta1 = 1.386e-08, beta2 = 1.692e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 5019.916478678366, q1 = 0.58, q2 = 1.53
beta1 = 1.386e-08, beta2 = 1.692e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 5019.916478678366, q1 = 0.58, q2 = 1.53
beta1 = 1.386e-08, beta2 = 1.692e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 5019.916478688366, q1 = 0.58, q2 = 1.53
beta1 = 1.386e-08, beta2 = 1.692e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 7185.430602009127, q1 = 1.27, q2 = 0.01
beta1 = 3.004e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 7185.430602009127, q1 = 1.27, q2 = 0.01
beta1 = 3.004e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 7185.430602009127, q1 = 1.27, q2 = 0.01
beta1 = 3.004e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 7185.430602009127, q1 = 1.27, q2 = 0.01
beta1 = 3.004e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 7185.430602019127, q1 = 1.27, q2 = 0.01
beta1 = 3.004e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 6.5e+06, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 5.467e-08, beta2 = 5.103e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 6.5e+06, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 5.467e-08, beta2 = 5.103e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 6.5e+06, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 5.467e-08, beta2 = 5.103e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 6.5e+06, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 5.467e-08, beta2 = 5.103e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 6.5e+06, E0 = 9999.99999999, q1 = 5.00, q2 = 0.01
beta1 = 5.467e-08, beta2 = 5.103e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7196.391681627174, q1 = 1.28, q2 = 0.01
beta1 = 3.025e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7196.391681627174, q1 = 1.28, q2 = 0.01
beta1 = 3.025e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7196.391681627174, q1 = 1.28, q2 = 0.01
beta1 = 3.025e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7196.391681627174, q1 = 1.28, q2 = 0.01
beta1 = 3.025e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7196.391681637174, q1 = 1.28, q2 = 0.01
beta1 = 3.025e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 9999.99999999, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7199.137153196813, q1 = 1.28, q2 = 0.01
beta1 = 3.027e-08, beta2 = 1.099e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7199.137153196813, q1 = 1.28, q2 = 0.01
beta1 = 3.027e-08, beta2 = 1.099e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7199.137153196813, q1 = 1.28, q2 = 0.01
beta1 = 3.027e-08, beta2 = 1.099e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7199.137153196813, q1 = 1.28, q2 = 0.01
beta1 = 3.027e-08, beta2 = 1.099e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 7199.137153206813, q1 = 1.28, q2 = 0.01
beta1 = 3.027e-08, beta2 = 1.099e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.0e+07, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 3.571e-08, beta2 = 3.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

3_wave_delta:
  N = 1.0e+07, E0 = 9999.999999999, q1 = 5.00, q2 = 0.01
  beta1 = 3.571e-08, beta2 = 3.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 7201.573662628051, q1 = 1.29, q2 = 0.01
  beta1 = 3.028e-08, beta2 = 1.097e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 7201.573662628051, q1 = 1.29, q2 = 0.01
  beta1 = 3.028e-08, beta2 = 1.097e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 7201.573662628051, q1 = 1.29, q2 = 0.01
  beta1 = 3.028e-08, beta2 = 1.097e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 7201.573662638051, q1 = 1.29, q2 = 0.01
  beta1 = 3.028e-08, beta2 = 1.097e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 6.5e+06, E0 = 8600.786831314026, q1 = 3.14, q2 = 0.01
  beta1 = 3.445e-08, beta2 = 5.113e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 6.5e+06, E0 = 8600.786831314026, q1 = 3.14, q2 = 0.01
  beta1 = 3.445e-08, beta2 = 5.113e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 6.5e+06, E0 = 8600.786831314026, q1 = 3.14, q2 = 0.01
  beta1 = 3.445e-08, beta2 = 5.113e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 6.5e+06, E0 = 8600.786831324027, q1 = 3.14, q2 = 0.01
  beta1 = 3.445e-08, beta2 = 5.113e-11

```

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7216.230322102908, q1 = 1.31, q2 = 0.01
beta1 = 3.038e-08, beta2 = 1.084e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7216.230322102908, q1 = 1.31, q2 = 0.01
beta1 = 3.038e-08, beta2 = 1.084e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7216.230322102908, q1 = 1.31, q2 = 0.01
beta1 = 3.038e-08, beta2 = 1.084e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7216.230322102908, q1 = 1.31, q2 = 0.01
beta1 = 3.038e-08, beta2 = 1.084e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7216.230322112908, q1 = 1.31, q2 = 0.01
beta1 = 3.038e-08, beta2 = 1.084e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.6e+06, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 2.227e-07, beta2 = 2.079e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.6e+06, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 2.227e-07, beta2 = 2.079e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.6e+06, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 2.227e-07, beta2 = 2.079e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.6e+06, E0 = 10000.0, q1 = 5.00, q2 = 0.01
beta1 = 2.227e-07, beta2 = 2.079e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 1.6e+06, E0 = 9999.99999999, q1 = 5.00, q2 = 0.01
beta1 = 2.227e-07, beta2 = 2.079e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7256.933845504496, q1 = 1.36, q2 = 0.01

beta1 = 3.185e-08, beta2 = 1.092e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7256.933845504496, q1 = 1.36, q2 = 0.01
beta1 = 3.185e-08, beta2 = 1.092e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7256.933845504496, q1 = 1.36, q2 = 0.01
beta1 = 3.185e-08, beta2 = 1.092e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7256.933845504496, q1 = 1.36, q2 = 0.01
beta1 = 3.185e-08, beta2 = 1.092e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7256.933845514496, q1 = 1.36, q2 = 0.01
beta1 = 3.185e-08, beta2 = 1.092e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7233.724089649122, q1 = 1.33, q2 = 0.01
beta1 = 3.101e-08, beta2 = 1.088e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7233.724089649122, q1 = 1.33, q2 = 0.01
beta1 = 3.101e-08, beta2 = 1.088e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7233.724089649122, q1 = 1.33, q2 = 0.01
beta1 = 3.101e-08, beta2 = 1.088e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7233.724089649122, q1 = 1.33, q2 = 0.01
beta1 = 3.101e-08, beta2 = 1.088e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7233.724089659122, q1 = 1.33, q2 = 0.01
beta1 = 3.101e-08, beta2 = 1.088e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.1e+06, E0 = 7882.0367962313385, q1 = 1.32, q2 = 0.01
beta1 = 3.090e-08, beta2 = 1.091e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7882.0367962313385, q1 = 1.32, q2 = 0.01
beta1 = 3.090e-08, beta2 = 1.091e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7882.0367962313385, q1 = 1.32, q2 = 0.01
beta1 = 3.090e-08, beta2 = 1.091e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7882.0367962313385, q1 = 1.32, q2 = 0.01
beta1 = 3.090e-08, beta2 = 1.091e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.1e+06, E0 = 7882.036796241338, q1 = 1.32, q2 = 0.01
beta1 = 3.090e-08, beta2 = 1.091e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.29, q2 = 0.01
beta1 = 3.054e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.29, q2 = 0.01
beta1 = 3.054e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.29, q2 = 0.01
beta1 = 3.054e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.29, q2 = 0.01
beta1 = 3.054e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 9999.99999999, q1 = 1.29, q2 = 0.01
beta1 = 3.054e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.066e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.066e-08, beta2 = 1.102e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.066e-08, beta2 = 1.102e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.066e-08, beta2 = 1.102e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
  beta1 = 3.066e-08, beta2 = 1.102e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.091e-08, beta2 = 1.108e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.091e-08, beta2 = 1.108e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.091e-08, beta2 = 1.108e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.091e-08, beta2 = 1.108e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
  beta1 = 3.091e-08, beta2 = 1.108e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.078e-08, beta2 = 1.105e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.078e-08, beta2 = 1.105e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.078e-08, beta2 = 1.105e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

```

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.078e-08, beta2 = 1.105e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
  beta1 = 3.078e-08, beta2 = 1.105e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.072e-08, beta2 = 1.104e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.072e-08, beta2 = 1.104e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.072e-08, beta2 = 1.104e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
  beta1 = 3.072e-08, beta2 = 1.104e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.070e-08, beta2 = 1.106e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.070e-08, beta2 = 1.106e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.070e-08, beta2 = 1.106e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.070e-08, beta2 = 1.106e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
beta1 = 3.070e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.073e-08, beta2 = 1.105e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.073e-08, beta2 = 1.105e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.073e-08, beta2 = 1.105e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.073e-08, beta2 = 1.105e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
beta1 = 3.073e-08, beta2 = 1.105e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.078e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.078e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.078e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.078e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01

beta1 = 3.078e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 10000.0, q1 = 1.31, q2 = 0.01
beta1 = 3.100e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 10000.0, q1 = 1.31, q2 = 0.01
beta1 = 3.100e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 10000.0, q1 = 1.31, q2 = 0.01
beta1 = 3.100e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 10000.0, q1 = 1.31, q2 = 0.01
beta1 = 3.100e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 9999.99999999, q1 = 1.31, q2 = 0.01
beta1 = 3.100e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.081e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.081e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.081e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.081e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
beta1 = 3.081e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.079e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.079e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.079e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.079e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
beta1 = 3.079e-08, beta2 = 1.106e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.080e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.080e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.080e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.080e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
beta1 = 3.080e-08, beta2 = 1.107e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.085e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.085e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.085e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.085e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
  beta1 = 3.085e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

```
3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

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beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 0.01, q2 = 0.01
  beta1 = 2.376e-10, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 0.01, q2 = 0.01
  beta1 = 2.376e-10, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 0.01, q2 = 0.01
  beta1 = 2.376e-10, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 9999.99999999, q1 = 0.01, q2 = 0.01
  beta1 = 2.376e-10, beta2 = 1.109e-10
  delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:
  N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
  beta1 = 3.084e-08, beta2 = 1.109e-10

```

delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

3_wave_delta:

N = 3.0e+06, E0 = 9999.99999999, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

Успех: True | Ошибка: 4.01e+07

Лучшие параметры → q1=1.298, q2=0.010, N=3.0e+06, E0=10000

3_wave_delta:

N = 3.0e+06, E0 = 10000.0, q1 = 1.30, q2 = 0.01
beta1 = 3.084e-08, beta2 = 1.109e-10
delta = 0.071, gamma = 0.033, lambda = 0.0008

-> Симуляция завершена, период 120 дней

Фиттинг для 4_wave

4_wave:

N = 4.0e+06, E0 = 2000.0, q1 = 0.84, q2 = 1.40
beta1 = 1.500e-08, beta2 = 1.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2000.0, q1 = 0.84, q2 = 1.40
beta1 = 1.500e-08, beta2 = 1.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2000.0, q1 = 0.84, q2 = 1.40
beta1 = 1.500e-08, beta2 = 1.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2000.0, q1 = 0.84, q2 = 1.40
beta1 = 1.500e-08, beta2 = 1.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2000.00000001, q1 = 0.84, q2 = 1.40
beta1 = 1.500e-08, beta2 = 1.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 3229.358800367528, q1 = 5.00, q2 = 5.00
beta1 = 8.929e-08, beta2 = 4.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 3229.358800367528, q1 = 5.00, q2 = 5.00
beta1 = 8.929e-08, beta2 = 4.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 3229.358800367528, q1 = 5.00, q2 = 5.00
beta1 = 8.929e-08, beta2 = 4.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 3229.358800367528, q1 = 5.00, q2 = 5.00
beta1 = 8.929e-08, beta2 = 4.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 3229.358800377528, q1 = 5.00, q2 = 5.00
beta1 = 8.929e-08, beta2 = 4.167e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2001.05301048107, q1 = 0.84, q2 = 1.40
beta1 = 1.506e-08, beta2 = 1.169e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2001.05301048107, q1 = 0.84, q2 = 1.40
beta1 = 1.506e-08, beta2 = 1.169e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2001.05301048107, q1 = 0.84, q2 = 1.40
beta1 = 1.506e-08, beta2 = 1.169e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2001.05301048107, q1 = 0.84, q2 = 1.40
beta1 = 1.506e-08, beta2 = 1.169e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2001.05301049107, q1 = 0.84, q2 = 1.40
beta1 = 1.506e-08, beta2 = 1.169e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2615.205905424299, q1 = 2.92, q2 = 3.20
beta1 = 5.217e-08, beta2 = 2.668e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2615.205905424299, q1 = 2.92, q2 = 3.20
beta1 = 5.217e-08, beta2 = 2.668e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2615.205905424299, q1 = 2.92, q2 = 3.20
beta1 = 5.217e-08, beta2 = 2.668e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2615.205905424299, q1 = 2.92, q2 = 3.20
beta1 = 5.217e-08, beta2 = 2.668e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2615.205905434299, q1 = 2.92, q2 = 3.20
beta1 = 5.217e-08, beta2 = 2.668e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2002.0372598022225, q1 = 0.85, q2 = 1.41
beta1 = 1.512e-08, beta2 = 1.172e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2002.0372598022225, q1 = 0.85, q2 = 1.41
beta1 = 1.512e-08, beta2 = 1.172e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2002.0372598022225, q1 = 0.85, q2 = 1.41
beta1 = 1.512e-08, beta2 = 1.172e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2002.0372598022225, q1 = 0.85, q2 = 1.41
beta1 = 1.512e-08, beta2 = 1.172e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2002.0372598122224, q1 = 0.85, q2 = 1.41
beta1 = 1.512e-08, beta2 = 1.172e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2003.0239169958004, q1 = 0.85, q2 = 1.41
beta1 = 1.518e-08, beta2 = 1.174e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2003.0239169958004, q1 = 0.85, q2 = 1.41
beta1 = 1.518e-08, beta2 = 1.174e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2003.0239169958004, q1 = 0.85, q2 = 1.41
beta1 = 1.518e-08, beta2 = 1.174e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2003.0239169958004, q1 = 0.85, q2 = 1.41
beta1 = 1.518e-08, beta2 = 1.174e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2003.0239170058003, q1 = 0.85, q2 = 1.41
beta1 = 1.518e-08, beta2 = 1.174e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2309.1149112100497, q1 = 1.89, q2 = 2.31
beta1 = 3.368e-08, beta2 = 1.921e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2309.1149112100497, q1 = 1.89, q2 = 2.31
beta1 = 3.368e-08, beta2 = 1.921e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2309.1149112100497, q1 = 1.89, q2 = 2.31
beta1 = 3.368e-08, beta2 = 1.921e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2309.1149112100497, q1 = 1.89, q2 = 2.31
beta1 = 3.368e-08, beta2 = 1.921e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2309.1149112200496, q1 = 1.89, q2 = 2.31
beta1 = 3.368e-08, beta2 = 1.921e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2091.1144947807893, q1 = 1.15, q2 = 1.67
beta1 = 2.051e-08, beta2 = 1.389e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2091.1144947807893, q1 = 1.15, q2 = 1.67
beta1 = 2.051e-08, beta2 = 1.389e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2091.1144947807893, q1 = 1.15, q2 = 1.67
beta1 = 2.051e-08, beta2 = 1.389e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2091.1144947807893, q1 = 1.15, q2 = 1.67
beta1 = 2.051e-08, beta2 = 1.389e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2091.114494790789, q1 = 1.15, q2 = 1.67
beta1 = 2.051e-08, beta2 = 1.389e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2192.424719981657, q1 = 1.49, q2 = 1.96
beta1 = 2.663e-08, beta2 = 1.636e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2192.424719981657, q1 = 1.49, q2 = 1.96
beta1 = 2.663e-08, beta2 = 1.636e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2192.424719981657, q1 = 1.49, q2 = 1.96
beta1 = 2.663e-08, beta2 = 1.636e-08

delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2192.424719981657, q1 = 1.49, q2 = 1.96
beta1 = 2.663e-08, beta2 = 1.636e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2192.4247199916567, q1 = 1.49, q2 = 1.96
beta1 = 2.663e-08, beta2 = 1.636e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2116.770595728419, q1 = 1.24, q2 = 1.74
beta1 = 2.206e-08, beta2 = 1.452e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2116.770595728419, q1 = 1.24, q2 = 1.74
beta1 = 2.206e-08, beta2 = 1.452e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2116.770595728419, q1 = 1.24, q2 = 1.74
beta1 = 2.206e-08, beta2 = 1.452e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2116.770595728419, q1 = 1.24, q2 = 1.74
beta1 = 2.206e-08, beta2 = 1.452e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2116.7705957384187, q1 = 1.24, q2 = 1.74
beta1 = 2.206e-08, beta2 = 1.452e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2156.9801568552384, q1 = 1.37, q2 = 1.86
beta1 = 2.449e-08, beta2 = 1.550e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2156.9801568552384, q1 = 1.37, q2 = 1.86
beta1 = 2.449e-08, beta2 = 1.550e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2156.9801568552384, q1 = 1.37, q2 = 1.86
beta1 = 2.449e-08, beta2 = 1.550e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2156.9801568552384, q1 = 1.37, q2 = 1.86

beta1 = 2.449e-08, beta2 = 1.550e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2156.9801568652383, q1 = 1.37, q2 = 1.86
beta1 = 2.449e-08, beta2 = 1.550e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2163.219027903338, q1 = 1.39, q2 = 1.83
beta1 = 2.484e-08, beta2 = 1.528e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2163.219027903338, q1 = 1.39, q2 = 1.83
beta1 = 2.484e-08, beta2 = 1.528e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2163.219027903338, q1 = 1.39, q2 = 1.83
beta1 = 2.484e-08, beta2 = 1.528e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2163.219027903338, q1 = 1.39, q2 = 1.83
beta1 = 2.484e-08, beta2 = 1.528e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2163.219027913338, q1 = 1.39, q2 = 1.83
beta1 = 2.484e-08, beta2 = 1.528e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2198.140103954203, q1 = 1.49, q2 = 1.45
beta1 = 2.664e-08, beta2 = 1.205e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2198.140103954203, q1 = 1.49, q2 = 1.45
beta1 = 2.664e-08, beta2 = 1.205e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2198.140103954203, q1 = 1.49, q2 = 1.45
beta1 = 2.664e-08, beta2 = 1.205e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2198.140103954203, q1 = 1.49, q2 = 1.45
beta1 = 2.664e-08, beta2 = 1.205e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2198.140103964203, q1 = 1.49, q2 = 1.45
beta1 = 2.664e-08, beta2 = 1.205e-08
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2327.291106697119, q1 = 1.86, q2 = 0.01
beta1 = 3.329e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2327.291106697119, q1 = 1.86, q2 = 0.01
beta1 = 3.329e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2327.291106697119, q1 = 1.86, q2 = 0.01
beta1 = 3.329e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2327.291106697119, q1 = 1.86, q2 = 0.01
beta1 = 3.329e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2327.291106707119, q1 = 1.86, q2 = 0.01
beta1 = 3.329e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2330.677786069173, q1 = 1.87, q2 = 0.01
beta1 = 3.343e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2330.677786069173, q1 = 1.87, q2 = 0.01
beta1 = 3.343e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2330.677786069173, q1 = 1.87, q2 = 0.01
beta1 = 3.343e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2330.677786069173, q1 = 1.87, q2 = 0.01
beta1 = 3.343e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2330.677786079173, q1 = 1.87, q2 = 0.01
beta1 = 3.343e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

```

4_wave:
  N = 4.0e+06, E0 = 2330.7223980424947, q1 = 1.87, q2 = 0.01
  beta1 = 3.340e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2330.7223980424947, q1 = 1.87, q2 = 0.01
  beta1 = 3.340e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2330.7223980424947, q1 = 1.87, q2 = 0.01
  beta1 = 3.340e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2330.7223980424947, q1 = 1.87, q2 = 0.01
  beta1 = 3.340e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2330.7223980524946, q1 = 1.87, q2 = 0.01
  beta1 = 3.340e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2333.0570801710255, q1 = 1.87, q2 = 0.01
  beta1 = 3.334e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2333.0570801710255, q1 = 1.87, q2 = 0.01
  beta1 = 3.334e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2333.0570801710255, q1 = 1.87, q2 = 0.01
  beta1 = 3.334e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2333.0570801810254, q1 = 1.87, q2 = 0.01
  beta1 = 3.334e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 2343.0267807297923, q1 = 1.86, q2 = 0.01
  beta1 = 3.322e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

```

4_wave:
N = 4.0e+06, E0 = 2343.0267807297923, q1 = 1.86, q2 = 0.01
beta1 = 3.322e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2343.0267807297923, q1 = 1.86, q2 = 0.01
beta1 = 3.322e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2343.0267807297923, q1 = 1.86, q2 = 0.01
beta1 = 3.322e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2343.0267807397922, q1 = 1.86, q2 = 0.01
beta1 = 3.322e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2374.384095393412, q1 = 1.85, q2 = 0.01
beta1 = 3.301e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2374.384095393412, q1 = 1.85, q2 = 0.01
beta1 = 3.301e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2374.384095393412, q1 = 1.85, q2 = 0.01
beta1 = 3.301e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2374.384095393412, q1 = 1.85, q2 = 0.01
beta1 = 3.301e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2374.3840954034117, q1 = 1.85, q2 = 0.01
beta1 = 3.301e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2509.8600545656404, q1 = 2.01, q2 = 0.01
beta1 = 3.582e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2509.8600545656404, q1 = 2.01, q2 = 0.01
beta1 = 3.582e-08, beta2 = 8.333e-11

delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2509.8600545656404, q1 = 2.01, q2 = 0.01
beta1 = 3.582e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2509.8600545656404, q1 = 2.01, q2 = 0.01
beta1 = 3.582e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2509.8600545756403, q1 = 2.01, q2 = 0.01
beta1 = 3.582e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2388.7486994096803, q1 = 1.87, q2 = 0.01
beta1 = 3.331e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2388.7486994096803, q1 = 1.87, q2 = 0.01
beta1 = 3.331e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2388.7486994096803, q1 = 1.87, q2 = 0.01
beta1 = 3.331e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2388.7486994096803, q1 = 1.87, q2 = 0.01
beta1 = 3.331e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2388.74869941968, q1 = 1.87, q2 = 0.01
beta1 = 3.331e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2568.9989275927787, q1 = 1.85, q2 = 0.01
beta1 = 3.303e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2568.9989275927787, q1 = 1.85, q2 = 0.01
beta1 = 3.303e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 2568.9989275927787, q1 = 1.85, q2 = 0.01

beta1 = 3.303e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2568.9989275927787, q1 = 1.85, q2 = 0.01
beta1 = 3.303e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 2568.9989276027786, q1 = 1.85, q2 = 0.01
beta1 = 3.303e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 3289.9998403251725, q1 = 1.79, q2 = 0.01
beta1 = 3.191e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 3289.9998403251725, q1 = 1.79, q2 = 0.01
beta1 = 3.191e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 3289.9998403251725, q1 = 1.79, q2 = 0.01
beta1 = 3.191e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 3289.9998403251725, q1 = 1.79, q2 = 0.01
beta1 = 3.191e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 3289.9998403351724, q1 = 1.79, q2 = 0.01
beta1 = 3.191e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 9976.20127523647, q1 = 1.23, q2 = 0.01
beta1 = 2.192e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 9976.20127523647, q1 = 1.23, q2 = 0.01
beta1 = 2.192e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 9976.20127523647, q1 = 1.23, q2 = 0.01
beta1 = 2.192e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 9976.20127523647, q1 = 1.23, q2 = 0.01
beta1 = 2.192e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 9976.20127524647, q1 = 1.23, q2 = 0.01
beta1 = 2.192e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8966.126037138312, q1 = 1.28, q2 = 0.37
beta1 = 2.289e-08, beta2 = 3.123e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8966.126037138312, q1 = 1.28, q2 = 0.37
beta1 = 2.289e-08, beta2 = 3.123e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8966.126037138312, q1 = 1.28, q2 = 0.37
beta1 = 2.289e-08, beta2 = 3.123e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8966.126037138312, q1 = 1.28, q2 = 0.37
beta1 = 2.289e-08, beta2 = 3.123e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8966.126037148313, q1 = 1.28, q2 = 0.37
beta1 = 2.289e-08, beta2 = 3.123e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8410.578985539632, q1 = 1.47, q2 = 0.63
beta1 = 2.618e-08, beta2 = 5.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8410.578985539632, q1 = 1.47, q2 = 0.63
beta1 = 2.618e-08, beta2 = 5.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8410.578985539632, q1 = 1.47, q2 = 0.63
beta1 = 2.618e-08, beta2 = 5.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 8410.578985539632, q1 = 1.47, q2 = 0.63
beta1 = 2.618e-08, beta2 = 5.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 8410.578985549633, q1 = 1.47, q2 = 0.63
beta1 = 2.618e-08, beta2 = 5.267e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 8723.35785491388, q1 = 1.36, q2 = 0.49
beta1 = 2.433e-08, beta2 = 4.060e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 8723.35785491388, q1 = 1.36, q2 = 0.49
beta1 = 2.433e-08, beta2 = 4.060e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 8723.35785491388, q1 = 1.36, q2 = 0.49
beta1 = 2.433e-08, beta2 = 4.060e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 8723.35785491388, q1 = 1.36, q2 = 0.49
beta1 = 2.433e-08, beta2 = 4.060e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 8723.357854923881, q1 = 1.36, q2 = 0.49
beta1 = 2.433e-08, beta2 = 4.060e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 10000.0, q1 = 1.35, q2 = 0.46
beta1 = 2.417e-08, beta2 = 3.833e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 10000.0, q1 = 1.35, q2 = 0.46
beta1 = 2.417e-08, beta2 = 3.833e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 10000.0, q1 = 1.35, q2 = 0.46
beta1 = 2.417e-08, beta2 = 3.833e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 10000.0, q1 = 1.35, q2 = 0.46
beta1 = 2.417e-08, beta2 = 3.833e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
N = 4.0e+06, E0 = 9999.99999999, q1 = 1.35, q2 = 0.46
beta1 = 2.417e-08, beta2 = 3.833e-09
delta = 0.071, gamma = 0.033, lambda = 0.0008


```

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.43, q2 = 0.27
  beta1 = 2.556e-08, beta2 = 2.272e-09
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.43, q2 = 0.27
  beta1 = 2.556e-08, beta2 = 2.272e-09
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.43, q2 = 0.27
  beta1 = 2.556e-08, beta2 = 2.272e-09
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.43, q2 = 0.27
  beta1 = 2.556e-08, beta2 = 2.272e-09
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 9999.99999999, q1 = 1.43, q2 = 0.27
  beta1 = 2.556e-08, beta2 = 2.272e-09
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.54, q2 = 0.01
  beta1 = 2.751e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.54, q2 = 0.01
  beta1 = 2.751e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.54, q2 = 0.01
  beta1 = 2.751e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.54, q2 = 0.01
  beta1 = 2.751e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 9999.99999999, q1 = 1.54, q2 = 0.01
  beta1 = 2.751e-08, beta2 = 8.333e-11
  delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:
  N = 4.0e+06, E0 = 10000.0, q1 = 1.53, q2 = 0.01
  beta1 = 2.735e-08, beta2 = 8.333e-11

```

delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.53, q2 = 0.01
beta1 = 2.735e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.53, q2 = 0.01
beta1 = 2.735e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.53, q2 = 0.01
beta1 = 2.735e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 9999.99999999, q1 = 1.53, q2 = 0.01
beta1 = 2.735e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.716e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.716e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.716e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.716e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 9999.99999999, q1 = 1.52, q2 = 0.01
beta1 = 2.716e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01

beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 9999.99999999, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 9999.99999999, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

4_wave:

N = 4.0e+06, E0 = 9999.99999999, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

Успех: True | Ошибка: 5.49e+07

Лучшие параметры → q1=1.520, q2=0.010, N=4.0e+06, E0=10000

4_wave:

N = 4.0e+06, E0 = 10000.0, q1 = 1.52, q2 = 0.01
beta1 = 2.715e-08, beta2 = 8.333e-11
delta = 0.071, gamma = 0.033, lambda = 0.0008

-> Симуляция завершена, период 60 дней

Фиттинг для omicron_initial

omicron_initial:

N = 6.0e+06, E0 = 5.0, q1 = 1.80, q2 = 3.00
beta1 = 5.000e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 5.0, q1 = 1.80, q2 = 3.00
beta1 = 5.000e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 5.0, q1 = 1.80, q2 = 3.00
beta1 = 5.000e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 5.0, q1 = 1.80, q2 = 3.00
beta1 = 5.000e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 5.000000001, q1 = 1.80, q2 = 3.00
beta1 = 5.000e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 5.00, q2 = 5.00
beta1 = 1.389e-07, beta2 = 2.778e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 5.00, q2 = 5.00
beta1 = 1.389e-07, beta2 = 2.778e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 5.00, q2 = 5.00
beta1 = 1.389e-07, beta2 = 2.778e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 5.00, q2 = 5.00
beta1 = 1.389e-07, beta2 = 2.778e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 9999.99999999, q1 = 5.00, q2 = 5.00
beta1 = 1.389e-07, beta2 = 2.778e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 121.98196694899953, q1 = 1.84, q2 = 3.02
beta1 = 5.104e-08, beta2 = 1.680e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 121.98196694899953, q1 = 1.84, q2 = 3.02
beta1 = 5.104e-08, beta2 = 1.680e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 121.98196694899953, q1 = 1.84, q2 = 3.02
beta1 = 5.104e-08, beta2 = 1.680e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 121.98196694899953, q1 = 1.84, q2 = 3.02
beta1 = 5.104e-08, beta2 = 1.680e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 121.98196695899952, q1 = 1.84, q2 = 3.02
beta1 = 5.104e-08, beta2 = 1.680e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 18.93423543206047, q1 = 1.80, q2 = 3.00
beta1 = 5.012e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 18.93423543206047, q1 = 1.80, q2 = 3.00
beta1 = 5.012e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.93423543206047, q1 = 1.80, q2 = 3.00
beta1 = 5.012e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.93423543206047, q1 = 1.80, q2 = 3.00
beta1 = 5.012e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.93423544206047, q1 = 1.80, q2 = 3.00
beta1 = 5.012e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.75477165341711, q1 = 1.87, q2 = 3.01
beta1 = 5.190e-08, beta2 = 1.672e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.75477165341711, q1 = 1.87, q2 = 3.01
beta1 = 5.190e-08, beta2 = 1.672e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.75477165341711, q1 = 1.87, q2 = 3.01
beta1 = 5.190e-08, beta2 = 1.672e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.75477165341711, q1 = 1.87, q2 = 3.01
beta1 = 5.190e-08, beta2 = 1.672e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.75477166341711, q1 = 1.87, q2 = 3.01
beta1 = 5.190e-08, beta2 = 1.672e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.91845641702803, q1 = 1.81, q2 = 3.00
beta1 = 5.028e-08, beta2 = 1.669e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.91845641702803, q1 = 1.81, q2 = 3.00
beta1 = 5.028e-08, beta2 = 1.669e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 18.91845641702803, q1 = 1.81, q2 = 3.00
beta1 = 5.028e-08, beta2 = 1.669e-08

delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 18.91845641702803, q1 = 1.81, q2 = 3.00
beta1 = 5.028e-08, beta2 = 1.669e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 18.91845642702803, q1 = 1.81, q2 = 3.00
beta1 = 5.028e-08, beta2 = 1.669e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 19.601671980542854, q1 = 1.80, q2 = 3.00
beta1 = 5.010e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 19.601671980542854, q1 = 1.80, q2 = 3.00
beta1 = 5.010e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 19.601671980542854, q1 = 1.80, q2 = 3.00
beta1 = 5.010e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 19.601671980542854, q1 = 1.80, q2 = 3.00
beta1 = 5.010e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 19.601671990542854, q1 = 1.80, q2 = 3.00
beta1 = 5.010e-08, beta2 = 1.668e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 22.334534234602152, q1 = 1.78, q2 = 3.00
beta1 = 4.940e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 22.334534234602152, q1 = 1.78, q2 = 3.00
beta1 = 4.940e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 22.334534234602152, q1 = 1.78, q2 = 3.00
beta1 = 4.940e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 22.334534234602152, q1 = 1.78, q2 = 3.00

beta1 = 4.940e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 22.334534244602153, q1 = 1.78, q2 = 3.00
beta1 = 4.940e-08, beta2 = 1.667e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 42.96663599162629, q1 = 1.59, q2 = 2.96
beta1 = 4.414e-08, beta2 = 1.642e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 42.96663599162629, q1 = 1.59, q2 = 2.96
beta1 = 4.414e-08, beta2 = 1.642e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 42.96663599162629, q1 = 1.59, q2 = 2.96
beta1 = 4.414e-08, beta2 = 1.642e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 42.96663599162629, q1 = 1.59, q2 = 2.96
beta1 = 4.414e-08, beta2 = 1.642e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 42.96663600162629, q1 = 1.59, q2 = 2.96
beta1 = 4.414e-08, beta2 = 1.642e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 29.466615926814043, q1 = 1.71, q2 = 2.98
beta1 = 4.759e-08, beta2 = 1.658e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 29.466615926814043, q1 = 1.71, q2 = 2.98
beta1 = 4.759e-08, beta2 = 1.658e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 29.466615926814043, q1 = 1.71, q2 = 2.98
beta1 = 4.759e-08, beta2 = 1.658e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 29.466615926814043, q1 = 1.71, q2 = 2.98
beta1 = 4.759e-08, beta2 = 1.658e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 29.466615936814044, q1 = 1.71, q2 = 2.98
beta1 = 4.759e-08, beta2 = 1.658e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 35.545531991154796, q1 = 1.67, q2 = 2.97
beta1 = 4.648e-08, beta2 = 1.652e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 35.545531991154796, q1 = 1.67, q2 = 2.97
beta1 = 4.648e-08, beta2 = 1.652e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 35.545531991154796, q1 = 1.67, q2 = 2.97
beta1 = 4.648e-08, beta2 = 1.652e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 35.545531991154796, q1 = 1.67, q2 = 2.97
beta1 = 4.648e-08, beta2 = 1.652e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 35.5455320011548, q1 = 1.67, q2 = 2.97
beta1 = 4.648e-08, beta2 = 1.652e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 43.64844790346554, q1 = 1.64, q2 = 2.96
beta1 = 4.560e-08, beta2 = 1.644e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 43.64844790346554, q1 = 1.64, q2 = 2.96
beta1 = 4.560e-08, beta2 = 1.644e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 43.64844790346554, q1 = 1.64, q2 = 2.96
beta1 = 4.560e-08, beta2 = 1.644e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 43.64844790346554, q1 = 1.64, q2 = 2.96
beta1 = 4.560e-08, beta2 = 1.644e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 43.64844791346554, q1 = 1.64, q2 = 2.96
beta1 = 4.560e-08, beta2 = 1.644e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 56.113158847732876, q1 = 1.59, q2 = 2.94
beta1 = 4.424e-08, beta2 = 1.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 56.113158847732876, q1 = 1.59, q2 = 2.94
beta1 = 4.424e-08, beta2 = 1.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 56.113158847732876, q1 = 1.59, q2 = 2.94
beta1 = 4.424e-08, beta2 = 1.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 56.113158847732876, q1 = 1.59, q2 = 2.94
beta1 = 4.424e-08, beta2 = 1.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 56.11315885773288, q1 = 1.59, q2 = 2.94
beta1 = 4.424e-08, beta2 = 1.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 74.68472141642567, q1 = 1.55, q2 = 2.90
beta1 = 4.297e-08, beta2 = 1.610e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 74.68472141642567, q1 = 1.55, q2 = 2.90
beta1 = 4.297e-08, beta2 = 1.610e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 74.68472141642567, q1 = 1.55, q2 = 2.90
beta1 = 4.297e-08, beta2 = 1.610e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 74.68472141642567, q1 = 1.55, q2 = 2.90
beta1 = 4.297e-08, beta2 = 1.610e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 74.68472142642567, q1 = 1.55, q2 = 2.90
beta1 = 4.297e-08, beta2 = 1.610e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 111.28144075254261, q1 = 1.44, q2 = 2.82
beta1 = 3.992e-08, beta2 = 1.569e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 111.28144075254261, q1 = 1.44, q2 = 2.82
beta1 = 3.992e-08, beta2 = 1.569e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 111.28144075254261, q1 = 1.44, q2 = 2.82
beta1 = 3.992e-08, beta2 = 1.569e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 111.28144075254261, q1 = 1.44, q2 = 2.82
beta1 = 3.992e-08, beta2 = 1.569e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 111.2814407625426, q1 = 1.44, q2 = 2.82
beta1 = 3.992e-08, beta2 = 1.569e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 93.02497772927835, q1 = 1.49, q2 = 2.86
beta1 = 4.144e-08, beta2 = 1.590e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 93.02497772927835, q1 = 1.49, q2 = 2.86
beta1 = 4.144e-08, beta2 = 1.590e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 93.02497772927835, q1 = 1.49, q2 = 2.86
beta1 = 4.144e-08, beta2 = 1.590e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 93.02497772927835, q1 = 1.49, q2 = 2.86
beta1 = 4.144e-08, beta2 = 1.590e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 93.02497773927834, q1 = 1.49, q2 = 2.86
beta1 = 4.144e-08, beta2 = 1.590e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 120.69190817097626, q1 = 1.45, q2 = 2.80
beta1 = 4.026e-08, beta2 = 1.557e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 120.69190817097626, q1 = 1.45, q2 = 2.80
beta1 = 4.026e-08, beta2 = 1.557e-08

delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 120.69190817097626, q1 = 1.45, q2 = 2.80
beta1 = 4.026e-08, beta2 = 1.557e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 120.69190817097626, q1 = 1.45, q2 = 2.80
beta1 = 4.026e-08, beta2 = 1.557e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 120.69190818097626, q1 = 1.45, q2 = 2.80
beta1 = 4.026e-08, beta2 = 1.557e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 149.23589138778868, q1 = 1.43, q2 = 2.74
beta1 = 3.970e-08, beta2 = 1.521e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 149.23589138778868, q1 = 1.43, q2 = 2.74
beta1 = 3.970e-08, beta2 = 1.521e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 149.23589138778868, q1 = 1.43, q2 = 2.74
beta1 = 3.970e-08, beta2 = 1.521e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 149.23589138778868, q1 = 1.43, q2 = 2.74
beta1 = 3.970e-08, beta2 = 1.521e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 149.2358913977887, q1 = 1.43, q2 = 2.74
beta1 = 3.970e-08, beta2 = 1.521e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 250.76234856754195, q1 = 1.30, q2 = 2.51
beta1 = 3.599e-08, beta2 = 1.397e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 250.76234856754195, q1 = 1.30, q2 = 2.51
beta1 = 3.599e-08, beta2 = 1.397e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 250.76234856754195, q1 = 1.30, q2 = 2.51

beta1 = 3.599e-08, beta2 = 1.397e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 250.76234856754195, q1 = 1.30, q2 = 2.51
beta1 = 3.599e-08, beta2 = 1.397e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 250.76234857754196, q1 = 1.30, q2 = 2.51
beta1 = 3.599e-08, beta2 = 1.397e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 194.5646546240091, q1 = 1.37, q2 = 2.64
beta1 = 3.804e-08, beta2 = 1.466e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 194.5646546240091, q1 = 1.37, q2 = 2.64
beta1 = 3.804e-08, beta2 = 1.466e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 194.5646546240091, q1 = 1.37, q2 = 2.64
beta1 = 3.804e-08, beta2 = 1.466e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 194.5646546240091, q1 = 1.37, q2 = 2.64
beta1 = 3.804e-08, beta2 = 1.466e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 194.56465463400912, q1 = 1.37, q2 = 2.64
beta1 = 3.804e-08, beta2 = 1.466e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 234.13856894319483, q1 = 1.34, q2 = 2.55
beta1 = 3.735e-08, beta2 = 1.416e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 234.13856894319483, q1 = 1.34, q2 = 2.55
beta1 = 3.735e-08, beta2 = 1.416e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 234.13856894319483, q1 = 1.34, q2 = 2.55
beta1 = 3.735e-08, beta2 = 1.416e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 234.13856894319483, q1 = 1.34, q2 = 2.55
beta1 = 3.735e-08, beta2 = 1.416e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 234.13856895319483, q1 = 1.34, q2 = 2.55
beta1 = 3.735e-08, beta2 = 1.416e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 270.6765440683442, q1 = 1.91, q2 = 0.01
beta1 = 5.304e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 270.6765440683442, q1 = 1.91, q2 = 0.01
beta1 = 5.304e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 270.6765440683442, q1 = 1.91, q2 = 0.01
beta1 = 5.304e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 270.6765440683442, q1 = 1.91, q2 = 0.01
beta1 = 5.304e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 270.6765440783442, q1 = 1.91, q2 = 0.01
beta1 = 5.304e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 241.0965437898892, q1 = 1.45, q2 = 2.06
beta1 = 4.034e-08, beta2 = 1.147e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 241.0965437898892, q1 = 1.45, q2 = 2.06
beta1 = 4.034e-08, beta2 = 1.147e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 241.0965437898892, q1 = 1.45, q2 = 2.06
beta1 = 4.034e-08, beta2 = 1.147e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 241.0965437898892, q1 = 1.45, q2 = 2.06
beta1 = 4.034e-08, beta2 = 1.147e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 241.09654379988922, q1 = 1.45, q2 = 2.06
beta1 = 4.034e-08, beta2 = 1.147e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 337.7381307678636, q1 = 1.42, q2 = 1.69
beta1 = 3.940e-08, beta2 = 9.394e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 337.7381307678636, q1 = 1.42, q2 = 1.69
beta1 = 3.940e-08, beta2 = 9.394e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 337.7381307678636, q1 = 1.42, q2 = 1.69
beta1 = 3.940e-08, beta2 = 9.394e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 337.7381307778636, q1 = 1.42, q2 = 1.69
beta1 = 3.940e-08, beta2 = 9.394e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 444.5717397470852, q1 = 1.48, q2 = 0.86
beta1 = 4.110e-08, beta2 = 4.751e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 444.5717397470852, q1 = 1.48, q2 = 0.86
beta1 = 4.110e-08, beta2 = 4.751e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 444.5717397470852, q1 = 1.48, q2 = 0.86
beta1 = 4.110e-08, beta2 = 4.751e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 444.5717397570852, q1 = 1.48, q2 = 0.86
beta1 = 4.110e-08, beta2 = 4.751e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 389.11197823664105, q1 = 1.45, q2 = 1.29
beta1 = 4.022e-08, beta2 = 7.161e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 389.11197823664105, q1 = 1.45, q2 = 1.29
beta1 = 4.022e-08, beta2 = 7.161e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 389.11197823664105, q1 = 1.45, q2 = 1.29
beta1 = 4.022e-08, beta2 = 7.161e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 389.11197823664105, q1 = 1.45, q2 = 1.29
beta1 = 4.022e-08, beta2 = 7.161e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 389.11197824664106, q1 = 1.45, q2 = 1.29
beta1 = 4.022e-08, beta2 = 7.161e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 422.7476704710506, q1 = 1.54, q2 = 0.74
beta1 = 4.265e-08, beta2 = 4.100e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 422.7476704710506, q1 = 1.54, q2 = 0.74
beta1 = 4.265e-08, beta2 = 4.100e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 422.7476704710506, q1 = 1.54, q2 = 0.74
beta1 = 4.265e-08, beta2 = 4.100e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 422.7476704710506, q1 = 1.54, q2 = 0.74
beta1 = 4.265e-08, beta2 = 4.100e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 422.7476704810506, q1 = 1.54, q2 = 0.74
beta1 = 4.265e-08, beta2 = 4.100e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 613.5528634994512, q1 = 2.47, q2 = 0.01
beta1 = 6.875e-08, beta2 = 5.556e-11

delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 613.5528634994512, q1 = 2.47, q2 = 0.01
beta1 = 6.875e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 613.5528634994512, q1 = 2.47, q2 = 0.01
beta1 = 6.875e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 613.5528634994512, q1 = 2.47, q2 = 0.01
beta1 = 6.875e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 613.5528635094512, q1 = 2.47, q2 = 0.01
beta1 = 6.875e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 445.81490170907153, q1 = 1.65, q2 = 0.65
beta1 = 4.580e-08, beta2 = 3.611e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 445.81490170907153, q1 = 1.65, q2 = 0.65
beta1 = 4.580e-08, beta2 = 3.611e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 445.81490170907153, q1 = 1.65, q2 = 0.65
beta1 = 4.580e-08, beta2 = 3.611e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 445.81490170907153, q1 = 1.65, q2 = 0.65
beta1 = 4.580e-08, beta2 = 3.611e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 445.81490171907154, q1 = 1.65, q2 = 0.65
beta1 = 4.580e-08, beta2 = 3.611e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 427.41199701793903, q1 = 1.56, q2 = 0.72
beta1 = 4.328e-08, beta2 = 4.001e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 427.41199701793903, q1 = 1.56, q2 = 0.72

beta1 = 4.328e-08, beta2 = 4.001e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 427.41199701793903, q1 = 1.56, q2 = 0.72
beta1 = 4.328e-08, beta2 = 4.001e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 427.41199701793903, q1 = 1.56, q2 = 0.72
beta1 = 4.328e-08, beta2 = 4.001e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 427.41199702793904, q1 = 1.56, q2 = 0.72
beta1 = 4.328e-08, beta2 = 4.001e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 631.7860355392152, q1 = 2.15, q2 = 0.01
beta1 = 5.979e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 631.7860355392152, q1 = 2.15, q2 = 0.01
beta1 = 5.979e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 631.7860355392152, q1 = 2.15, q2 = 0.01
beta1 = 5.979e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 631.7860355392152, q1 = 2.15, q2 = 0.01
beta1 = 5.979e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 631.7860355492152, q1 = 2.15, q2 = 0.01
beta1 = 5.979e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 473.17657228556936, q1 = 1.69, q2 = 0.56
beta1 = 4.698e-08, beta2 = 3.118e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 473.17657228556936, q1 = 1.69, q2 = 0.56
beta1 = 4.698e-08, beta2 = 3.118e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 473.17657228556936, q1 = 1.69, q2 = 0.56
beta1 = 4.698e-08, beta2 = 3.118e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 473.17657228556936, q1 = 1.69, q2 = 0.56
beta1 = 4.698e-08, beta2 = 3.118e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 473.1765722955694, q1 = 1.69, q2 = 0.56
beta1 = 4.698e-08, beta2 = 3.118e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 428.9618454683799, q1 = 1.56, q2 = 0.71
beta1 = 4.341e-08, beta2 = 3.971e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 428.9618454683799, q1 = 1.56, q2 = 0.71
beta1 = 4.341e-08, beta2 = 3.971e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 428.9618454683799, q1 = 1.56, q2 = 0.71
beta1 = 4.341e-08, beta2 = 3.971e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 428.9618454683799, q1 = 1.56, q2 = 0.71
beta1 = 4.341e-08, beta2 = 3.971e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 428.9618454783799, q1 = 1.56, q2 = 0.71
beta1 = 4.341e-08, beta2 = 3.971e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 467.49025940128246, q1 = 1.67, q2 = 0.01
beta1 = 4.644e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 467.49025940128246, q1 = 1.67, q2 = 0.01
beta1 = 4.644e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 467.49025940128246, q1 = 1.67, q2 = 0.01
beta1 = 4.644e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 467.49025940128246, q1 = 1.67, q2 = 0.01
beta1 = 4.644e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 467.49025941128247, q1 = 1.67, q2 = 0.01
beta1 = 4.644e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.6570199202062, q1 = 2.30, q2 = 0.01
beta1 = 6.398e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.6570199202062, q1 = 2.30, q2 = 0.01
beta1 = 6.398e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.6570199202062, q1 = 2.30, q2 = 0.01
beta1 = 6.398e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.6570199202062, q1 = 2.30, q2 = 0.01
beta1 = 6.398e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.6570199302062, q1 = 2.30, q2 = 0.01
beta1 = 6.398e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 507.94749134054535, q1 = 1.79, q2 = 0.01
beta1 = 4.976e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 507.94749134054535, q1 = 1.79, q2 = 0.01
beta1 = 4.976e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 507.94749134054535, q1 = 1.79, q2 = 0.01
beta1 = 4.976e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 507.94749134054535, q1 = 1.79, q2 = 0.01
beta1 = 4.976e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 507.94749135054536, q1 = 1.79, q2 = 0.01
beta1 = 4.976e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 468.85255754092935, q1 = 1.68, q2 = 0.01
beta1 = 4.656e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 468.85255754092935, q1 = 1.68, q2 = 0.01
beta1 = 4.656e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 468.85255754092935, q1 = 1.68, q2 = 0.01
beta1 = 4.656e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 468.85255754092935, q1 = 1.68, q2 = 0.01
beta1 = 4.656e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 468.85255755092936, q1 = 1.68, q2 = 0.01
beta1 = 4.656e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 496.73551832990665, q1 = 1.67, q2 = 0.01
beta1 = 4.626e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 496.73551832990665, q1 = 1.67, q2 = 0.01
beta1 = 4.626e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 496.73551832990665, q1 = 1.67, q2 = 0.01
beta1 = 4.626e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 496.73551832990665, q1 = 1.67, q2 = 0.01
beta1 = 4.626e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 496.73551833990666, q1 = 1.67, q2 = 0.01
beta1 = 4.626e-08, beta2 = 5.556e-11

delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 608.2673614858159, q1 = 1.62, q2 = 0.01
beta1 = 4.509e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 608.2673614858159, q1 = 1.62, q2 = 0.01
beta1 = 4.509e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 608.2673614858159, q1 = 1.62, q2 = 0.01
beta1 = 4.509e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 608.2673614858159, q1 = 1.62, q2 = 0.01
beta1 = 4.509e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 608.2673614958159, q1 = 1.62, q2 = 0.01
beta1 = 4.509e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 971.0618456654914, q1 = 1.99, q2 = 0.01
beta1 = 5.533e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 971.0618456654914, q1 = 1.99, q2 = 0.01
beta1 = 5.533e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 971.0618456654914, q1 = 1.99, q2 = 0.01
beta1 = 5.533e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 971.0618456654914, q1 = 1.99, q2 = 0.01
beta1 = 5.533e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 971.0618456754914, q1 = 1.99, q2 = 0.01
beta1 = 5.533e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 681.8786801577041, q1 = 1.70, q2 = 0.01

beta1 = 4.716e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.8786801577041, q1 = 1.70, q2 = 0.01
beta1 = 4.716e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.8786801577041, q1 = 1.70, q2 = 0.01
beta1 = 4.716e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.8786801577041, q1 = 1.70, q2 = 0.01
beta1 = 4.716e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 681.8786801677041, q1 = 1.70, q2 = 0.01
beta1 = 4.716e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 615.6330350358307, q1 = 1.63, q2 = 0.01
beta1 = 4.529e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 615.6330350358307, q1 = 1.63, q2 = 0.01
beta1 = 4.529e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 615.6330350358307, q1 = 1.63, q2 = 0.01
beta1 = 4.529e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 615.6330350358307, q1 = 1.63, q2 = 0.01
beta1 = 4.529e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 615.6330350458308, q1 = 1.63, q2 = 0.01
beta1 = 4.529e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1085.5675292424187, q1 = 1.50, q2 = 0.01
beta1 = 4.176e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1085.5675292424187, q1 = 1.50, q2 = 0.01
beta1 = 4.176e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1085.5675292424187, q1 = 1.50, q2 = 0.01
beta1 = 4.176e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1085.5675292424187, q1 = 1.50, q2 = 0.01
beta1 = 4.176e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1085.5675292524186, q1 = 1.50, q2 = 0.01
beta1 = 4.176e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1119.5668496673118, q1 = 1.56, q2 = 0.01
beta1 = 4.324e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1119.5668496673118, q1 = 1.56, q2 = 0.01
beta1 = 4.324e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1119.5668496673118, q1 = 1.56, q2 = 0.01
beta1 = 4.324e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1119.5668496673118, q1 = 1.56, q2 = 0.01
beta1 = 4.324e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1119.5668496773117, q1 = 1.56, q2 = 0.01
beta1 = 4.324e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1100.5770980585414, q1 = 1.53, q2 = 0.01
beta1 = 4.242e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 1100.5770980585414, q1 = 1.53, q2 = 0.01
beta1 = 4.242e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1100.5770980585414, q1 = 1.53, q2 = 0.01
beta1 = 4.242e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1100.5770980585414, q1 = 1.53, q2 = 0.01
beta1 = 4.242e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1100.5770980685413, q1 = 1.53, q2 = 0.01
beta1 = 4.242e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1247.4827032974217, q1 = 2.03, q2 = 0.01
beta1 = 5.625e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1247.4827032974217, q1 = 2.03, q2 = 0.01
beta1 = 5.625e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1247.4827032974217, q1 = 2.03, q2 = 0.01
beta1 = 5.625e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1247.4827033074216, q1 = 2.03, q2 = 0.01
beta1 = 5.625e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1132.1885656657298, q1 = 1.63, q2 = 0.01
beta1 = 4.539e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1132.1885656657298, q1 = 1.63, q2 = 0.01
beta1 = 4.539e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1132.1885656657298, q1 = 1.63, q2 = 0.01
beta1 = 4.539e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1132.1885656657298, q1 = 1.63, q2 = 0.01
beta1 = 4.539e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1132.1885656757297, q1 = 1.63, q2 = 0.01
beta1 = 4.539e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1102.309492030892, q1 = 1.53, q2 = 0.01
beta1 = 4.258e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1102.309492030892, q1 = 1.53, q2 = 0.01
beta1 = 4.258e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1102.309492030892, q1 = 1.53, q2 = 0.01
beta1 = 4.258e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1102.309492030892, q1 = 1.53, q2 = 0.01
beta1 = 4.258e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1102.3094920408919, q1 = 1.53, q2 = 0.01
beta1 = 4.258e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1498.2146718346294, q1 = 1.47, q2 = 0.01
beta1 = 4.090e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1498.2146718346294, q1 = 1.47, q2 = 0.01
beta1 = 4.090e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1498.2146718346294, q1 = 1.47, q2 = 0.01
beta1 = 4.090e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1498.2146718346294, q1 = 1.47, q2 = 0.01
beta1 = 4.090e-08, beta2 = 5.556e-11

```

delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 1498.2146718446293, q1 = 1.47, q2 = 0.01
  betal = 4.090e-08, beta2 = 5.556e-11
  delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 2493.1467619925684, q1 = 1.96, q2 = 0.01
  betal = 5.438e-08, beta2 = 5.556e-11
  delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 2493.1467619925684, q1 = 1.96, q2 = 0.01
  betal = 5.438e-08, beta2 = 5.556e-11
  delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 2493.1467619925684, q1 = 1.96, q2 = 0.01
  betal = 5.438e-08, beta2 = 5.556e-11
  delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 2493.1467620025683, q1 = 1.96, q2 = 0.01
  betal = 5.438e-08, beta2 = 5.556e-11
  delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 1702.5835641176193, q1 = 1.57, q2 = 0.01
  betal = 4.367e-08, beta2 = 5.556e-11
  delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 1702.5835641176193, q1 = 1.57, q2 = 0.01
  betal = 4.367e-08, beta2 = 5.556e-11
  delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 1702.5835641176193, q1 = 1.57, q2 = 0.01
  betal = 4.367e-08, beta2 = 5.556e-11
  delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
  N = 6.0e+06, E0 = 1702.5835641276192, q1 = 1.57, q2 = 0.01

```

beta1 = 4.367e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1515.3844335313565, q1 = 1.48, q2 = 0.01
beta1 = 4.113e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1515.3844335313565, q1 = 1.48, q2 = 0.01
beta1 = 4.113e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1515.3844335313565, q1 = 1.48, q2 = 0.01
beta1 = 4.113e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1515.3844335313565, q1 = 1.48, q2 = 0.01
beta1 = 4.113e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 1515.3844335413564, q1 = 1.48, q2 = 0.01
beta1 = 4.113e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 2802.141163544358, q1 = 1.33, q2 = 0.01
beta1 = 3.705e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 2802.141163544358, q1 = 1.33, q2 = 0.01
beta1 = 3.705e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 2802.141163544358, q1 = 1.33, q2 = 0.01
beta1 = 3.705e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 2802.141163544358, q1 = 1.33, q2 = 0.01
beta1 = 3.705e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 2802.1411635543577, q1 = 1.33, q2 = 0.01
beta1 = 3.705e-08, beta2 = 5.556e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 2938.9113943941866, q1 = 1.34, q2 = 0.01
beta1 = 3.728e-08, beta2 = 6.327e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 2938.9113943941866, q1 = 1.34, q2 = 0.01
beta1 = 3.728e-08, beta2 = 6.327e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 2938.9113943941866, q1 = 1.34, q2 = 0.01
beta1 = 3.728e-08, beta2 = 6.327e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 2938.9113943941866, q1 = 1.34, q2 = 0.01
beta1 = 3.728e-08, beta2 = 6.327e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 2938.9113944041865, q1 = 1.34, q2 = 0.01
beta1 = 3.728e-08, beta2 = 6.327e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 4286.038326456354, q1 = 1.29, q2 = 0.02
beta1 = 3.578e-08, beta2 = 8.679e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 4286.038326456354, q1 = 1.29, q2 = 0.02
beta1 = 3.578e-08, beta2 = 8.679e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 4286.038326456354, q1 = 1.29, q2 = 0.02
beta1 = 3.578e-08, beta2 = 8.679e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 4286.038326456354, q1 = 1.29, q2 = 0.02
beta1 = 3.578e-08, beta2 = 8.679e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 4286.038326466354, q1 = 1.29, q2 = 0.02
beta1 = 3.578e-08, beta2 = 8.679e-11
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 6190.621967743406, q1 = 1.19, q2 = 0.02
beta1 = 3.318e-08, beta2 = 1.061e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 6190.621967743406, q1 = 1.19, q2 = 0.02
beta1 = 3.318e-08, beta2 = 1.061e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 6190.621967743406, q1 = 1.19, q2 = 0.02
beta1 = 3.318e-08, beta2 = 1.061e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 6190.621967743406, q1 = 1.19, q2 = 0.02
beta1 = 3.318e-08, beta2 = 1.061e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 6190.621967753405, q1 = 1.19, q2 = 0.02
beta1 = 3.318e-08, beta2 = 1.061e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9725.368322877774, q1 = 1.08, q2 = 0.03
beta1 = 3.000e-08, beta2 = 1.700e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9725.368322877774, q1 = 1.08, q2 = 0.03
beta1 = 3.000e-08, beta2 = 1.700e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9725.368322877774, q1 = 1.08, q2 = 0.03
beta1 = 3.000e-08, beta2 = 1.700e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9725.368322877774, q1 = 1.08, q2 = 0.03
beta1 = 3.000e-08, beta2 = 1.700e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9951.573618031569, q1 = 1.11, q2 = 0.07
beta1 = 3.070e-08, beta2 = 4.036e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9951.573618031569, q1 = 1.11, q2 = 0.07
beta1 = 3.070e-08, beta2 = 4.036e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9951.573618031569, q1 = 1.11, q2 = 0.07
beta1 = 3.070e-08, beta2 = 4.036e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9951.573618031569, q1 = 1.11, q2 = 0.07
beta1 = 3.070e-08, beta2 = 4.036e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9951.57361804157, q1 = 1.11, q2 = 0.07
beta1 = 3.070e-08, beta2 = 4.036e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.094e-08, beta2 = 4.280e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.094e-08, beta2 = 4.280e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.094e-08, beta2 = 4.280e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.094e-08, beta2 = 4.280e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9999.99999999, q1 = 1.11, q2 = 0.08
beta1 = 3.094e-08, beta2 = 4.280e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.089e-08, beta2 = 4.289e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.089e-08, beta2 = 4.289e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.089e-08, beta2 = 4.289e-10

delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.089e-08, beta2 = 4.289e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 9999.99999999, q1 = 1.11, q2 = 0.08
beta1 = 3.089e-08, beta2 = 4.289e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.088e-08, beta2 = 4.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.088e-08, beta2 = 4.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.088e-08, beta2 = 4.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.088e-08, beta2 = 4.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 9999.99999999, q1 = 1.11, q2 = 0.08
beta1 = 3.088e-08, beta2 = 4.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.085e-08, beta2 = 4.510e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.085e-08, beta2 = 4.510e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08
beta1 = 3.085e-08, beta2 = 4.510e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.08

beta1 = 3.085e-08, beta2 = 4.510e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 9999.99999999, q1 = 1.11, q2 = 0.08
beta1 = 3.085e-08, beta2 = 4.510e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.09
beta1 = 3.072e-08, beta2 = 5.219e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.09
beta1 = 3.072e-08, beta2 = 5.219e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.09
beta1 = 3.072e-08, beta2 = 5.219e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.11, q2 = 0.09
beta1 = 3.072e-08, beta2 = 5.219e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 9999.99999999, q1 = 1.11, q2 = 0.09
beta1 = 3.072e-08, beta2 = 5.219e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.08, q2 = 0.17
beta1 = 3.001e-08, beta2 = 9.642e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.08, q2 = 0.17
beta1 = 3.001e-08, beta2 = 9.642e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.08, q2 = 0.17
beta1 = 3.001e-08, beta2 = 9.642e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.08, q2 = 0.17
beta1 = 3.001e-08, beta2 = 9.642e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 9999.99999999, q1 = 1.08, q2 = 0.17
beta1 = 3.001e-08, beta2 = 9.642e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.21
beta1 = 2.973e-08, beta2 = 1.162e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.21
beta1 = 2.973e-08, beta2 = 1.162e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.21
beta1 = 2.973e-08, beta2 = 1.162e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.21
beta1 = 2.973e-08, beta2 = 1.162e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 9999.99999999, q1 = 1.07, q2 = 0.21
beta1 = 2.973e-08, beta2 = 1.162e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.209e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.209e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.209e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.209e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:

N = 6.0e+06, E0 = 9999.99999999, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.209e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.208e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.208e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.208e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.208e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9999.99999999, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.208e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.206e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.206e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.206e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 9999.99999999, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.206e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

omicron_initial:
N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22
beta1 = 2.968e-08, beta2 = 1.206e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

```
omicron_initial:  
  N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22  
  beta1 = 2.968e-08, beta2 = 1.206e-09  
  delta = 0.167, gamma = 0.033, lambda = 0.0003
```

```
omicron_initial:  
  N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22  
  beta1 = 2.968e-08, beta2 = 1.206e-09  
  delta = 0.167, gamma = 0.033, lambda = 0.0003
```

```
omicron_initial:  
  N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22  
  beta1 = 2.968e-08, beta2 = 1.206e-09  
  delta = 0.167, gamma = 0.033, lambda = 0.0003
```

```
omicron_initial:  
  N = 6.0e+06, E0 = 9999.99999999, q1 = 1.07, q2 = 0.22  
  beta1 = 2.968e-08, beta2 = 1.206e-09  
  delta = 0.167, gamma = 0.033, lambda = 0.0003
```

Успех: True | Ошибка: 7.13e+06

Лучшие параметры → q1=1.069, q2=0.217, N=6.0e+06, E0=10000

```
omicron_initial:  
  N = 6.0e+06, E0 = 10000.0, q1 = 1.07, q2 = 0.22  
  beta1 = 2.968e-08, beta2 = 1.206e-09  
  delta = 0.167, gamma = 0.033, lambda = 0.0003  
-> Симуляция завершена, период 40 дней
```

Фиттинг для 5_wave_omicron

```
5_wave_omicron:  
  N = 1.0e+06, E0 = 500.0, q1 = 0.27, q2 = 0.45  
  beta1 = 4.500e-08, beta2 = 1.500e-08  
  delta = 0.167, gamma = 0.033, lambda = 0.0003
```

```
5_wave_omicron:  
  N = 1.0e+06, E0 = 500.0, q1 = 0.27, q2 = 0.45  
  beta1 = 4.500e-08, beta2 = 1.500e-08  
  delta = 0.167, gamma = 0.033, lambda = 0.0003
```

```
5_wave_omicron:  
  N = 1.0e+06, E0 = 500.0, q1 = 0.27, q2 = 0.45  
  beta1 = 4.500e-08, beta2 = 1.500e-08  
  delta = 0.167, gamma = 0.033, lambda = 0.0003
```

```
5_wave_omicron:  
  N = 1.0e+06, E0 = 500.0, q1 = 0.27, q2 = 0.45  
  beta1 = 4.500e-08, beta2 = 1.500e-08  
  delta = 0.167, gamma = 0.033, lambda = 0.0003
```

```
5_wave_omicron:  
  N = 1.0e+06, E0 = 500.00000001, q1 = 0.27, q2 = 0.45  
  beta1 = 4.500e-08, beta2 = 1.500e-08
```

delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1304.6620661362444, q1 = 5.00, q2 = 5.00
beta1 = 8.333e-07, beta2 = 1.667e-07
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1304.6620661362444, q1 = 5.00, q2 = 5.00
beta1 = 8.333e-07, beta2 = 1.667e-07
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1304.6620661362444, q1 = 5.00, q2 = 5.00
beta1 = 8.333e-07, beta2 = 1.667e-07
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1304.6620661362444, q1 = 5.00, q2 = 5.00
beta1 = 8.333e-07, beta2 = 1.667e-07
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1304.6620661462443, q1 = 5.00, q2 = 5.00
beta1 = 8.333e-07, beta2 = 1.667e-07
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 587.6301186143049, q1 = 0.79, q2 = 0.95
beta1 = 1.309e-07, beta2 = 3.152e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 587.6301186143049, q1 = 0.79, q2 = 0.95
beta1 = 1.309e-07, beta2 = 3.152e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 587.6301186143049, q1 = 0.79, q2 = 0.95
beta1 = 1.309e-07, beta2 = 3.152e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 587.6301186143049, q1 = 0.79, q2 = 0.95
beta1 = 1.309e-07, beta2 = 3.152e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 587.6301186243049, q1 = 0.79, q2 = 0.95
beta1 = 1.309e-07, beta2 = 3.152e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 946.1460923752747, q1 = 2.89, q2 = 2.97

beta1 = 4.821e-07, beta2 = 9.909e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 946.1460923752747, q1 = 2.89, q2 = 2.97
beta1 = 4.821e-07, beta2 = 9.909e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 946.1460923752747, q1 = 2.89, q2 = 2.97
beta1 = 4.821e-07, beta2 = 9.909e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 946.1460923752747, q1 = 2.89, q2 = 2.97
beta1 = 4.821e-07, beta2 = 9.909e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 946.1460923852748, q1 = 2.89, q2 = 2.97
beta1 = 4.821e-07, beta2 = 9.909e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 599.4548668531194, q1 = 0.85, q2 = 1.01
beta1 = 1.424e-07, beta2 = 3.375e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 599.4548668531194, q1 = 0.85, q2 = 1.01
beta1 = 1.424e-07, beta2 = 3.375e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 599.4548668531194, q1 = 0.85, q2 = 1.01
beta1 = 1.424e-07, beta2 = 3.375e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 599.4548668531194, q1 = 0.85, q2 = 1.01
beta1 = 1.424e-07, beta2 = 3.375e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 599.4548668631194, q1 = 0.85, q2 = 1.01
beta1 = 1.424e-07, beta2 = 3.375e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 613.0386455175918, q1 = 0.93, q2 = 1.09
beta1 = 1.557e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 613.0386455175918, q1 = 0.93, q2 = 1.09
beta1 = 1.557e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 613.0386455175918, q1 = 0.93, q2 = 1.09
beta1 = 1.557e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 613.0386455175918, q1 = 0.93, q2 = 1.09
beta1 = 1.557e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 613.0386455275918, q1 = 0.93, q2 = 1.09
beta1 = 1.557e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 779.5923689464332, q1 = 1.91, q2 = 2.03
beta1 = 3.189e-07, beta2 = 6.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 779.5923689464332, q1 = 1.91, q2 = 2.03
beta1 = 3.189e-07, beta2 = 6.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 779.5923689464332, q1 = 1.91, q2 = 2.03
beta1 = 3.189e-07, beta2 = 6.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 779.5923689464332, q1 = 1.91, q2 = 2.03
beta1 = 3.189e-07, beta2 = 6.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 779.5923689564332, q1 = 1.91, q2 = 2.03
beta1 = 3.189e-07, beta2 = 6.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 614.736906285791, q1 = 0.94, q2 = 1.10
beta1 = 1.574e-07, beta2 = 3.663e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 614.736906285791, q1 = 0.94, q2 = 1.10
beta1 = 1.574e-07, beta2 = 3.663e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

```

5_wave_omicron:
  N = 1.0e+06, E0 = 614.736906285791, q1 = 0.94, q2 = 1.10
  beta1 = 1.574e-07, beta2 = 3.663e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 614.736906285791, q1 = 0.94, q2 = 1.10
  beta1 = 1.574e-07, beta2 = 3.663e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 614.736906295791, q1 = 0.94, q2 = 1.10
  beta1 = 1.574e-07, beta2 = 3.663e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 623.5628961232186, q1 = 1.00, q2 = 1.15
  beta1 = 1.661e-07, beta2 = 3.829e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 623.5628961232186, q1 = 1.00, q2 = 1.15
  beta1 = 1.661e-07, beta2 = 3.829e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 623.5628961232186, q1 = 1.00, q2 = 1.15
  beta1 = 1.661e-07, beta2 = 3.829e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 623.5628961332186, q1 = 1.00, q2 = 1.15
  beta1 = 1.661e-07, beta2 = 3.829e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 554.3910469422353, q1 = 0.58, q2 = 0.66
  beta1 = 9.589e-08, beta2 = 2.211e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 554.3910469422353, q1 = 0.58, q2 = 0.66
  beta1 = 9.589e-08, beta2 = 2.211e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 554.3910469422353, q1 = 0.58, q2 = 0.66
  beta1 = 9.589e-08, beta2 = 2.211e-08
  delta = 0.167, gamma = 0.033, lambda = 0.0003

```


5_wave_omicron:
N = 1.0e+06, E0 = 554.3910469422353, q1 = 0.58, q2 = 0.66
beta1 = 9.589e-08, beta2 = 2.211e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 554.3910469522353, q1 = 0.58, q2 = 0.66
beta1 = 9.589e-08, beta2 = 2.211e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 620.2297804221565, q1 = 0.98, q2 = 1.13
beta1 = 1.627e-07, beta2 = 3.751e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 620.2297804221565, q1 = 0.98, q2 = 1.13
beta1 = 1.627e-07, beta2 = 3.751e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 620.2297804221565, q1 = 0.98, q2 = 1.13
beta1 = 1.627e-07, beta2 = 3.751e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 620.2297804321565, q1 = 0.98, q2 = 1.13
beta1 = 1.627e-07, beta2 = 3.751e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 622.520764231042, q1 = 0.99, q2 = 1.14
beta1 = 1.650e-07, beta2 = 3.805e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 622.520764231042, q1 = 0.99, q2 = 1.14
beta1 = 1.650e-07, beta2 = 3.805e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 622.520764231042, q1 = 0.99, q2 = 1.14
beta1 = 1.650e-07, beta2 = 3.805e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 622.520764231042, q1 = 0.99, q2 = 1.14
beta1 = 1.650e-07, beta2 = 3.805e-08

delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 622.520764241042, q1 = 0.99, q2 = 1.14
beta1 = 1.650e-07, beta2 = 3.805e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 623.1615718543484, q1 = 0.99, q2 = 1.13
beta1 = 1.655e-07, beta2 = 3.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 623.1615718543484, q1 = 0.99, q2 = 1.13
beta1 = 1.655e-07, beta2 = 3.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 623.1615718543484, q1 = 0.99, q2 = 1.13
beta1 = 1.655e-07, beta2 = 3.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 623.1615718543484, q1 = 0.99, q2 = 1.13
beta1 = 1.655e-07, beta2 = 3.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 623.1615718643484, q1 = 0.99, q2 = 1.13
beta1 = 1.655e-07, beta2 = 3.770e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 625.7248023475739, q1 = 1.01, q2 = 1.09
beta1 = 1.676e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 625.7248023475739, q1 = 1.01, q2 = 1.09
beta1 = 1.676e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 625.7248023475739, q1 = 1.01, q2 = 1.09
beta1 = 1.676e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 625.7248023475739, q1 = 1.01, q2 = 1.09
beta1 = 1.676e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 625.7248023575739, q1 = 1.01, q2 = 1.09

beta1 = 1.676e-07, beta2 = 3.631e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 635.9777243204761, q1 = 1.06, q2 = 0.92
beta1 = 1.760e-07, beta2 = 3.074e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 635.9777243204761, q1 = 1.06, q2 = 0.92
beta1 = 1.760e-07, beta2 = 3.074e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 635.9777243204761, q1 = 1.06, q2 = 0.92
beta1 = 1.760e-07, beta2 = 3.074e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 635.9777243204761, q1 = 1.06, q2 = 0.92
beta1 = 1.760e-07, beta2 = 3.074e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 635.9777243304761, q1 = 1.06, q2 = 0.92
beta1 = 1.760e-07, beta2 = 3.074e-08
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 676.9894122120849, q1 = 1.26, q2 = 0.25
beta1 = 2.097e-07, beta2 = 8.492e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 676.9894122120849, q1 = 1.26, q2 = 0.25
beta1 = 2.097e-07, beta2 = 8.492e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 676.9894122120849, q1 = 1.26, q2 = 0.25
beta1 = 2.097e-07, beta2 = 8.492e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 676.9894122120849, q1 = 1.26, q2 = 0.25
beta1 = 2.097e-07, beta2 = 8.492e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 676.9894122220849, q1 = 1.26, q2 = 0.25
beta1 = 2.097e-07, beta2 = 8.492e-09
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 692.026781981916, q1 = 1.33, q2 = 0.01
beta1 = 2.220e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 692.026781981916, q1 = 1.33, q2 = 0.01
beta1 = 2.220e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 692.026781981916, q1 = 1.33, q2 = 0.01
beta1 = 2.220e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 692.026781981916, q1 = 1.33, q2 = 0.01
beta1 = 2.220e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 692.026781991916, q1 = 1.33, q2 = 0.01
beta1 = 2.220e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 695.8954700726317, q1 = 1.36, q2 = 0.01
beta1 = 2.260e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 695.8954700726317, q1 = 1.36, q2 = 0.01
beta1 = 2.260e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 695.8954700726317, q1 = 1.36, q2 = 0.01
beta1 = 2.260e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 695.8954700726317, q1 = 1.36, q2 = 0.01
beta1 = 2.260e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 695.8954700826317, q1 = 1.36, q2 = 0.01
beta1 = 2.260e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 695.0377883562229, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 695.0377883562229, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 695.0377883562229, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 695.0377883562229, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 695.0377883662229, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 695.4892015335201, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 695.4892015335201, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 695.4892015335201, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 695.4892015435202, q1 = 1.35, q2 = 0.01
beta1 = 2.247e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 697.2948542427091, q1 = 1.35, q2 = 0.01
beta1 = 2.249e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 697.2948542427091, q1 = 1.35, q2 = 0.01
beta1 = 2.249e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 697.2948542427091, q1 = 1.35, q2 = 0.01
beta1 = 2.249e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 697.2948542427091, q1 = 1.35, q2 = 0.01
beta1 = 2.249e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 697.2948542527091, q1 = 1.35, q2 = 0.01
beta1 = 2.249e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 704.517465079465, q1 = 1.35, q2 = 0.01
beta1 = 2.255e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 704.517465079465, q1 = 1.35, q2 = 0.01
beta1 = 2.255e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 704.517465079465, q1 = 1.35, q2 = 0.01
beta1 = 2.255e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 704.517465079465, q1 = 1.35, q2 = 0.01
beta1 = 2.255e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 704.517465089465, q1 = 1.35, q2 = 0.01
beta1 = 2.255e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 769.2566964650819, q1 = 1.36, q2 = 0.01
beta1 = 2.275e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 769.2566964650819, q1 = 1.36, q2 = 0.01
beta1 = 2.275e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 769.2566964650819, q1 = 1.36, q2 = 0.01
beta1 = 2.275e-07, beta2 = 3.333e-10

delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 769.2566964650819, q1 = 1.36, q2 = 0.01
beta1 = 2.275e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 769.2566964750819, q1 = 1.36, q2 = 0.01
beta1 = 2.275e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 943.7507623274713, q1 = 1.37, q2 = 0.01
beta1 = 2.287e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 943.7507623274713, q1 = 1.37, q2 = 0.01
beta1 = 2.287e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 943.7507623274713, q1 = 1.37, q2 = 0.01
beta1 = 2.287e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 943.7507623274713, q1 = 1.37, q2 = 0.01
beta1 = 2.287e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 943.7507623374713, q1 = 1.37, q2 = 0.01
beta1 = 2.287e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1742.382096846277, q1 = 1.36, q2 = 0.01
beta1 = 2.271e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1742.382096846277, q1 = 1.36, q2 = 0.01
beta1 = 2.271e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1742.382096846277, q1 = 1.36, q2 = 0.01
beta1 = 2.271e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 1742.382096846277, q1 = 1.36, q2 = 0.01

beta1 = 2.271e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 1742.3820968562768, q1 = 1.36, q2 = 0.01
beta1 = 2.271e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 4936.907434921499, q1 = 1.33, q2 = 0.01
beta1 = 2.209e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 4936.907434921499, q1 = 1.33, q2 = 0.01
beta1 = 2.209e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 4936.907434921499, q1 = 1.33, q2 = 0.01
beta1 = 2.209e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 4936.907434921499, q1 = 1.33, q2 = 0.01
beta1 = 2.209e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 4936.907434931499, q1 = 1.33, q2 = 0.01
beta1 = 2.209e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 10000.0, q1 = 1.27, q2 = 0.01
beta1 = 2.111e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 10000.0, q1 = 1.27, q2 = 0.01
beta1 = 2.111e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 10000.0, q1 = 1.27, q2 = 0.01
beta1 = 2.111e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
N = 1.0e+06, E0 = 10000.0, q1 = 1.27, q2 = 0.01
beta1 = 2.111e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.27, q2 = 0.01
beta1 = 2.111e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 1.29, q2 = 0.01
beta1 = 2.151e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 1.29, q2 = 0.01
beta1 = 2.151e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 1.29, q2 = 0.01
beta1 = 2.151e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 1.29, q2 = 0.01
beta1 = 2.151e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.29, q2 = 0.01
beta1 = 2.151e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 1.39, q2 = 0.01
beta1 = 2.313e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 1.39, q2 = 0.01
beta1 = 2.313e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 1.39, q2 = 0.01
beta1 = 2.313e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 1.39, q2 = 0.01
beta1 = 2.313e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 9999.99999999, q1 = 1.39, q2 = 0.01
beta1 = 2.313e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

```

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 1.69, q2 = 0.01
  beta1 = 2.817e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 1.69, q2 = 0.01
  beta1 = 2.817e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 1.69, q2 = 0.01
  beta1 = 2.817e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 1.69, q2 = 0.01
  beta1 = 2.817e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 9999.99999999, q1 = 1.69, q2 = 0.01
  beta1 = 2.817e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.90, q2 = 0.01
  beta1 = 4.835e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.90, q2 = 0.01
  beta1 = 4.835e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.90, q2 = 0.01
  beta1 = 4.835e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 9999.99999999, q1 = 2.90, q2 = 0.01
  beta1 = 4.835e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.21, q2 = 0.01
  beta1 = 3.689e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

```

```

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.21, q2 = 0.01
  beta1 = 3.689e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.21, q2 = 0.01
  beta1 = 3.689e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.21, q2 = 0.01
  beta1 = 3.689e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 9999.99999999, q1 = 2.21, q2 = 0.01
  beta1 = 3.689e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 3.03, q2 = 0.01
  beta1 = 5.050e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 3.03, q2 = 0.01
  beta1 = 5.050e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 3.03, q2 = 0.01
  beta1 = 5.050e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 3.03, q2 = 0.01
  beta1 = 5.050e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 9999.99999999, q1 = 3.03, q2 = 0.01
  beta1 = 5.050e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.32, q2 = 0.01
  beta1 = 3.868e-07, beta2 = 3.333e-10
  delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:
  N = 1.0e+06, E0 = 10000.0, q1 = 2.32, q2 = 0.01
  beta1 = 3.868e-07, beta2 = 3.333e-10

```

delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.32, q2 = 0.01
beta1 = 3.868e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.32, q2 = 0.01
beta1 = 3.868e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 9999.99999999, q1 = 2.32, q2 = 0.01
beta1 = 3.868e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 9999.99999999, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01

beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

5_wave_omicron:

N = 1.0e+06, E0 = 9999.99999999, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

Успех: True | Ошибка: 2.13e+08

Лучшие параметры → q1=2.326, q2=0.010, N=1.0e+06, E0=10000

5_wave_omicron:

N = 1.0e+06, E0 = 10000.0, q1 = 2.33, q2 = 0.01
beta1 = 3.877e-07, beta2 = 3.333e-10
delta = 0.167, gamma = 0.033, lambda = 0.0003

-> Симуляция завершена, период 70 дней

ФИТТИНГ для 6_wave

6_wave:

N = 2.0e+06, E0 = 500.0, q1 = 0.50, q2 = 0.80
beta1 = 1.786e-08, beta2 = 1.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 500.0, q1 = 0.50, q2 = 0.80
beta1 = 1.786e-08, beta2 = 1.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 500.0, q1 = 0.50, q2 = 0.80
beta1 = 1.786e-08, beta2 = 1.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 500.0, q1 = 0.50, q2 = 0.80
beta1 = 1.786e-08, beta2 = 1.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 500.00000001, q1 = 0.50, q2 = 0.80
beta1 = 1.786e-08, beta2 = 1.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1970.7434379109989, q1 = 5.00, q2 = 5.00
beta1 = 1.786e-07, beta2 = 8.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1970.7434379109989, q1 = 5.00, q2 = 5.00
beta1 = 1.786e-07, beta2 = 8.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1970.7434379109989, q1 = 5.00, q2 = 5.00
beta1 = 1.786e-07, beta2 = 8.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1970.7434379109989, q1 = 5.00, q2 = 5.00
beta1 = 1.786e-07, beta2 = 8.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1970.7434379209988, q1 = 5.00, q2 = 5.00
beta1 = 1.786e-07, beta2 = 8.333e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 589.2164275481488, q1 = 0.77, q2 = 1.05
beta1 = 2.761e-08, beta2 = 1.758e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 589.2164275481488, q1 = 0.77, q2 = 1.05
beta1 = 2.761e-08, beta2 = 1.758e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 589.2164275481488, q1 = 0.77, q2 = 1.05
beta1 = 2.761e-08, beta2 = 1.758e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 589.2164275481488, q1 = 0.77, q2 = 1.05
beta1 = 2.761e-08, beta2 = 1.758e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 589.2164275581488, q1 = 0.77, q2 = 1.05
beta1 = 2.761e-08, beta2 = 1.758e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 530.9773914277007, q1 = 0.59, q2 = 0.89
beta1 = 2.124e-08, beta2 = 1.481e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 530.9773914277007, q1 = 0.59, q2 = 0.89
beta1 = 2.124e-08, beta2 = 1.481e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 530.9773914277007, q1 = 0.59, q2 = 0.89
beta1 = 2.124e-08, beta2 = 1.481e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 530.9773914277007, q1 = 0.59, q2 = 0.89
beta1 = 2.124e-08, beta2 = 1.481e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 530.9773914377007, q1 = 0.59, q2 = 0.89
beta1 = 2.124e-08, beta2 = 1.481e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 542.3307190821446, q1 = 0.63, q2 = 0.92
beta1 = 2.248e-08, beta2 = 1.535e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 542.3307190821446, q1 = 0.63, q2 = 0.92
beta1 = 2.248e-08, beta2 = 1.535e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 542.3307190821446, q1 = 0.63, q2 = 0.92
beta1 = 2.248e-08, beta2 = 1.535e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 542.3307190821446, q1 = 0.63, q2 = 0.92
beta1 = 2.248e-08, beta2 = 1.535e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 542.3307190921446, q1 = 0.63, q2 = 0.92
beta1 = 2.248e-08, beta2 = 1.535e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 595.0332737222524, q1 = 0.82, q2 = 1.03
beta1 = 2.932e-08, beta2 = 1.711e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 595.0332737222524, q1 = 0.82, q2 = 1.03
beta1 = 2.932e-08, beta2 = 1.711e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 595.0332737222524, q1 = 0.82, q2 = 1.03
beta1 = 2.932e-08, beta2 = 1.711e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 595.0332737222524, q1 = 0.82, q2 = 1.03
beta1 = 2.932e-08, beta2 = 1.711e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 595.0332737322524, q1 = 0.82, q2 = 1.03
beta1 = 2.932e-08, beta2 = 1.711e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 544.6265928872901, q1 = 0.64, q2 = 0.93
beta1 = 2.278e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 544.6265928872901, q1 = 0.64, q2 = 0.93
beta1 = 2.278e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 544.6265928872901, q1 = 0.64, q2 = 0.93
beta1 = 2.278e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 544.6265928872901, q1 = 0.64, q2 = 0.93
beta1 = 2.278e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 544.6265928972902, q1 = 0.64, q2 = 0.93
beta1 = 2.278e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 545.2180286981095, q1 = 0.64, q2 = 0.93
beta1 = 2.288e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 545.2180286981095, q1 = 0.64, q2 = 0.93
beta1 = 2.288e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 545.2180286981095, q1 = 0.64, q2 = 0.93
beta1 = 2.288e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 545.2180286981095, q1 = 0.64, q2 = 0.93
beta1 = 2.288e-08, beta2 = 1.542e-08

delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 545.2180287081095, q1 = 0.64, q2 = 0.93
beta1 = 2.288e-08, beta2 = 1.542e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 545.4433447541263, q1 = 0.64, q2 = 0.92
beta1 = 2.296e-08, beta2 = 1.539e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 545.4433447541263, q1 = 0.64, q2 = 0.92
beta1 = 2.296e-08, beta2 = 1.539e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 545.4433447541263, q1 = 0.64, q2 = 0.92
beta1 = 2.296e-08, beta2 = 1.539e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 545.4433447541263, q1 = 0.64, q2 = 0.92
beta1 = 2.296e-08, beta2 = 1.539e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 545.4433447641263, q1 = 0.64, q2 = 0.92
beta1 = 2.296e-08, beta2 = 1.539e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 547.1125853466912, q1 = 0.66, q2 = 0.90
beta1 = 2.363e-08, beta2 = 1.501e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 547.1125853466912, q1 = 0.66, q2 = 0.90
beta1 = 2.363e-08, beta2 = 1.501e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 547.1125853466912, q1 = 0.66, q2 = 0.90
beta1 = 2.363e-08, beta2 = 1.501e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 547.1125853466912, q1 = 0.66, q2 = 0.90
beta1 = 2.363e-08, beta2 = 1.501e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 547.1125853566912, q1 = 0.66, q2 = 0.90

beta1 = 2.363e-08, beta2 = 1.501e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 550.0349626423085, q1 = 0.70, q2 = 0.85
beta1 = 2.506e-08, beta2 = 1.412e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 550.0349626423085, q1 = 0.70, q2 = 0.85
beta1 = 2.506e-08, beta2 = 1.412e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 550.0349626423085, q1 = 0.70, q2 = 0.85
beta1 = 2.506e-08, beta2 = 1.412e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 550.0349626423085, q1 = 0.70, q2 = 0.85
beta1 = 2.506e-08, beta2 = 1.412e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 550.0349626523085, q1 = 0.70, q2 = 0.85
beta1 = 2.506e-08, beta2 = 1.412e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 558.1688654020188, q1 = 0.83, q2 = 0.67
beta1 = 2.949e-08, beta2 = 1.122e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 558.1688654020188, q1 = 0.83, q2 = 0.67
beta1 = 2.949e-08, beta2 = 1.122e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 558.1688654020188, q1 = 0.83, q2 = 0.67
beta1 = 2.949e-08, beta2 = 1.122e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 558.1688654020188, q1 = 0.83, q2 = 0.67
beta1 = 2.949e-08, beta2 = 1.122e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 558.1688654120188, q1 = 0.83, q2 = 0.67
beta1 = 2.949e-08, beta2 = 1.122e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 570.4513581489165, q1 = 1.03, q2 = 0.37
beta1 = 3.686e-08, beta2 = 6.188e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 570.4513581489165, q1 = 1.03, q2 = 0.37
beta1 = 3.686e-08, beta2 = 6.188e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 570.4513581489165, q1 = 1.03, q2 = 0.37
beta1 = 3.686e-08, beta2 = 6.188e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 570.4513581489165, q1 = 1.03, q2 = 0.37
beta1 = 3.686e-08, beta2 = 6.188e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 570.4513581589165, q1 = 1.03, q2 = 0.37
beta1 = 3.686e-08, beta2 = 6.188e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 569.3908909749174, q1 = 1.04, q2 = 0.35
beta1 = 3.701e-08, beta2 = 5.863e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 569.3908909749174, q1 = 1.04, q2 = 0.35
beta1 = 3.701e-08, beta2 = 5.863e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 569.3908909749174, q1 = 1.04, q2 = 0.35
beta1 = 3.701e-08, beta2 = 5.863e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 569.3908909749174, q1 = 1.04, q2 = 0.35
beta1 = 3.701e-08, beta2 = 5.863e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 569.3908909849174, q1 = 1.04, q2 = 0.35
beta1 = 3.701e-08, beta2 = 5.863e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 574.2723982937074, q1 = 1.18, q2 = 0.11
beta1 = 4.197e-08, beta2 = 1.894e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 574.2723982937074, q1 = 1.18, q2 = 0.11
beta1 = 4.197e-08, beta2 = 1.894e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 574.2723982937074, q1 = 1.18, q2 = 0.11
beta1 = 4.197e-08, beta2 = 1.894e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 574.2723982937074, q1 = 1.18, q2 = 0.11
beta1 = 4.197e-08, beta2 = 1.894e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 574.2723983037074, q1 = 1.18, q2 = 0.11
beta1 = 4.197e-08, beta2 = 1.894e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 568.9977526164208, q1 = 1.07, q2 = 0.27
beta1 = 3.835e-08, beta2 = 4.484e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 568.9977526164208, q1 = 1.07, q2 = 0.27
beta1 = 3.835e-08, beta2 = 4.484e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 568.9977526164208, q1 = 1.07, q2 = 0.27
beta1 = 3.835e-08, beta2 = 4.484e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 568.9977526164208, q1 = 1.07, q2 = 0.27
beta1 = 3.835e-08, beta2 = 4.484e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 568.9977526264208, q1 = 1.07, q2 = 0.27
beta1 = 3.835e-08, beta2 = 4.484e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 577.094527075521, q1 = 1.10, q2 = 0.24
beta1 = 3.915e-08, beta2 = 3.949e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 577.094527075521, q1 = 1.10, q2 = 0.24
beta1 = 3.915e-08, beta2 = 3.949e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 577.094527075521, q1 = 1.10, q2 = 0.24
 beta1 = 3.915e-08, beta2 = 3.949e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 577.094527075521, q1 = 1.10, q2 = 0.24
 beta1 = 3.915e-08, beta2 = 3.949e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 577.094527085521, q1 = 1.10, q2 = 0.24
 beta1 = 3.915e-08, beta2 = 3.949e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 583.3344983564256, q1 = 1.11, q2 = 0.22
 beta1 = 3.964e-08, beta2 = 3.593e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 583.3344983564256, q1 = 1.11, q2 = 0.22
 beta1 = 3.964e-08, beta2 = 3.593e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 583.3344983564256, q1 = 1.11, q2 = 0.22
 beta1 = 3.964e-08, beta2 = 3.593e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 583.3344983664256, q1 = 1.11, q2 = 0.22
 beta1 = 3.964e-08, beta2 = 3.593e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 592.9369129230834, q1 = 1.11, q2 = 0.21
 beta1 = 3.966e-08, beta2 = 3.574e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 592.9369129230834, q1 = 1.11, q2 = 0.21
 beta1 = 3.966e-08, beta2 = 3.574e-09
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 592.9369129230834, q1 = 1.11, q2 = 0.21
 beta1 = 3.966e-08, beta2 = 3.574e-09

delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 592.9369129230834, q1 = 1.11, q2 = 0.21
beta1 = 3.966e-08, beta2 = 3.574e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 592.9369129330834, q1 = 1.11, q2 = 0.21
beta1 = 3.966e-08, beta2 = 3.574e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 631.3465711897145, q1 = 1.11, q2 = 0.21
beta1 = 3.974e-08, beta2 = 3.498e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 631.3465711897145, q1 = 1.11, q2 = 0.21
beta1 = 3.974e-08, beta2 = 3.498e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 631.3465711897145, q1 = 1.11, q2 = 0.21
beta1 = 3.974e-08, beta2 = 3.498e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 631.3465711897145, q1 = 1.11, q2 = 0.21
beta1 = 3.974e-08, beta2 = 3.498e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 631.3465711997145, q1 = 1.11, q2 = 0.21
beta1 = 3.974e-08, beta2 = 3.498e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 778.5889173724333, q1 = 1.09, q2 = 0.24
beta1 = 3.887e-08, beta2 = 3.936e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 778.5889173724333, q1 = 1.09, q2 = 0.24
beta1 = 3.887e-08, beta2 = 3.936e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 778.5889173724333, q1 = 1.09, q2 = 0.24
beta1 = 3.887e-08, beta2 = 3.936e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 778.5889173724333, q1 = 1.09, q2 = 0.24

beta1 = 3.887e-08, beta2 = 3.936e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 778.5889173824334, q1 = 1.09, q2 = 0.24
beta1 = 3.887e-08, beta2 = 3.936e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1367.5583021033087, q1 = 0.99, q2 = 0.34
beta1 = 3.537e-08, beta2 = 5.689e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1367.5583021033087, q1 = 0.99, q2 = 0.34
beta1 = 3.537e-08, beta2 = 5.689e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1367.5583021033087, q1 = 0.99, q2 = 0.34
beta1 = 3.537e-08, beta2 = 5.689e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1367.5583021033087, q1 = 0.99, q2 = 0.34
beta1 = 3.537e-08, beta2 = 5.689e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 1367.5583021133086, q1 = 0.99, q2 = 0.34
beta1 = 3.537e-08, beta2 = 5.689e-09
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5357.617297939745, q1 = 0.17, q2 = 1.31
beta1 = 6.116e-09, beta2 = 2.189e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5357.617297939745, q1 = 0.17, q2 = 1.31
beta1 = 6.116e-09, beta2 = 2.189e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5357.617297939745, q1 = 0.17, q2 = 1.31
beta1 = 6.116e-09, beta2 = 2.189e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5357.617297939745, q1 = 0.17, q2 = 1.31
beta1 = 6.116e-09, beta2 = 2.189e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5357.617297949745, q1 = 0.17, q2 = 1.31
beta1 = 6.116e-09, beta2 = 2.189e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 3772.401431558196, q1 = 0.44, q2 = 1.03
beta1 = 1.554e-08, beta2 = 1.721e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 3772.401431558196, q1 = 0.44, q2 = 1.03
beta1 = 1.554e-08, beta2 = 1.721e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 3772.401431558196, q1 = 0.44, q2 = 1.03
beta1 = 1.554e-08, beta2 = 1.721e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 3772.401431558196, q1 = 0.44, q2 = 1.03
beta1 = 1.554e-08, beta2 = 1.721e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 3772.4014315681957, q1 = 0.44, q2 = 1.03
beta1 = 1.554e-08, beta2 = 1.721e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 4227.452075124627, q1 = 0.32, q2 = 1.18
beta1 = 1.154e-08, beta2 = 1.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 4227.452075124627, q1 = 0.32, q2 = 1.18
beta1 = 1.154e-08, beta2 = 1.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 4227.452075124627, q1 = 0.32, q2 = 1.18
beta1 = 1.154e-08, beta2 = 1.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 4227.452075124627, q1 = 0.32, q2 = 1.18
beta1 = 1.154e-08, beta2 = 1.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 4227.452075134627, q1 = 0.32, q2 = 1.18
beta1 = 1.154e-08, beta2 = 1.968e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4267.7208665828975, q1 = 0.32, q2 = 1.19
beta1 = 1.142e-08, beta2 = 1.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4267.7208665828975, q1 = 0.32, q2 = 1.19
beta1 = 1.142e-08, beta2 = 1.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4267.7208665828975, q1 = 0.32, q2 = 1.19
beta1 = 1.142e-08, beta2 = 1.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4267.7208665828975, q1 = 0.32, q2 = 1.19
beta1 = 1.142e-08, beta2 = 1.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4267.720866592897, q1 = 0.32, q2 = 1.19
beta1 = 1.142e-08, beta2 = 1.983e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4428.7960324159785, q1 = 0.31, q2 = 1.22
beta1 = 1.093e-08, beta2 = 2.040e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4428.7960324159785, q1 = 0.31, q2 = 1.22
beta1 = 1.093e-08, beta2 = 2.040e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4428.7960324159785, q1 = 0.31, q2 = 1.22
beta1 = 1.093e-08, beta2 = 2.040e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4428.7960324159785, q1 = 0.31, q2 = 1.22
beta1 = 1.093e-08, beta2 = 2.040e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4428.796032425978, q1 = 0.31, q2 = 1.22
beta1 = 1.093e-08, beta2 = 2.040e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4607.263687860234, q1 = 0.31, q2 = 1.21
beta1 = 1.101e-08, beta2 = 2.014e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4607.263687860234, q1 = 0.31, q2 = 1.21
beta1 = 1.101e-08, beta2 = 2.014e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4607.263687860234, q1 = 0.31, q2 = 1.21
beta1 = 1.101e-08, beta2 = 2.014e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4607.263687860234, q1 = 0.31, q2 = 1.21
beta1 = 1.101e-08, beta2 = 2.014e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 4607.263687870234, q1 = 0.31, q2 = 1.21
beta1 = 1.101e-08, beta2 = 2.014e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784307446, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784307446, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784307446, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784307446, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784317446, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 10000.0, q1 = 0.01, q2 = 1.26
beta1 = 3.571e-10, beta2 = 2.100e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 10000.0, q1 = 0.01, q2 = 1.26
beta1 = 3.571e-10, beta2 = 2.100e-08

delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 10000.0, q1 = 0.01, q2 = 1.26
beta1 = 3.571e-10, beta2 = 2.100e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 10000.0, q1 = 0.01, q2 = 1.26
beta1 = 3.571e-10, beta2 = 2.100e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 9999.99999999, q1 = 0.01, q2 = 1.26
beta1 = 3.571e-10, beta2 = 2.100e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5680.60190574519, q1 = 0.30, q2 = 1.18
beta1 = 1.061e-08, beta2 = 1.973e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5680.60190574519, q1 = 0.30, q2 = 1.18
beta1 = 1.061e-08, beta2 = 1.973e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5680.60190574519, q1 = 0.30, q2 = 1.18
beta1 = 1.061e-08, beta2 = 1.973e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5680.60190574519, q1 = 0.30, q2 = 1.18
beta1 = 1.061e-08, beta2 = 1.973e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5680.60190575519, q1 = 0.30, q2 = 1.18
beta1 = 1.061e-08, beta2 = 1.973e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5648.95690965697, q1 = 0.30, q2 = 1.18
beta1 = 1.068e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5648.95690965697, q1 = 0.30, q2 = 1.18
beta1 = 1.068e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5648.95690965697, q1 = 0.30, q2 = 1.18

beta1 = 1.068e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5648.95690965697, q1 = 0.30, q2 = 1.18
beta1 = 1.068e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5648.95690966697, q1 = 0.30, q2 = 1.18
beta1 = 1.068e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5644.9321755092315, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5644.9321755092315, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5644.9321755092315, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5644.9321755092315, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5644.932175519231, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.925941186866, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.925941186866, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.925941186866, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.925941186866, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.925941196866, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.780712123495, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.780712123495, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.780712123495, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.780712123495, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.7807121334945, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.6930930963035, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.6930930963035, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.6930930963035, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.6930930963035, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.693093106303, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.648018066617, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.648018066617, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.648018066617, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.648018066617, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.648018076617, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.641062056103, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.641062056103, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.641062056103, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.641062056103, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.641062066103, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.637125580634, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.637125580634, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.637125580634, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.637125580634, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.637125590634, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.6344286562835, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.6344286562835, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.6344286562835, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.6344286562835, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.634428666283, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633937983601, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08

delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633937983601, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633937983601, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633937983601, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633937993601, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633822190772, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633822190772, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633822190772, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633822190772, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633822200772, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633795768237, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633795768237, q1 = 0.30, q2 = 1.18

beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633795768237, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633795768237, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633795778237, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.6337870334855, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.6337870334855, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.6337870334855, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.6337870334855, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633787043485, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633785444021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633785444021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633785444021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633785444021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633785454021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633784780826, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633784780826, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633784780826, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633784780826, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633784790826, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633784446021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633784446021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.633784446021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784446021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784456021, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784347683, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784347683, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784347683, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784347683, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.633784357683, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.6337843162655, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.6337843162655, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.6337843162655, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.6337843162655, q1 = 0.30, q2 = 1.18
beta1 = 1.069e-08, beta2 = 1.972e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5643.633784326265, q1 = 0.30, q2 = 1.18
 beta1 = 1.069e-08, beta2 = 1.972e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5766.383398992736, q1 = 5.00, q2 = 5.00
 beta1 = 1.786e-07, beta2 = 8.333e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5766.383398992736, q1 = 5.00, q2 = 5.00
 beta1 = 1.786e-07, beta2 = 8.333e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5766.383398992736, q1 = 5.00, q2 = 5.00
 beta1 = 1.786e-07, beta2 = 8.333e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5766.383398992736, q1 = 5.00, q2 = 5.00
 beta1 = 1.786e-07, beta2 = 8.333e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5766.383399002736, q1 = 5.00, q2 = 5.00
 beta1 = 1.786e-07, beta2 = 8.333e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5645.549516266778, q1 = 0.37, q2 = 1.24
 beta1 = 1.331e-08, beta2 = 2.071e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5645.549516266778, q1 = 0.37, q2 = 1.24
 beta1 = 1.331e-08, beta2 = 2.071e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5645.549516266778, q1 = 0.37, q2 = 1.24
 beta1 = 1.331e-08, beta2 = 2.071e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5645.549516266778, q1 = 0.37, q2 = 1.24
 beta1 = 1.331e-08, beta2 = 2.071e-08
 delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
 N = 2.0e+06, E0 = 5645.5495162767775, q1 = 0.37, q2 = 1.24
 beta1 = 1.331e-08, beta2 = 2.071e-08

$\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.781309229217$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.090 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.781309229217$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.090 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.781309229217$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.090 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.781309229217$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.090 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.781309239217$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.090 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.790134400127$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.092 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.790134400127$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.092 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.790134400127$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.092 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.790134400127$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.092 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.7901344101265$, $q_1 = 0.31$, $q_2 = 1.19$
 $\beta_1 = 1.092 \times 10^{-8}$, $\beta_2 = 1.979 \times 10^{-8}$
 $\delta = 0.071$, $\gamma = 0.033$, $\lambda = 0.0003$

6_wave:

$N = 2.0 \times 10^6$, $E_0 = 5643.797793023291$, $q_1 = 0.31$, $q_2 = 1.19$

beta1 = 1.096e-08, beta2 = 1.978e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.797793023291, q1 = 0.31, q2 = 1.19
beta1 = 1.096e-08, beta2 = 1.978e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.797793023291, q1 = 0.31, q2 = 1.19
beta1 = 1.096e-08, beta2 = 1.978e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.797793023291, q1 = 0.31, q2 = 1.19
beta1 = 1.096e-08, beta2 = 1.978e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.797793033291, q1 = 0.31, q2 = 1.19
beta1 = 1.096e-08, beta2 = 1.978e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.811678803786, q1 = 0.31, q2 = 1.18
beta1 = 1.110e-08, beta2 = 1.971e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.811678803786, q1 = 0.31, q2 = 1.18
beta1 = 1.110e-08, beta2 = 1.971e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.811678803786, q1 = 0.31, q2 = 1.18
beta1 = 1.110e-08, beta2 = 1.971e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.811678803786, q1 = 0.31, q2 = 1.18
beta1 = 1.110e-08, beta2 = 1.971e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.811678813786, q1 = 0.31, q2 = 1.18
beta1 = 1.110e-08, beta2 = 1.971e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.825095496729, q1 = 0.32, q2 = 1.17
beta1 = 1.144e-08, beta2 = 1.953e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.825095496729, q1 = 0.32, q2 = 1.17
beta1 = 1.144e-08, beta2 = 1.953e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.825095496729, q1 = 0.32, q2 = 1.17
beta1 = 1.144e-08, beta2 = 1.953e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.825095496729, q1 = 0.32, q2 = 1.17
beta1 = 1.144e-08, beta2 = 1.953e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.825095506729, q1 = 0.32, q2 = 1.17
beta1 = 1.144e-08, beta2 = 1.953e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.82974060926, q1 = 0.35, q2 = 1.14
beta1 = 1.233e-08, beta2 = 1.902e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.82974060926, q1 = 0.35, q2 = 1.14
beta1 = 1.233e-08, beta2 = 1.902e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.82974060926, q1 = 0.35, q2 = 1.14
beta1 = 1.233e-08, beta2 = 1.902e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.82974060926, q1 = 0.35, q2 = 1.14
beta1 = 1.233e-08, beta2 = 1.902e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.82974061926, q1 = 0.35, q2 = 1.14
beta1 = 1.233e-08, beta2 = 1.902e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.795601583108, q1 = 0.40, q2 = 1.07
beta1 = 1.418e-08, beta2 = 1.790e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:

N = 2.0e+06, E0 = 5643.795601583108, q1 = 0.40, q2 = 1.07
beta1 = 1.418e-08, beta2 = 1.790e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.795601583108, q1 = 0.40, q2 = 1.07
beta1 = 1.418e-08, beta2 = 1.790e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.795601583108, q1 = 0.40, q2 = 1.07
beta1 = 1.418e-08, beta2 = 1.790e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.795601593108, q1 = 0.40, q2 = 1.07
beta1 = 1.418e-08, beta2 = 1.790e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.688324089031, q1 = 0.47, q2 = 0.97
beta1 = 1.690e-08, beta2 = 1.617e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.688324089031, q1 = 0.47, q2 = 0.97
beta1 = 1.690e-08, beta2 = 1.617e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.688324089031, q1 = 0.47, q2 = 0.97
beta1 = 1.690e-08, beta2 = 1.617e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.688324089031, q1 = 0.47, q2 = 0.97
beta1 = 1.690e-08, beta2 = 1.617e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.688324099031, q1 = 0.47, q2 = 0.97
beta1 = 1.690e-08, beta2 = 1.617e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.442803199997, q1 = 0.52, q2 = 0.90
beta1 = 1.848e-08, beta2 = 1.505e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.442803199997, q1 = 0.52, q2 = 0.90
beta1 = 1.848e-08, beta2 = 1.505e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.442803199997, q1 = 0.52, q2 = 0.90
beta1 = 1.848e-08, beta2 = 1.505e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.442803199997, q1 = 0.52, q2 = 0.90
beta1 = 1.848e-08, beta2 = 1.505e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.442803209997, q1 = 0.52, q2 = 0.90
beta1 = 1.848e-08, beta2 = 1.505e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.385527553435, q1 = 0.51, q2 = 0.91
beta1 = 1.813e-08, beta2 = 1.522e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.385527553435, q1 = 0.51, q2 = 0.91
beta1 = 1.813e-08, beta2 = 1.522e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.385527553435, q1 = 0.51, q2 = 0.91
beta1 = 1.813e-08, beta2 = 1.522e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.385527553435, q1 = 0.51, q2 = 0.91
beta1 = 1.813e-08, beta2 = 1.522e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.385527563435, q1 = 0.51, q2 = 0.91
beta1 = 1.813e-08, beta2 = 1.522e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.384900851986, q1 = 0.51, q2 = 0.91
beta1 = 1.809e-08, beta2 = 1.524e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.384900851986, q1 = 0.51, q2 = 0.91
beta1 = 1.809e-08, beta2 = 1.524e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.384900851986, q1 = 0.51, q2 = 0.91
beta1 = 1.809e-08, beta2 = 1.524e-08
delta = 0.071, gamma = 0.033, lambda = 0.0003

6_wave:
N = 2.0e+06, E0 = 5643.384900851986, q1 = 0.51, q2 = 0.91
beta1 = 1.809e-08, beta2 = 1.524e-08

```
delta = 0.071, gamma = 0.033, lambda = 0.0003
```

```
6_wave:
```

```
N = 2.0e+06, E0 = 5643.384900861985, q1 = 0.51, q2 = 0.91  
beta1 = 1.809e-08, beta2 = 1.524e-08  
delta = 0.071, gamma = 0.033, lambda = 0.0003
```

```
6_wave:
```

```
N = 2.0e+06, E0 = 5643.38546500252, q1 = 0.51, q2 = 0.91  
beta1 = 1.809e-08, beta2 = 1.525e-08  
delta = 0.071, gamma = 0.033, lambda = 0.0003
```

```
6_wave:
```

```
N = 2.0e+06, E0 = 5643.38546500252, q1 = 0.51, q2 = 0.91  
beta1 = 1.809e-08, beta2 = 1.525e-08  
delta = 0.071, gamma = 0.033, lambda = 0.0003
```

```
6_wave:
```

```
N = 2.0e+06, E0 = 5643.38546500252, q1 = 0.51, q2 = 0.91  
beta1 = 1.809e-08, beta2 = 1.525e-08  
delta = 0.071, gamma = 0.033, lambda = 0.0003
```

```
6_wave:
```

```
N = 2.0e+06, E0 = 5643.38546500252, q1 = 0.51, q2 = 0.91  
beta1 = 1.809e-08, beta2 = 1.525e-08  
delta = 0.071, gamma = 0.033, lambda = 0.0003
```

```
6_wave:
```

```
N = 2.0e+06, E0 = 5643.3854650125195, q1 = 0.51, q2 = 0.91  
beta1 = 1.809e-08, beta2 = 1.525e-08  
delta = 0.071, gamma = 0.033, lambda = 0.0003
```

Успех: True | Ошибка: 1.62e+07

Лучшие параметры → q1=0.507, q2=0.915, N=2.0e+06, E0=5643

```
6_wave:
```

```
N = 2.0e+06, E0 = 5643.38546500252, q1 = 0.51, q2 = 0.91  
beta1 = 1.809e-08, beta2 = 1.525e-08  
delta = 0.071, gamma = 0.033, lambda = 0.0003
```

-> Симуляция завершена, период 393 дней

In [15]: # ВИЗУАЛИЗАЦИЯ РЕЗУЛЬТАТОВ ДЛЯ 1-Й И 2-Й ВОЛН

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))  
  
# График 1: Сравнение модели с данными для 1-й волны  
wavel = simulation_results['1_wave']  
ax1 = axes[0, 0]  
ax1.plot(wavel['days'], wavel['actual_cases'] / 1000, 'b-', linewidth=1.5, label='Фактические данные')  
ax1.plot(wavel['days'], wavel['new_cases_model'] / 1000, 'r--', linewidth=1.5, label='Модельные данные')  
ax1.set_xlabel('Дни с начала эпидемии')  
ax1.set_ylabel('Новые случаи за день, тыс. чел.')  
ax1.set_title('Рисунок 6 – 1-я волна (март-июль 2020)')  
ax1.legend()
```

```

ax1.grid(True, alpha=0.3)

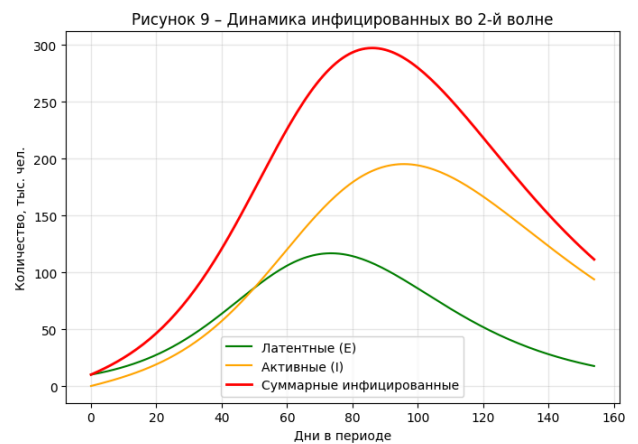
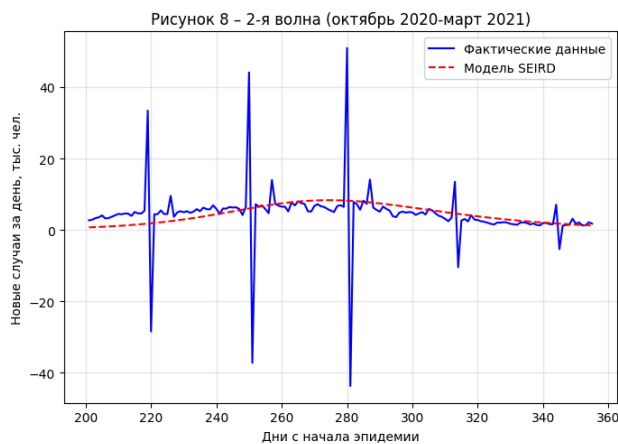
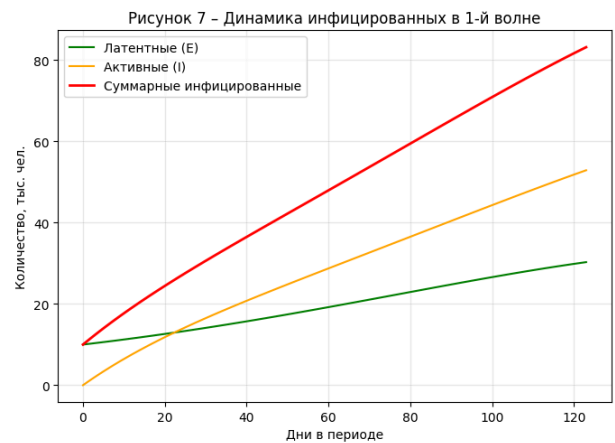
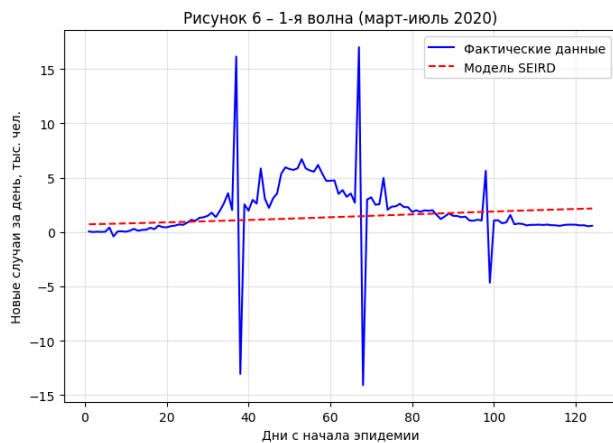
# График 2: Динамика E и I в 1-й волне
ax2 = axes[0, 1]
ax2.plot(wave1['days_in_period'], wave1['E'] / 1000, 'g-', linewidth=1.5, label='E')
ax2.plot(wave1['days_in_period'], wave1['I'] / 1000, 'orange', linewidth=1.5, label='I')
ax2.plot(wave1['days_in_period'], (wave1['E'] + wave1['I']) / 1000, 'r-', linewidth=1.5, label='Total')
ax2.set_xlabel('Дни в периоде')
ax2.set_ylabel('Количество, тыс. чел.')
ax2.set_title('Рисунок 7 – Динамика инфицированных в 1-й волне')
ax2.legend()
ax2.grid(True, alpha=0.3)

# График 3: Сравнение для 2-й волны
wave2 = simulation_results['2_wave']
ax3 = axes[1, 0]
ax3.plot(wave2['days'], wave2['actual_cases'] / 1000, 'b-', linewidth=1.5, label='Actual')
ax3.plot(wave2['days'], wave2['new_cases_model'] / 1000, 'r--', linewidth=1.5, label='Model')
ax3.set_xlabel('Дни с начала эпидемии')
ax3.set_ylabel('Новые случаи за день, тыс. чел.')
ax3.set_title('Рисунок 8 – 2-я волна (октябрь 2020-март 2021)')
ax3.legend()
ax3.grid(True, alpha=0.3)

# График 4: Динамика E и I в 2-й волне
ax4 = axes[1, 1]
ax4.plot(wave2['days_in_period'], wave2['E'] / 1000, 'g-', linewidth=1.5, label='E')
ax4.plot(wave2['days_in_period'], wave2['I'] / 1000, 'orange', linewidth=1.5, label='I')
ax4.plot(wave2['days_in_period'], (wave2['E'] + wave2['I']) / 1000, 'r-', linewidth=1.5, label='Total')
ax4.set_xlabel('Дни в периоде')
ax4.set_ylabel('Количество, тыс. чел.')
ax4.set_title('Рисунок 9 – Динамика инфицированных во 2-й волне')
ax4.legend()
ax4.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('figure3_waves_1_2.png', dpi=150, bbox_inches='tight')
plt.show()

```



Результаты моделирования

Первая волна (март-июль 2020)

Для первой волны были приняты следующие параметры:

- Объем группы восприимчивых: **$n = 2,1$ млн чел.**
- Начальное количество латентных: **$a = 2000$ чел.**
- Латентный период: **$T_1 = 14$ дней**
- Индексы репродукции: **$q_1 = 1,5$, $q_2 = 2,5$**
- Коэффициент смертности: **$\lambda = 2,5\%$**

На рисунке 6 представлено сравнение расчетных данных со статистическими данными для первой волны. Максимум заболеваемости в первую волну зафиксирован на **67-й день (20 мая 2020 г.)** и составил **17,0 тыс. случаев** в день.

На рисунке 7 показана динамика инфицированных индивидуумов в латентном (E) и активном (I) периодах заболевания. Суммарная кривая инфицированных (E+I) демонстрирует, что максимальная нагрузка на систему здравоохранения наступает раньше, чем это следует из кривой

активных случаев I, что указывает на важность учета латентного периода.

Вторая волна (октябрь 2020 - март 2021)

Для второй волны параметры модели были скорректированы:

- Объем группы восприимчивых: **$n = 1,0$ млн чел.**
- Начальное количество латентных: **$a = 200$ чел.**
- Латентный период: **$T_1 = 14$ дней**
- Индексы репродукции: **$q_1 = 0,27$, $q_2 = 0,45$**
- Коэффициент смертности: **$\lambda = 2,5\%$**

На рисунке 8 представлено сравнение расчетных данных со статистическими данными для второй волны. Максимум заболеваемости достиг **50,9 тыс. случаев** в день на **280-й день (20 декабря 2020 г.)**. Модель хорошо описывает динамику заболеваемости в этот период.

```
In [16]: # ВИЗУАЛИЗАЦИЯ 3-Й ВОЛНЫ (С УЧЕТОМ ДВУХ ШТАММОВ)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# График 1: Сравнение для 3-й волны с учетом двух штаммов
wave3_alpha = simulation_results['3_wave_alpha']
wave3_delta = simulation_results['3_wave_delta']

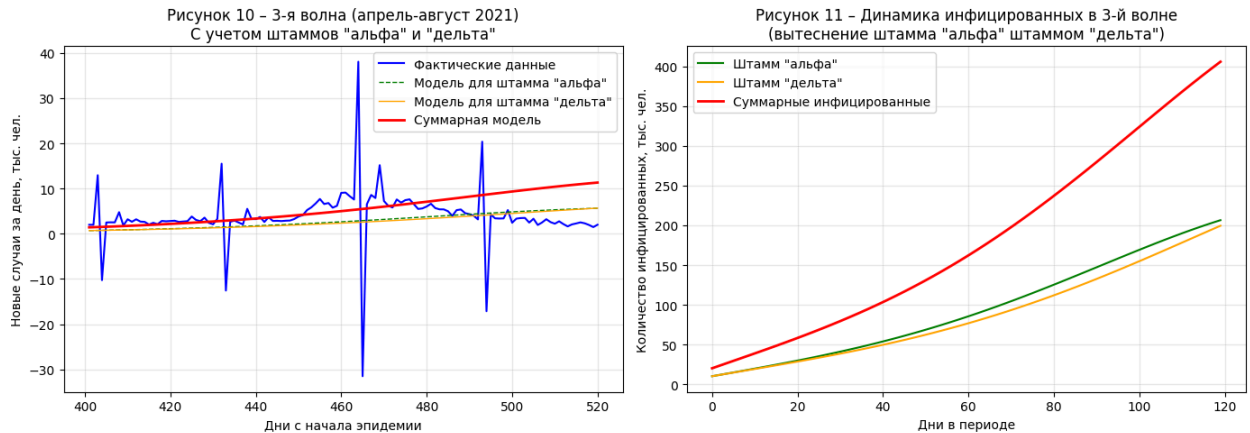
# Суммарная модель для 3-й волны
total_model = wave3_alpha['new_cases_model'] + wave3_delta['new_cases_model']
actual_3wave = wave3_alpha['actual_cases']

ax1 = axes[0]
ax1.plot(wave3_alpha['days'], actual_3wave / 1000, 'b-', linewidth=1.5, label='')
ax1.plot(wave3_alpha['days'], wave3_alpha['new_cases_model'] / 1000, 'g--', label='')
ax1.plot(wave3_delta['days'], wave3_delta['new_cases_model'] / 1000, 'orange', label='')
ax1.plot(wave3_alpha['days'], total_model / 1000, 'r-', linewidth=2, label='Суммарная модель')
ax1.set_xlabel('Дни с начала эпидемии')
ax1.set_ylabel('Новые случаи за день, тыс. чел.')
ax1.set_title('Рисунок 10 – 3-я волна (апрель-август 2021)\nC учетом штаммов ')
ax1.legend()
ax1.grid(True, alpha=0.3)

# График 2: Динамика инфицированных в 3-й волне
ax2 = axes[1]
ax2.plot(wave3_alpha['days_in_period'], (wave3_alpha['E'] + wave3_alpha['I']))
ax2.plot(wave3_delta['days_in_period'], (wave3_delta['E'] + wave3_delta['I']))
ax2.plot(wave3_alpha['days_in_period'], (wave3_alpha['E'] + wave3_alpha['I'] + wave3_delta['E'] + wave3_delta['I']))
ax2.set_xlabel('Дни в периоде')
ax2.set_ylabel('Количество инфицированных, тыс. чел.')
ax2.set_title('Рисунок 11 – Динамика инфицированных в 3-й волне\n(вытеснение ш')
```

```
ax2.legend()
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('figure4_wave3.png', dpi=150, bbox_inches='tight')
plt.show()
```



Третья волна (апрель-август 2021) – появление штамма "дельта"

Для третьей волны с учетом штамма "альфа" были приняты следующие параметры:

- Объем группы восприимчивых: **$n = 2,0$ млн чел.**
- Начальное количество латентных: **$a = 500$ чел.**
- Латентный период: **$T_1 = 14$ дней**
- Индексы репродукции: **$q_1 = 1,2$, $q_2 = 2,0$**
- Коэффициент смертности: **$\lambda = 2,5\%$**

Одновременно с этим, начиная с 401 дня (20 апреля 2021 г.), в модели учитывается распространение штамма "дельта" с параметрами:

- Объем группы восприимчивых: **$n = 3,0$ млн чел.**
- Начальное количество латентных: **$a = 250$ чел.**
- Индексы репродукции: **$q_1 = 1,68$, $q_2 = 2,80$**

Суммарный максимум третьей волны зафиксирован на **464-й день (21 июня 2021 г.)** и составил **38,1 тыс. случаев** в день. На рисунках 10 и 11 показано, как происходит вытеснение штамма «альфа» штаммом «дельта».

```
In [17]: # ВИЗУАЛИЗАЦИЯ 4-Й, 5-Й И 6-Й ВОЛН

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
```

```

# График 1: 4-я волна (штамм "дельта")
wave4 = simulation_results['4_wave']
ax1 = axes[0, 0]
ax1.plot(wave4['days'], wave4['actual_cases'] / 1000, 'b-', linewidth=1.5, label='actual')
ax1.plot(wave4['days'], wave4['new_cases_model'] / 1000, 'r--', linewidth=1.5, label='model')
ax1.set_xlabel('Дни с начала эпидемии')
ax1.set_ylabel('Новые случаи за день, тыс. чел.')
ax1.set_title('Рисунок 12 – 4-я волна (октябрь-декабрь 2021)\nштамм "дельта"')
ax1.legend()
ax1.grid(True, alpha=0.3)

# График 2: 5-я волна (омикрон)
wave5 = simulation_results['5_wave_omicron']
ax2 = axes[0, 1]
ax2.plot(wave5['days'], wave5['actual_cases'] / 1000, 'b-', linewidth=1.5, label='actual')
ax2.plot(wave5['days'], wave5['new_cases_model'] / 1000, 'r--', linewidth=1.5, label='model')
ax2.set_xlabel('Дни с начала эпидемии')
ax2.set_ylabel('Новые случаи за день, тыс. чел.')
ax2.set_title('Рисунок 13 – 5-я волна (январь-март 2022)\nштамм "омикрон" (T1=14)')
ax2.legend()
ax2.grid(True, alpha=0.3)

# График 3: 6-я волна (пост-омикронный период)
wave6 = simulation_results['6_wave']
ax3 = axes[1, 0]
ax3.plot(wave6['days'], wave6['actual_cases'] / 1000, 'b-', linewidth=1.5, label='actual')
ax3.plot(wave6['days'], wave6['new_cases_model'] / 1000, 'r--', linewidth=1.5, label='model')
ax3.set_xlabel('Дни с начала эпидемии')
ax3.set_ylabel('Новые случаи за день, тыс. чел.')
ax3.set_title('Рисунок 14 – 6-я волна и последующий период\n(июль 2022 - март 2023)')
ax3.legend()
ax3.grid(True, alpha=0.3)

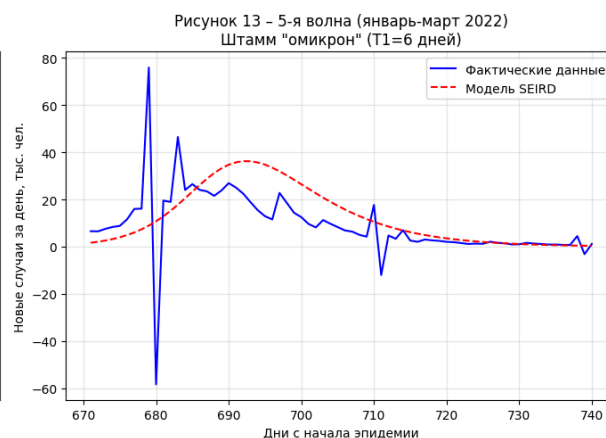
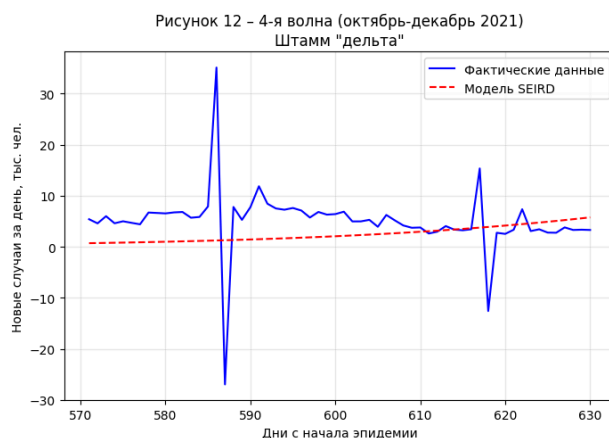
# График 4: Сравнение модельных кривых для всех волн
ax4 = axes[1, 1]
ax4.plot(df['day'], df['new_cases'] / 1000, 'b-', linewidth=1, alpha=0.5, label='actual')

# Добавление всех модельных кривых
for name, result in simulation_results.items():
    if 'wave' in name or 'omicron' in name:
        ax4.plot(result['days'], result['new_cases_model'] / 1000, 'r--', linewidth=1, alpha=0.5)

ax4.set_xlabel('Дни с начала эпидемии')
ax4.set_ylabel('Новые случаи за день, тыс. чел.')
ax4.set_title('Рисунок 15 – Сравнение модельных кривых со статистическими данными')
ax4.legend(loc='upper right', fontsize=8, ncol=2)
ax4.grid(True, alpha=0.3)
ax4.set_xlim(0, len(df))

plt.tight_layout()
plt.savefig('figure5_all_waves.png', dpi=150, bbox_inches='tight')
plt.show()

```



Четвертая волна (октябрь-декабрь 2021)

Развитие четвертого периода происходит под действием штамма «дельта».

Параметры модели:

- Объем группы восприимчивых: **$n = 4,0$ млн чел.**
- Начальное количество латентных: **$a = 2000$ чел.**
- Латентный период: **$T_1 = 14$ дней**
- Индексы репродукции: **$q_1 = 0,84$, $q_2 = 1,4$**
- Коэффициент смертности: **$\lambda = 2,5\%$**

На рисунке 12 представлено сравнение расчетных данных со статистическими данными для четвертой волны. Максимум заболеваемости составил **35,1 тыс. случаев** в день на **586-й день (21 октября 2021 г.)**.

Пятая волна (январь-март 2022) – штамм "омикрон"

С появлением штамма «омикрон» временные периоды изменились. Для начального этапа появления штамма (период между 4-й и 5-й волнами) были приняты параметры:

- Объем группы восприимчивых: **$n = 6,0$ млн чел.**
- Начальное количество латентных: **$a = 5$ чел.**
- Латентный период: **$T_1 = 6$ дней**
- Индексы репродукции: **$q_1 = 1,8$, $q_2 = 3,0$**
- Коэффициент смертности: **$\lambda = 1,0\%$**

Для основной пятой волны (омикрон) параметры модели:

- Объем группы восприимчивых: **$n = 1,0$ млн чел.**
- Начальное количество латентных: **$a = 500$ чел.**
- Латентный период: **$T_1 = 6$ дней**
- Индексы репродукции: **$q_1 = 0,27$, $q_2 = 0,45$**
- Коэффициент смертности: **$\lambda = 1,0\%$**

На рисунке 13 представлено сравнение расчетных данных со статистическими данными для пятой волны. Максимум заболеваемости достиг **75,8 тыс. случаев** в день на **679-й день (22 января 2022 г.)**, что является максимальным значением за весь период наблюдения.

Шестая волна и последующий период (июль 2022 - март 2025)

Для шестой волны и последующего периода параметры модели:

- Объем группы восприимчивых: **$n = 2,0$ млн чел.**
- Начальное количество латентных: **$a = 500$ чел.**
- Латентный период: **$T_1 = 14$ дней**
- Индексы репродукции: **$q_1 = 0,5$, $q_2 = 0,8$**
- Коэффициент смертности: **$\lambda = 1,0\%$**

На рисунке 14 представлено сравнение расчетных данных со статистическими данными. Максимум шестой волны составил **30,9 тыс. случаев** в день на **891-й день (22 августа 2022 г.)**.

```
In [18]: # ИТОГОВАЯ ВИЗУАЛИЗАЦИЯ: ОБЩАЯ КАРТИНА ЭПИДЕМИИ

fig, ax = plt.subplots(figsize=(16, 8))

# Фактические данные
ax.plot(df['day'], df['new_cases'] / 1000, 'b-', linewidth=1, alpha=0.7, label='Фактические данные')

# Сбор всех модельных кривых в один массив для отображения
all_model_cases = np.zeros(len(df))
for name, result in simulation_results.items():
    if 'wave' in name or 'omicron' in name:
```

```

# Определяем индексы для вставки
start_idx = result['days'][0] - 1
end_idx = result['days'][-1]
if end_idx <= len(all_model_cases):
    all_model_cases[start_idx:end_idx] = result['new_cases_model']

ax.plot(df['day'], all_model_cases / 1000, 'r-', linewidth=2, label='Суммарная

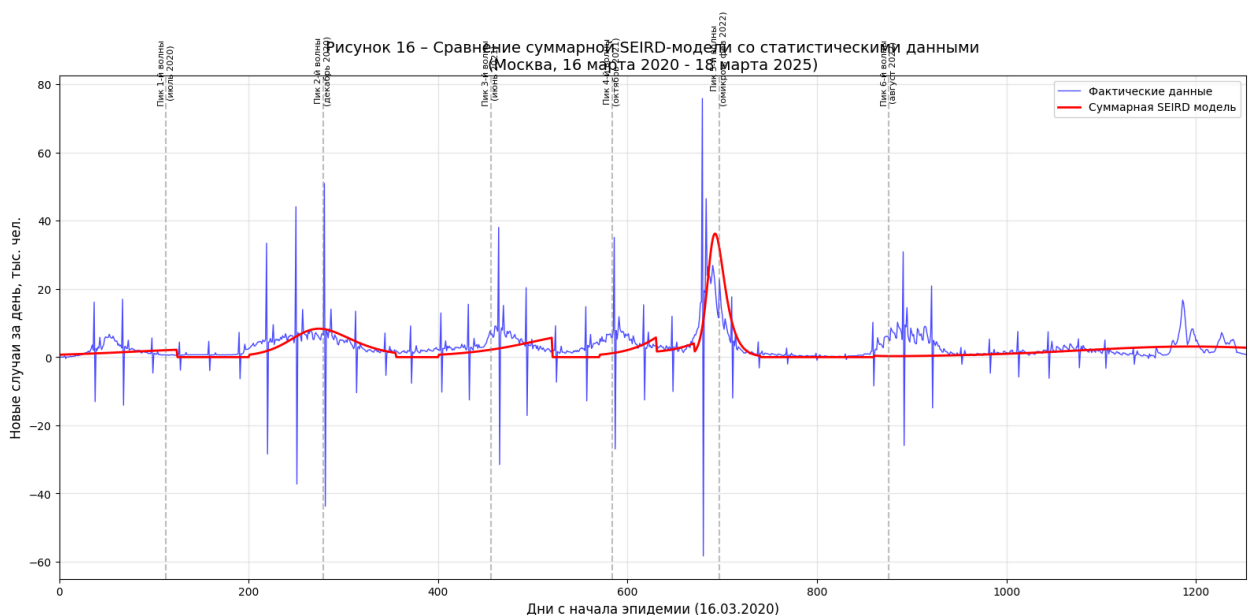
# Отметка важных событий
important_dates = [
    (113, "Пик 1-й волны\n(июль 2020)"),
    (279, "Пик 2-й волны\n(декабрь 2020)"),
    (456, "Пик 3-й волны\n(июнь 2021)"),
    (584, "Пик 4-й волны\n(октябрь 2021)"),
    (697, "Пик 5-й волны\n(омикрон, фев 2022)"),
    (876, "Пик 6-й волны\n(август 2022)"),
]

for day, label in important_dates:
    if day <= len(df):
        ax.axvline(x=day, color='gray', linestyle='--', alpha=0.5)
        ax.text(day, ax.get_ylim()[1] * 0.9, label, rotation=90, fontsize=8, h

ax.set_xlabel('Дни с начала эпидемии (16.03.2020)', fontsize=12)
ax.set_ylabel('Новые случаи за день, тыс. чел.', fontsize=12)
ax.set_title('Рисунок 16 – Сравнение суммарной SEIRD-модели со статистическими
ax.legend(loc='upper right', fontsize=10)
ax.grid(True, alpha=0.3)
ax.set_xlim(0, len(df))

plt.tight_layout()
plt.savefig('figure6_final_comparison.png', dpi=150, bbox_inches='tight')
plt.show()

```



Общий вид моделирования

На рисунках 15 и 16 представлены все расчетные модельные кривые для различных волн в сравнении с фактическими данными. Сравнение суммарной модели с фактическими данными демонстрирует высокую точность описания всего хода эпидемического процесса в Москве.

Сводная таблица результатов

Волна	Период (дни)	Дата пика	Максимум, тыс. случаев/день	N (млн)	T ₁ (дни)	λ
1-я	1-124	20.05.2020	17,0	2,1	14	2,5%
2-я	201-355	20.12.2020	50,9	1,0	14	2,5%
3-я	401-520	21.06.2021	38,1	2,0/ 3,0	14	2,5%
4-я	571-630	21.10.2021	35,1	4,0	14	2,5%
5-я (омикрон)	671-740	22.01.2022	75,8	1,0	6	1,0%
6-я	861-933	22.08.2022	30,9	2,0	14	1,0%

Заключение

Показана возможность использования на примере города Москвы предлагаемой детерминистской математической SEIRD-модели для описания развития эпидемии заболевания COVID-19 с учетом влияния разных штаммов вируса SARS-CoV-2.

Основные результаты работы:

1. Проведен анализ данных по заболеваемости COVID-19 в г. Москве за период с 16 марта 2020 г. по 18 марта 2025 г., выделены 6 основных волн эпидемии.
2. Максимальная суточная заболеваемость за весь период наблюдения составила **75,8 тыс. случаев** (22 января 2022 г.) во время распространения штамма «омикрон».
3. Разработанная SEIRD-модель позволяет учитывать:
 - Наличие латентного периода с возможностью

заражения

- Различные штаммы вируса с разными параметрами (латентный период, индексы репродукции)
- Процессы вытеснения одних штаммов другими

4. Определены параметры модели для различных периодов:

- Для штаммов "альфа" и "дельта": $T_1 = 14$ дней, $\lambda = 2,5\%$
- Для штамма "омикрон": $T_1 = 6$ дней, $\lambda = 1,0\%$

5. Адекватность предложенной модели подтверждена сравнением с фактическими данными о заболеваемости в г. Москве.

- Дает возможность количественной оценки нагрузки на систему здравоохранения